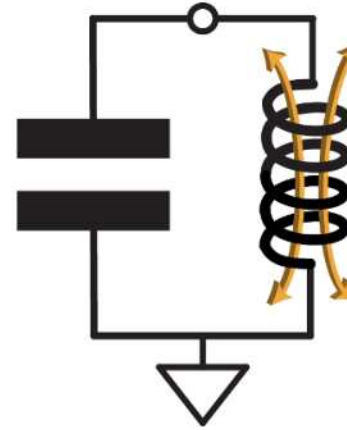


Advancing Quantum Sensing via Quantum Control

Superconducting



Zlatko K. Minev

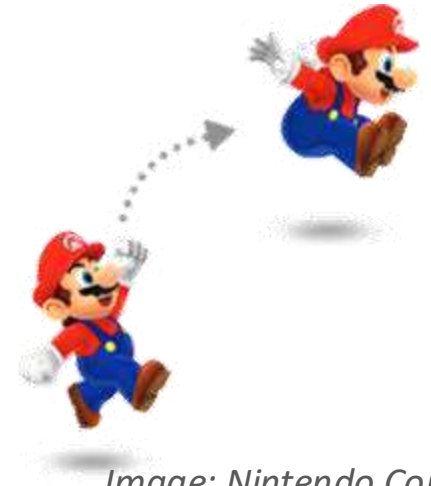


Image: Nintendo Corp.



Until recently
Devoret Group
Yale University

IBM Q[™]

Presently
IBM-Q Research (Gambetta)
Yorktown, NY



@zlatko_minev



zlatko-minev.com

Outline

What is quantum sensing?

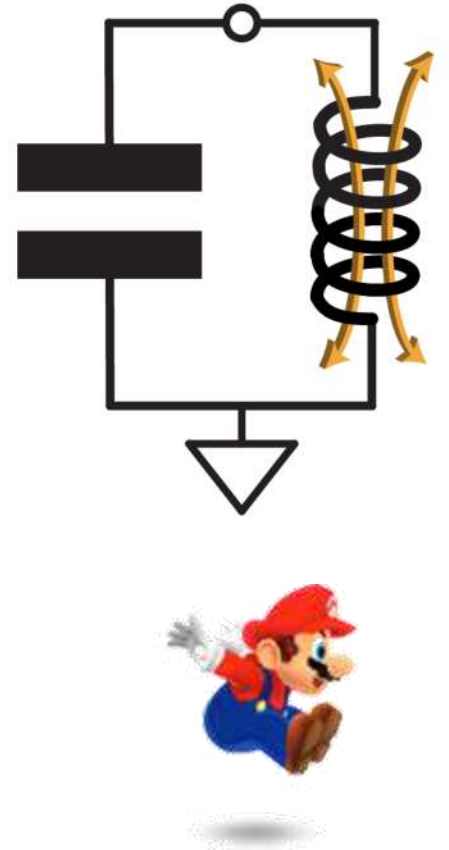
Measurements

Circuits

Sensing with circuits: survey

Case study

Sensing photons with nearly 100% efficiency
To catch and reverse a quantum jump mid-flight



What is quantum sensing?

(a discussion)

What is quantum sensing?

“Quantum sensing” describes the use of a quantum system, quantum properties, or quantum phenomena to perform a measurement of a physical quantity.

(close to application)

Degen, Reinhard, and Cappellaro, RMP (2017)

a physical quantity is defined by prescribing the operations that are carried out in order to measure it.

Cook, A.H., Cambridge (1994)

“I know it when I see it.”

378 U.S. at 197 (Stewart, J., concurring)

Principal element of sensing: the measurement

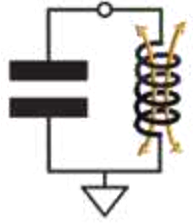
Dilbert.com DilbertCartoonist@gmail.com



4-17-12 ©2012 Scott Adams, Inc./Dist. by Universal Uclick



Advancing Quantum Sensing via Quantum Control with



Good at measurements

(principal element of sensing)

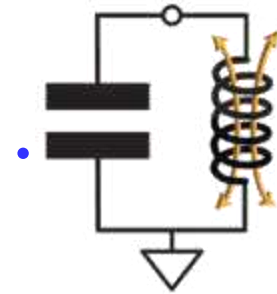
Good at real-time control

Versatile

Thesis of the talk

Quantum measurements: a primer

with emphasis on sensing for quantum



continuous

generally dyne detection (diffusive)

time-resolved

near-unit efficiency

resolve the collapse dynamics

A classical measurement example

Are there fumes in the oil barrel?



Example based on Wiseman and Milburn (2010)

image:freepik.com

Two basic classes of measurements

Demolition



Non-demolition



Caution: Classical
measurements can
also be disturbing!

e.g., photon absorption

e.g., dispersive cavity

Both accessible in circuits, more on this later

Dynamics of interrogating a qubit

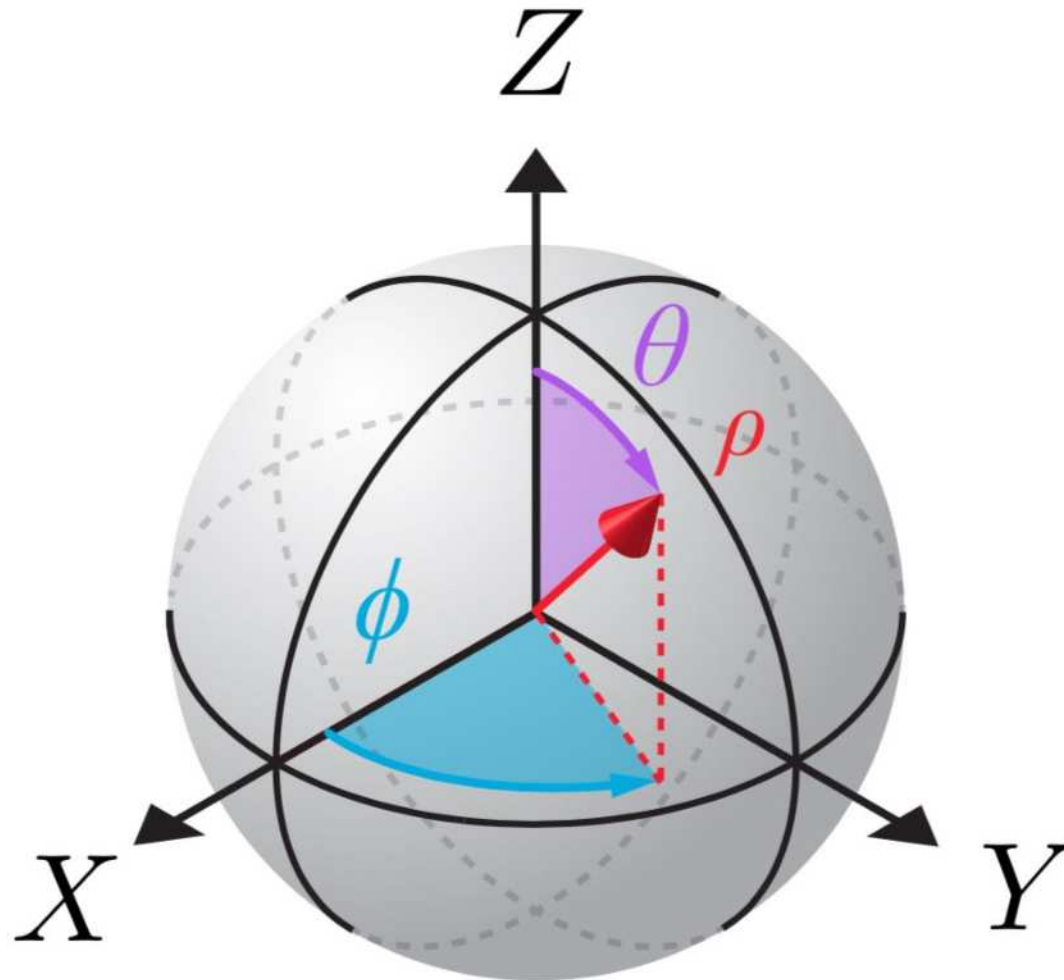
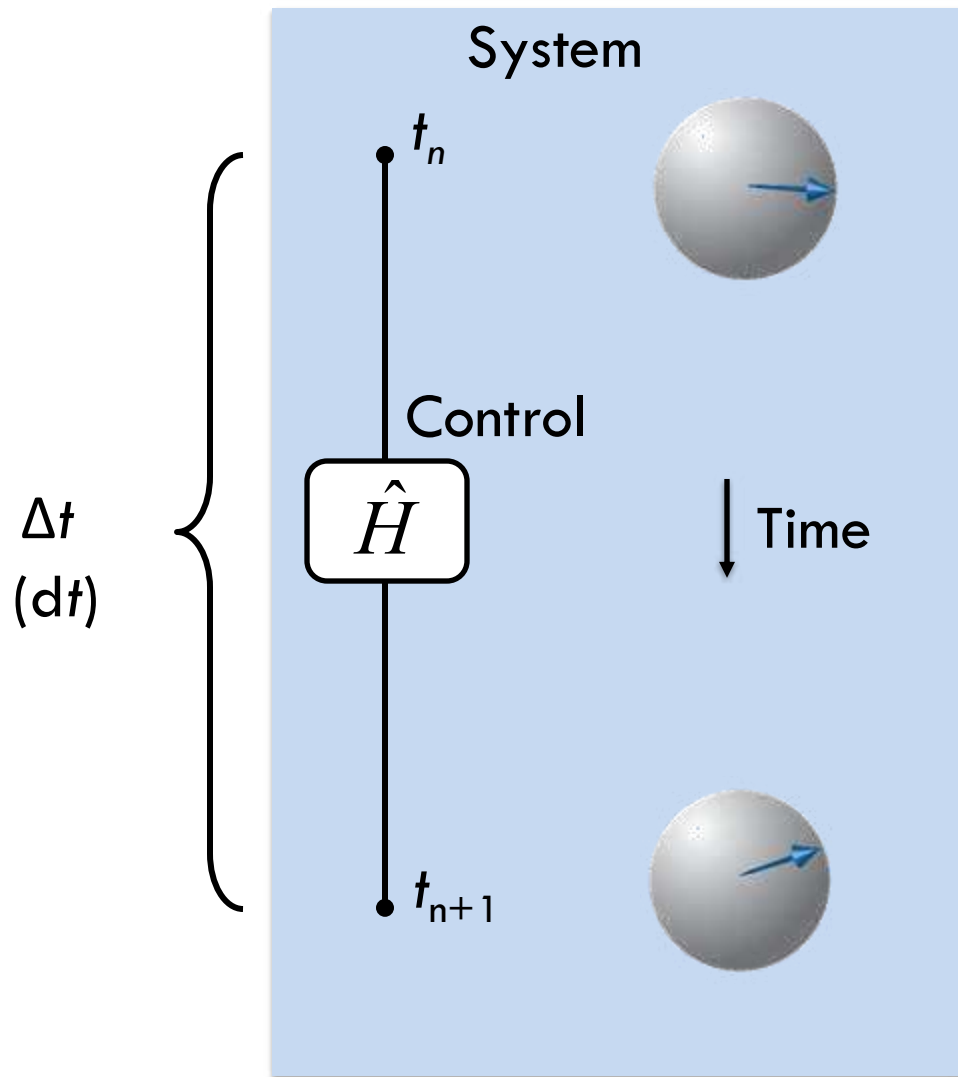
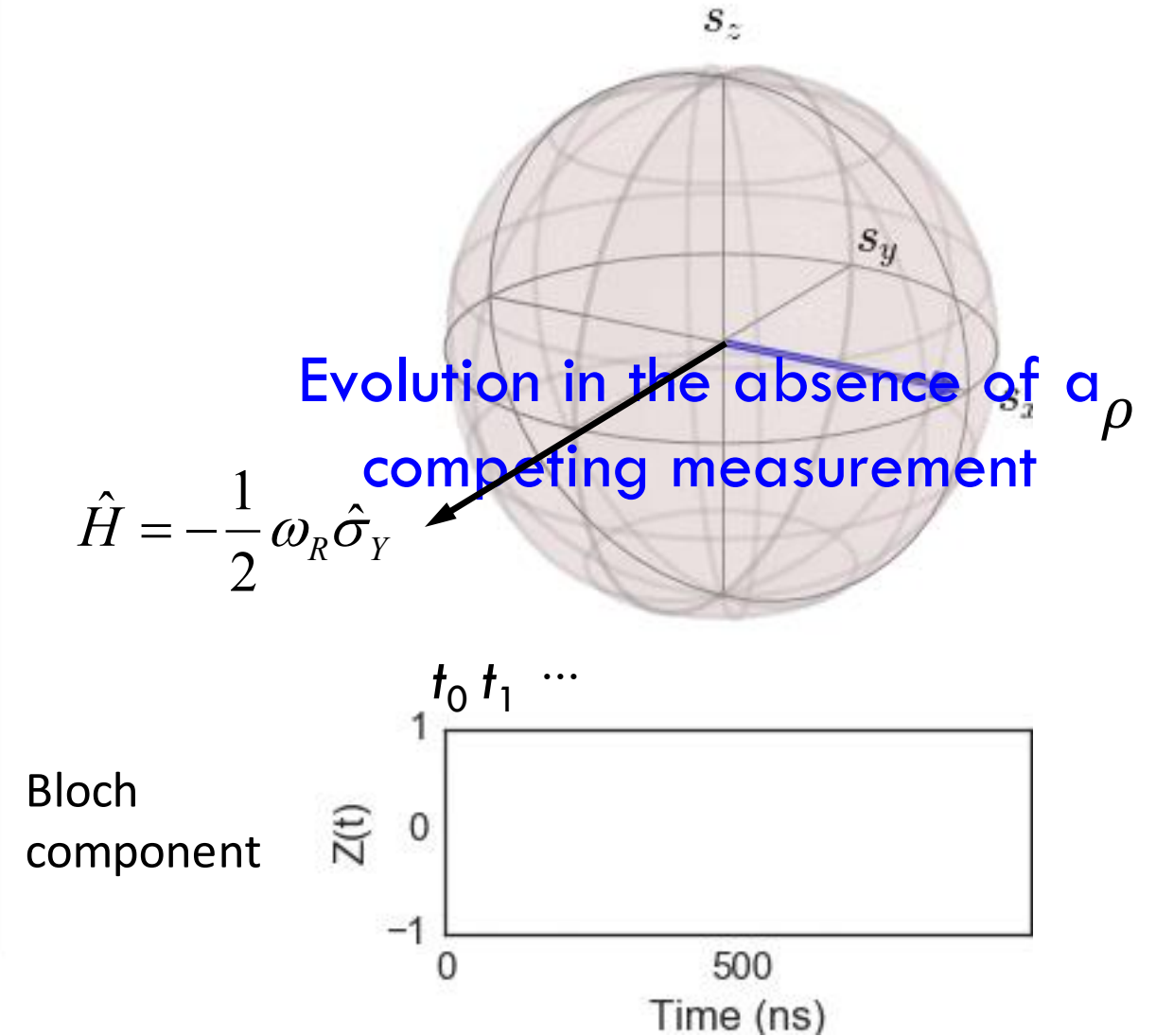


image: Mineev, Yale Thesis (2018)

Isolated system



Example: Rabi drive



Open quantum system

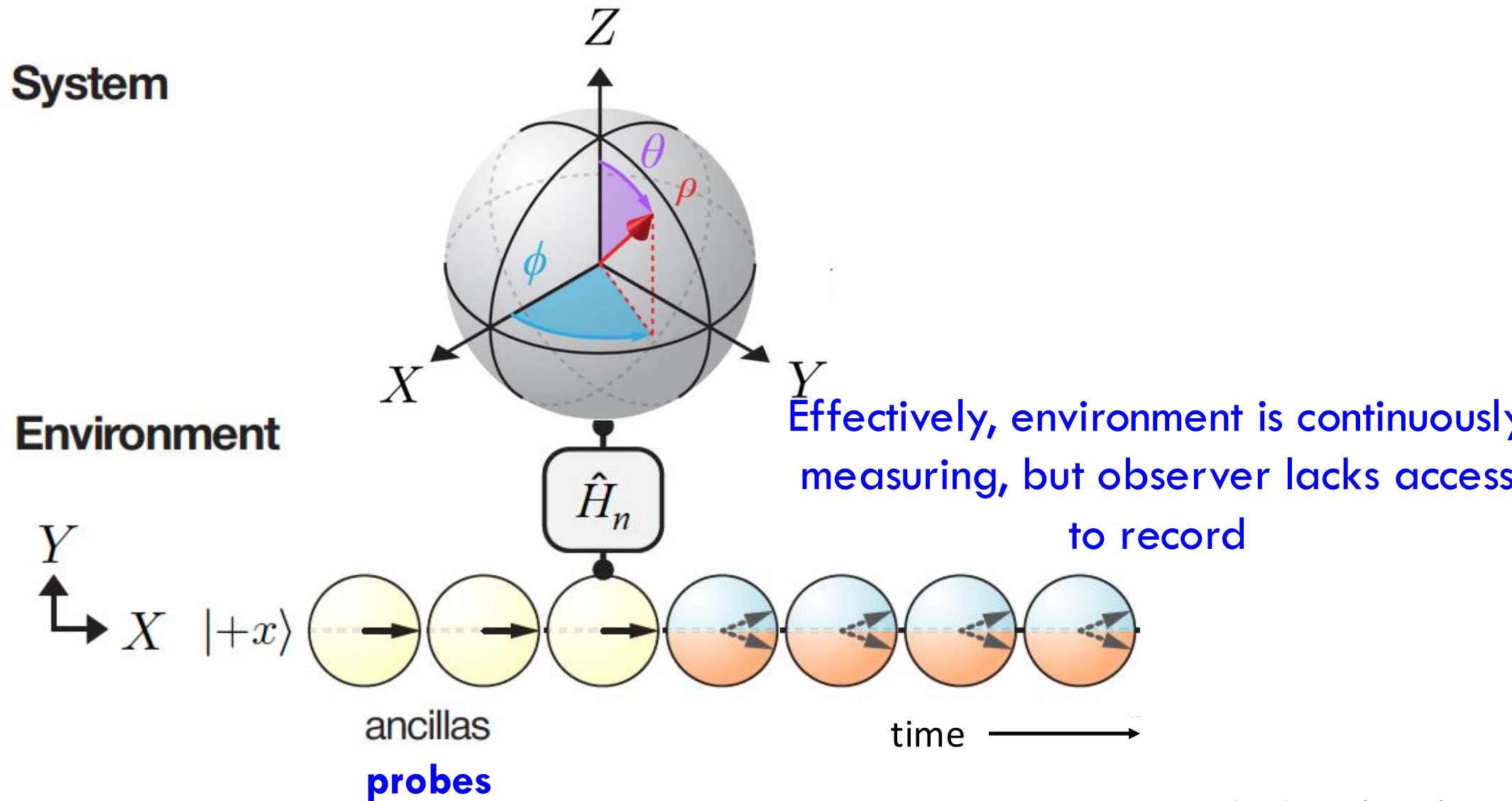
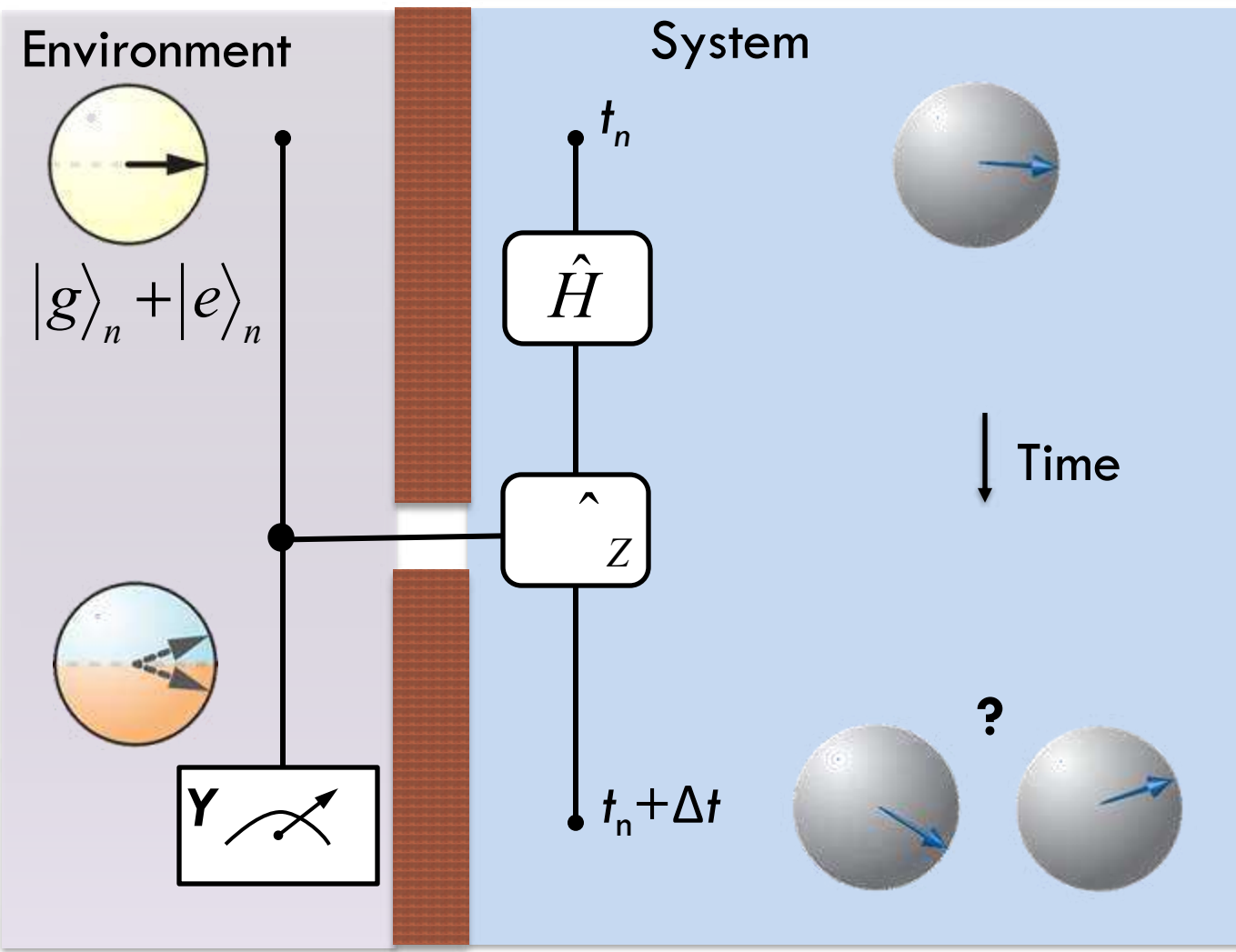
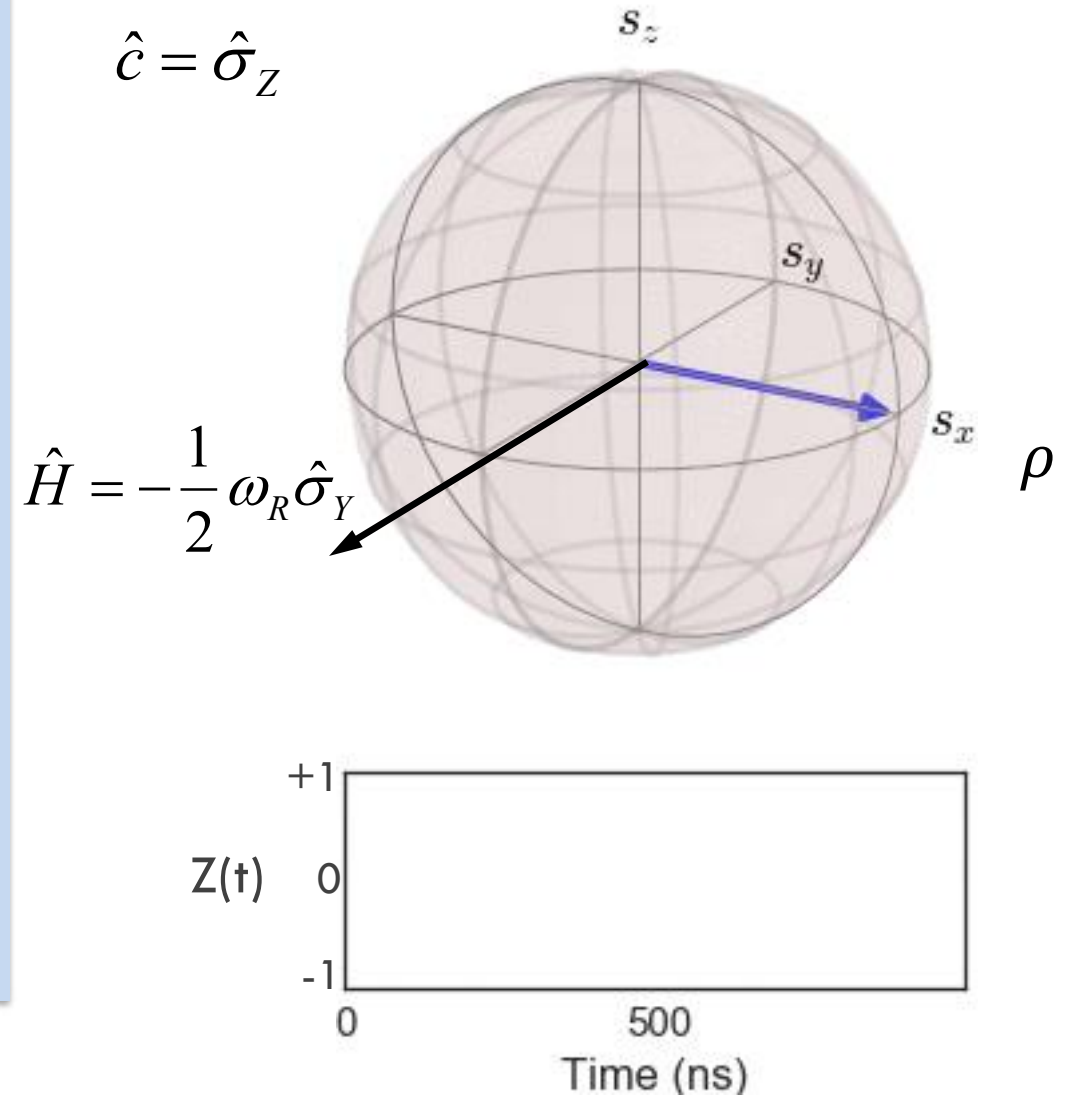


image: Mineev, Yale Thesis (2018)

Open system

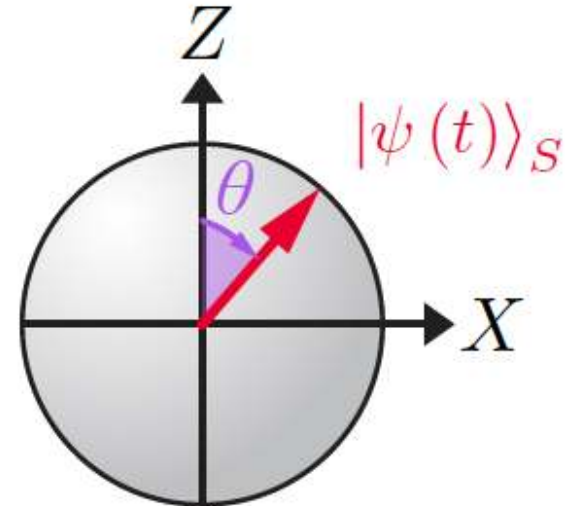


Example

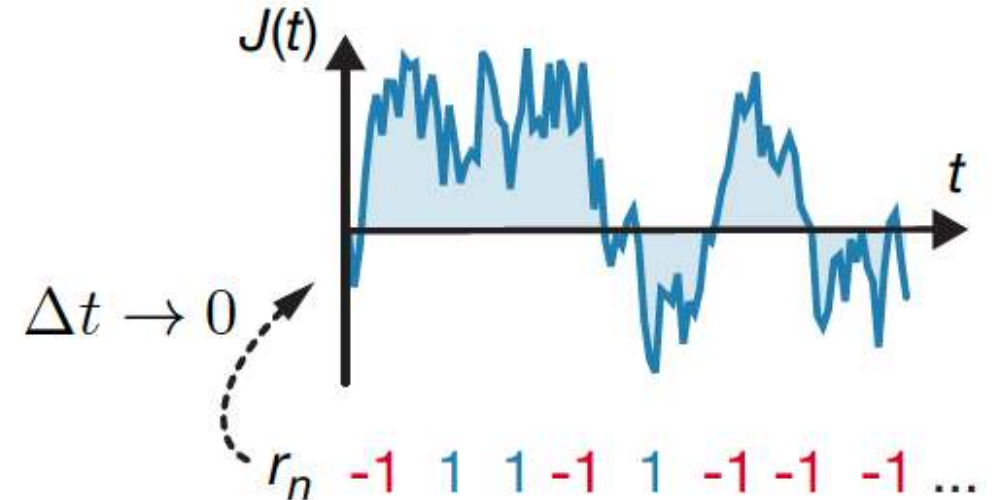


Model of a monitored quantum system

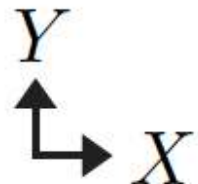
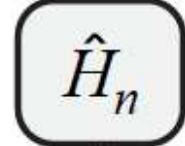
System



Measurement record

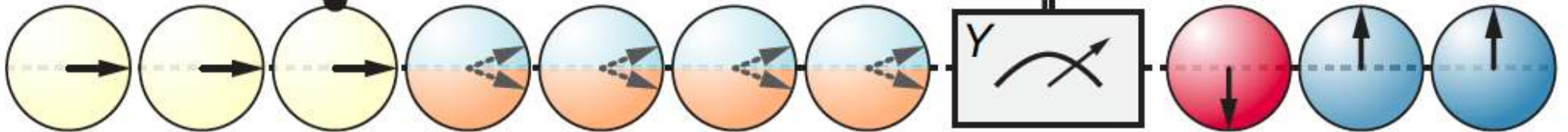


Environment + observer



$|+x\rangle$

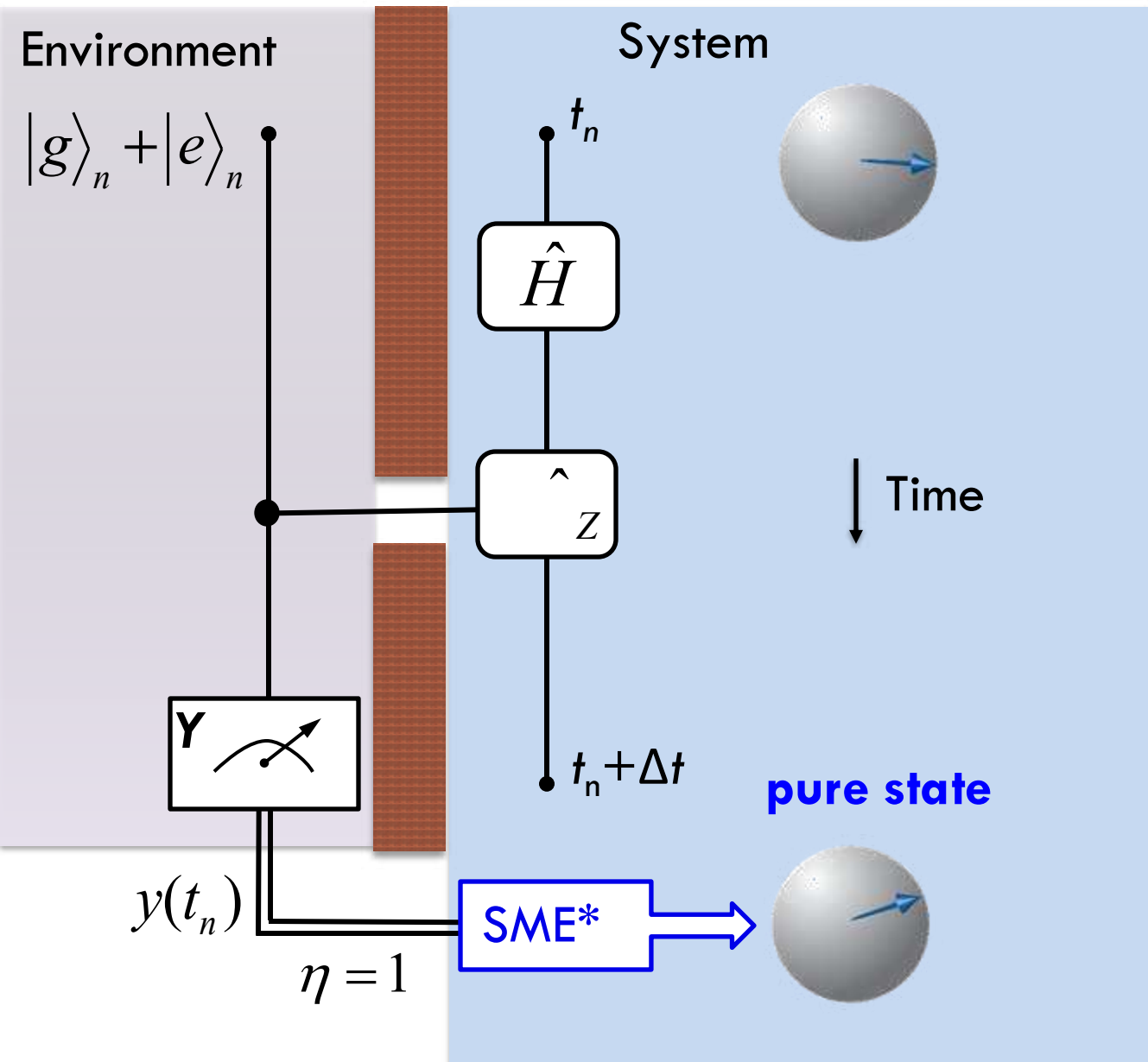
Measurement probes



time $\longrightarrow$

image: Mineev, arXiv:1902.10355 (2018)

System + omniscient observer

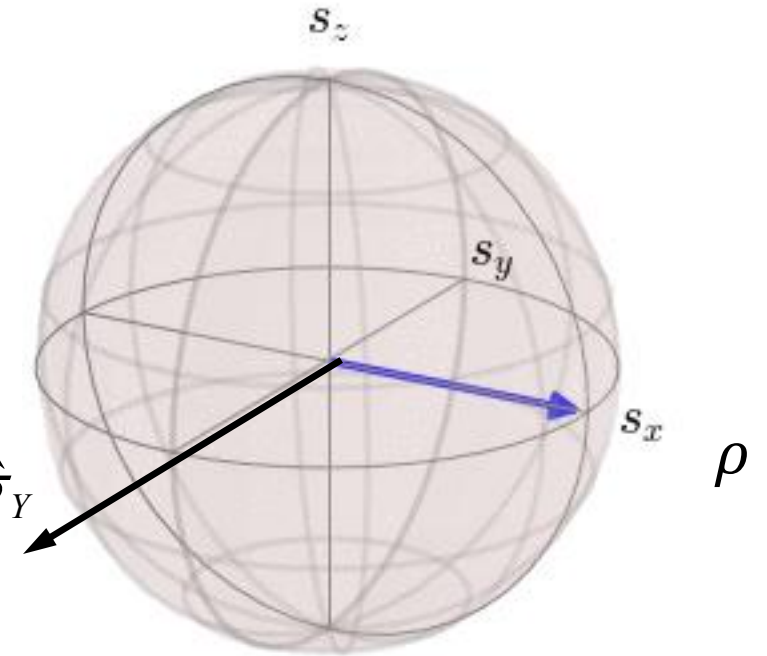


Example

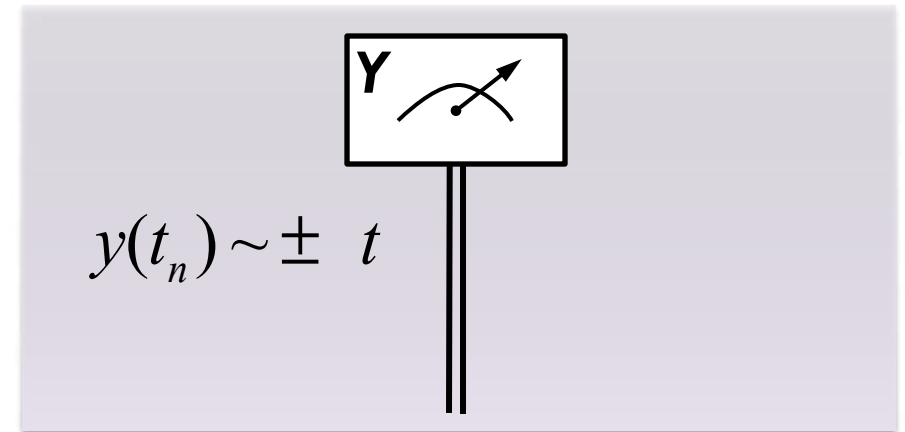
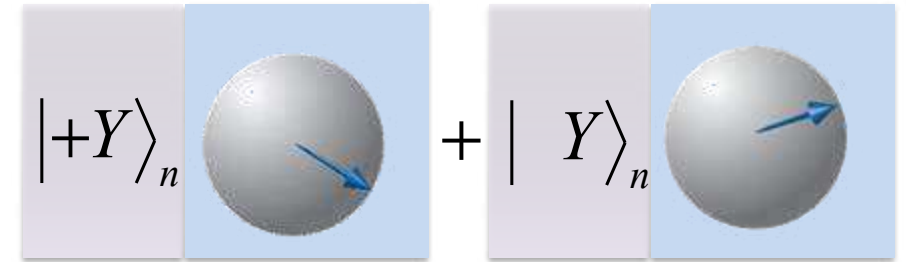
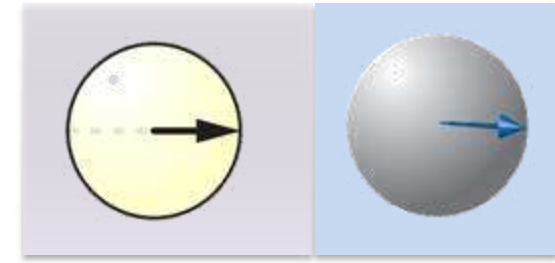
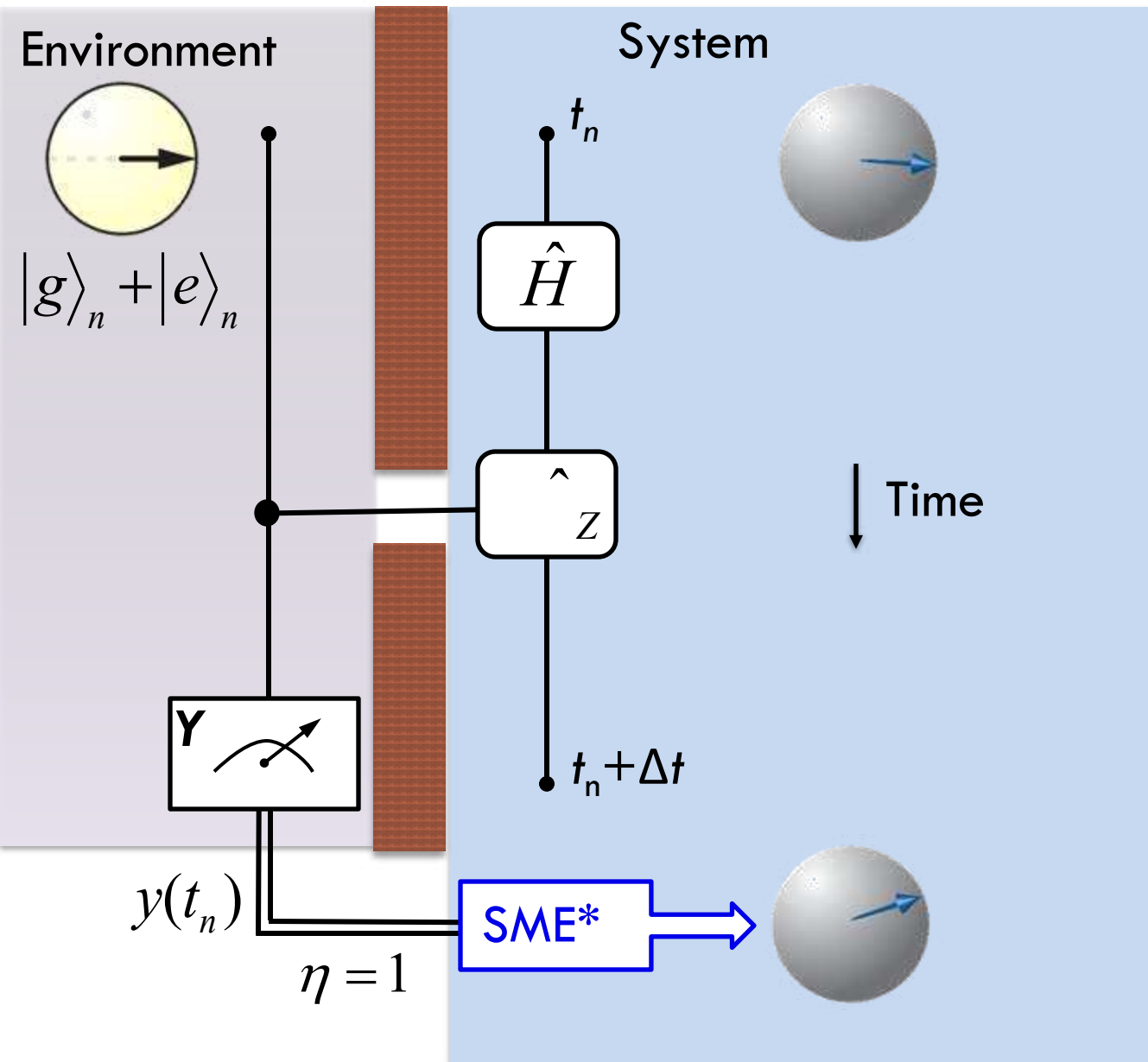
$$\hat{c} = \hat{\sigma}_Z$$

homodyne

$$\hat{H} = -\frac{1}{2} \omega_R \hat{\sigma}_Y$$

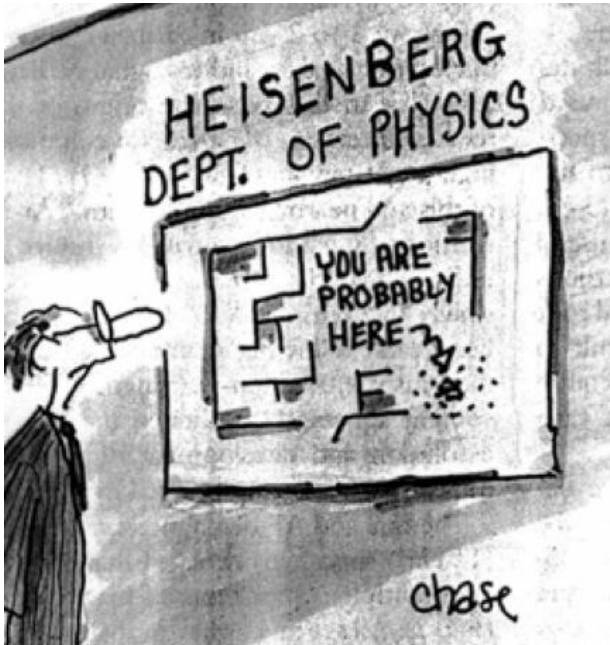


System + omniscient observer



see for instance, Carmichael, *An Open Systems Approach to Quantum Optics* (1993)

Takeaways: Basic character of quantum measurements



Necessarily *disturb* system (back-action)

non-commuting

Heisenberg uncertainty

fundamental limits to precision, e.g., SQL, ...

no joint probability distribution (over x , p)

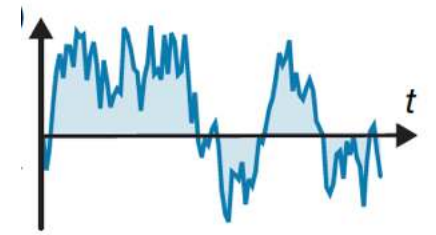
quasi-probabilities (Wigner functions)

no classical Fisher information

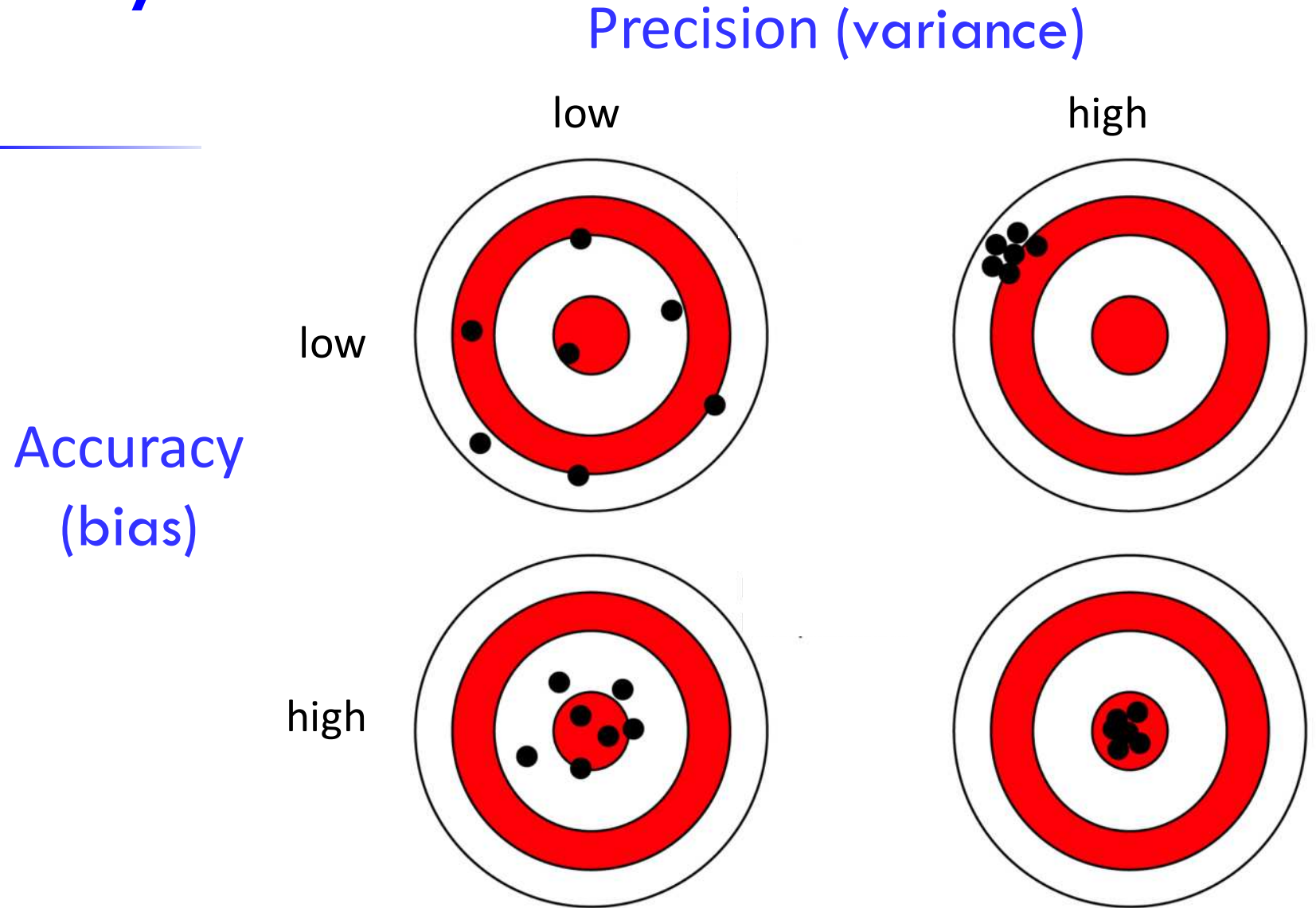
entropy-increasing

contextuality

....

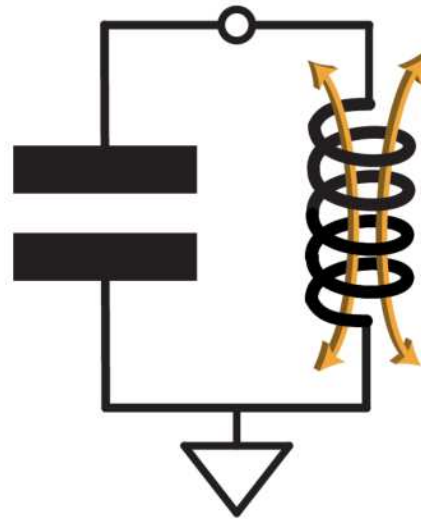


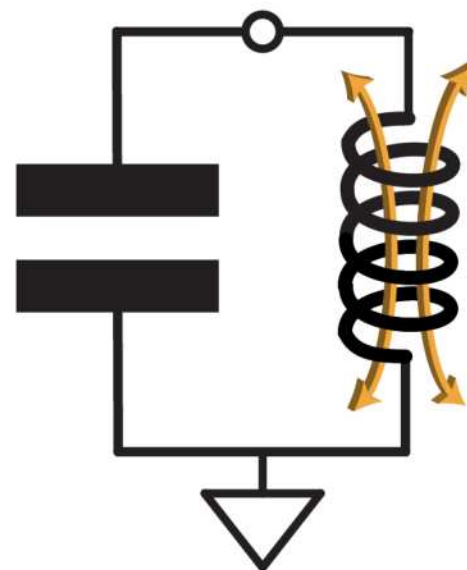
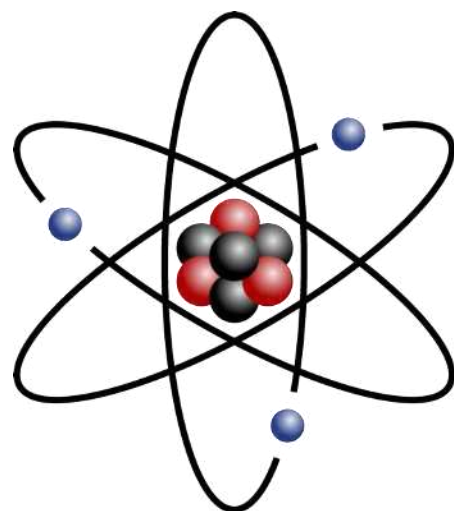
Measurement: accuracy vs. precision

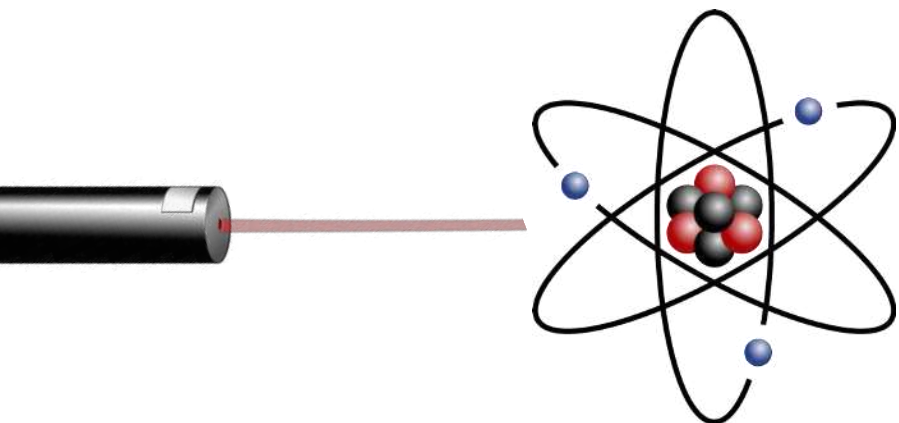


Caution: Within the ISO (5725) framework, there is a slight but important shift in the meaning of these terms

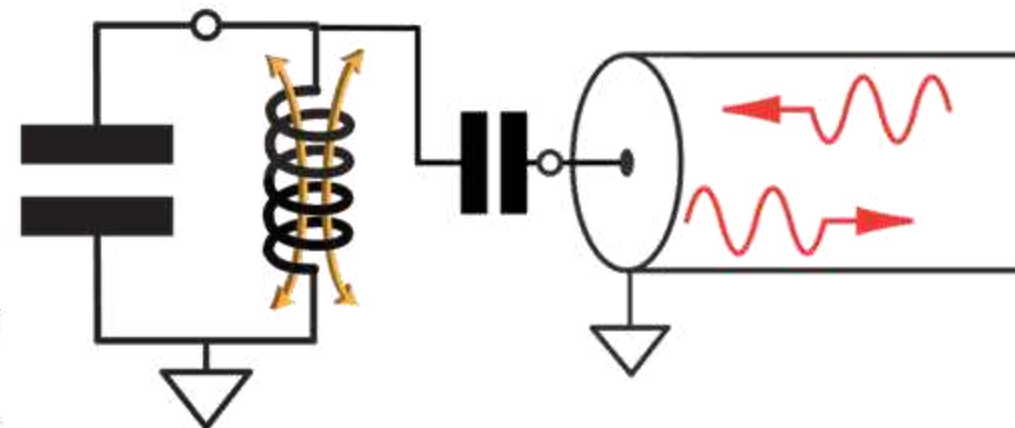
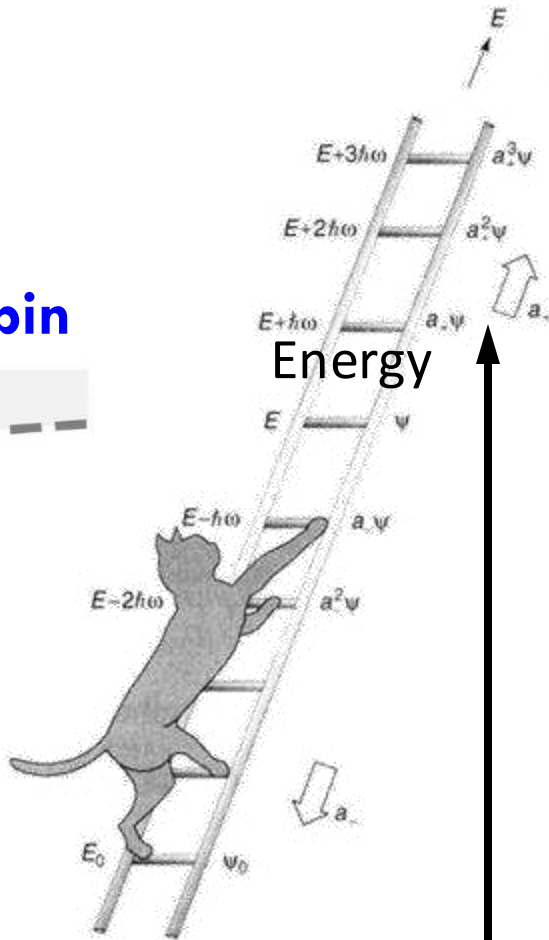
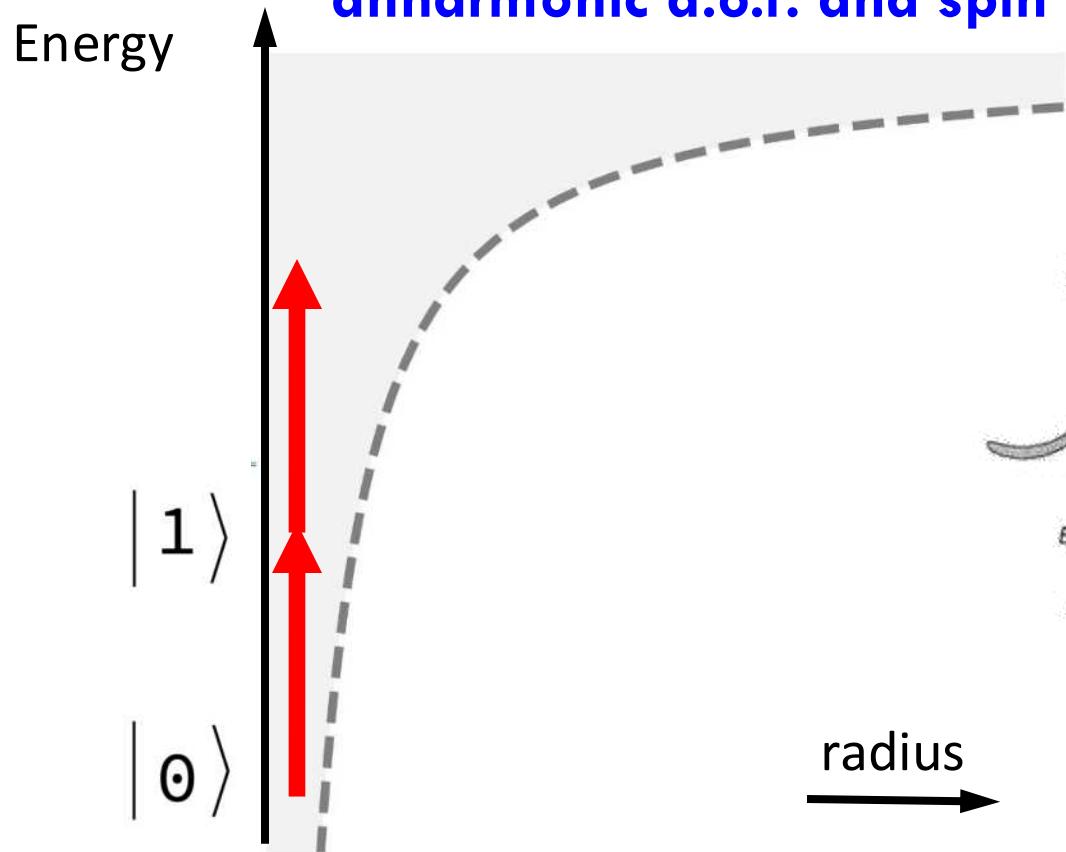
Superconducting quantum circuits



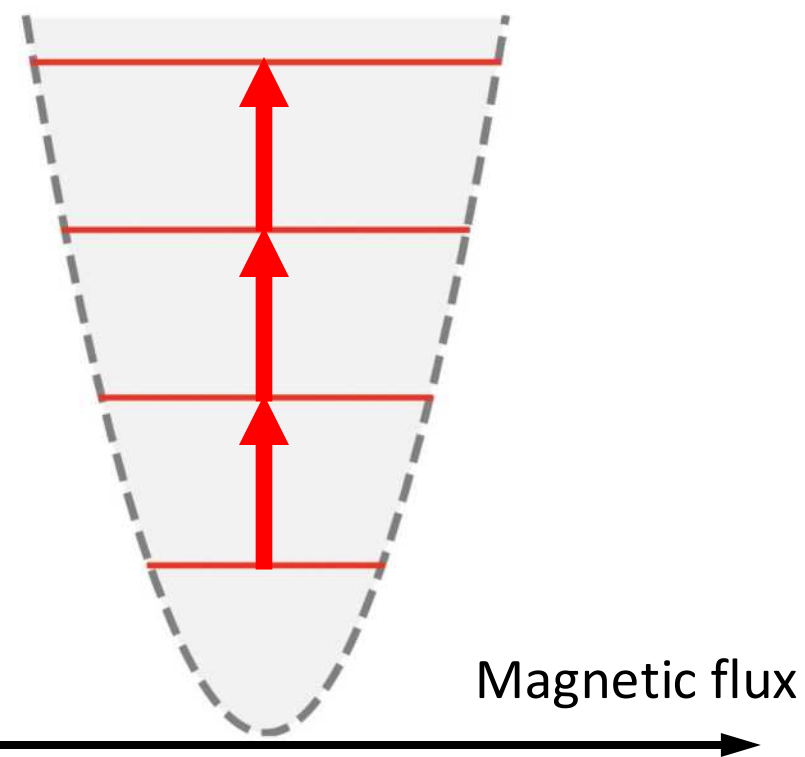


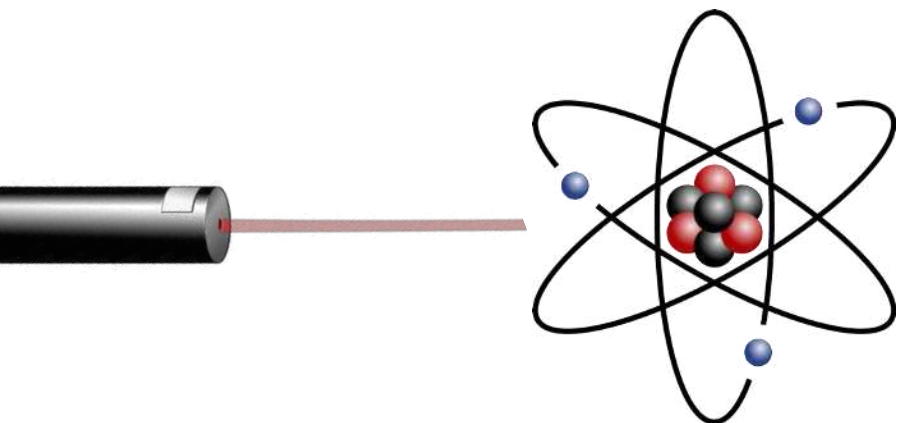


anharmonic d.o.f. and spin

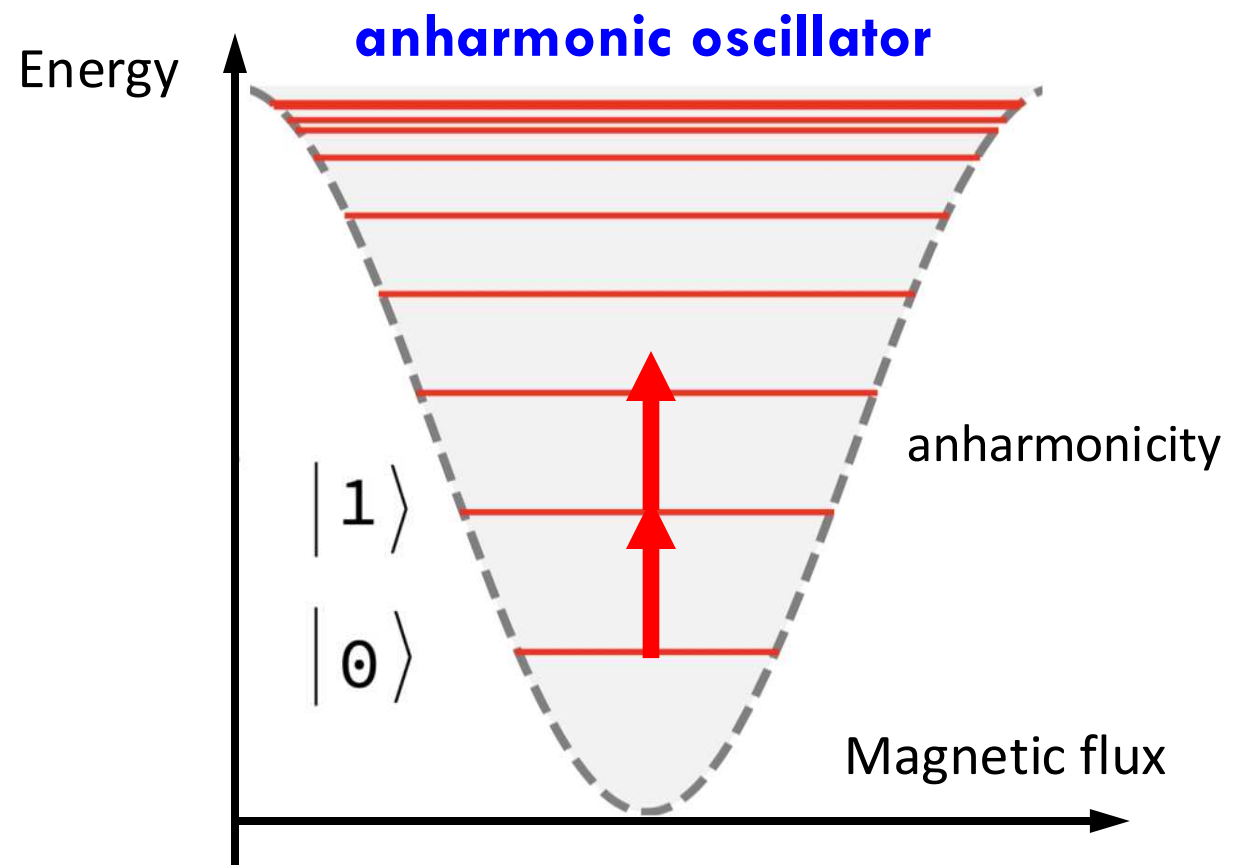
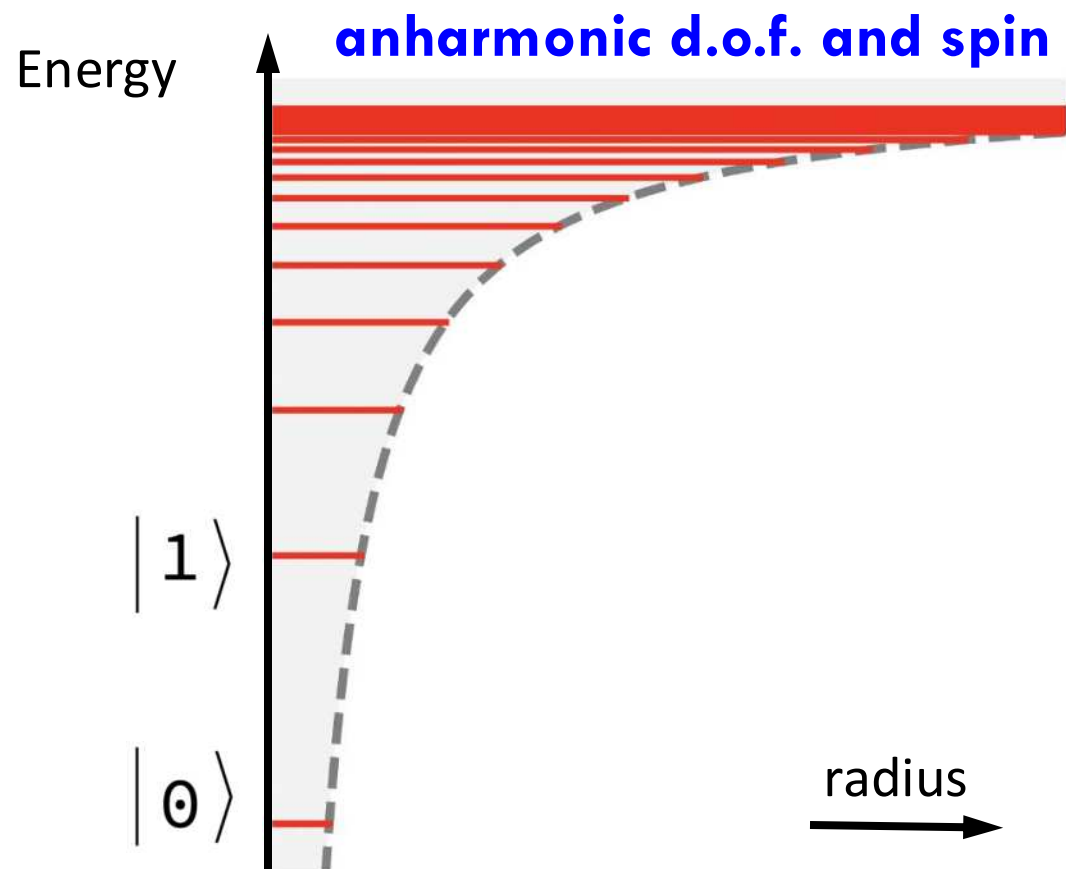
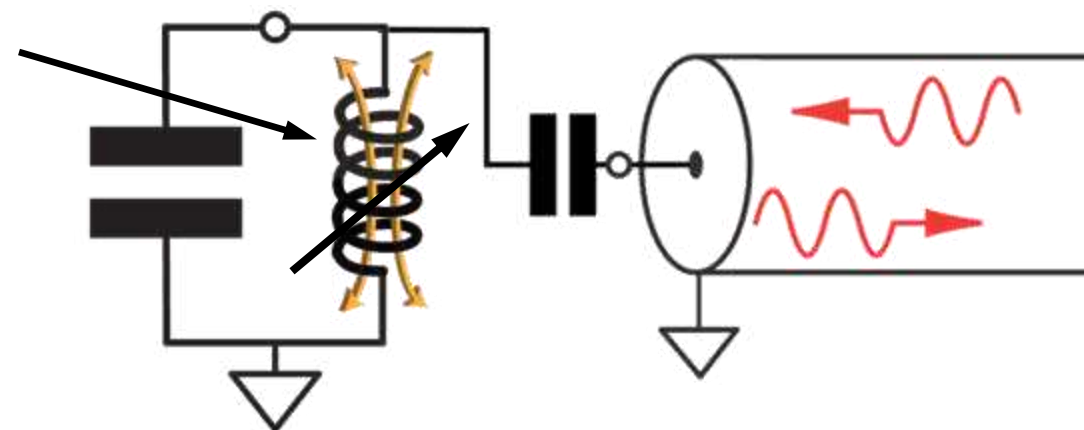


Harmonic oscillator

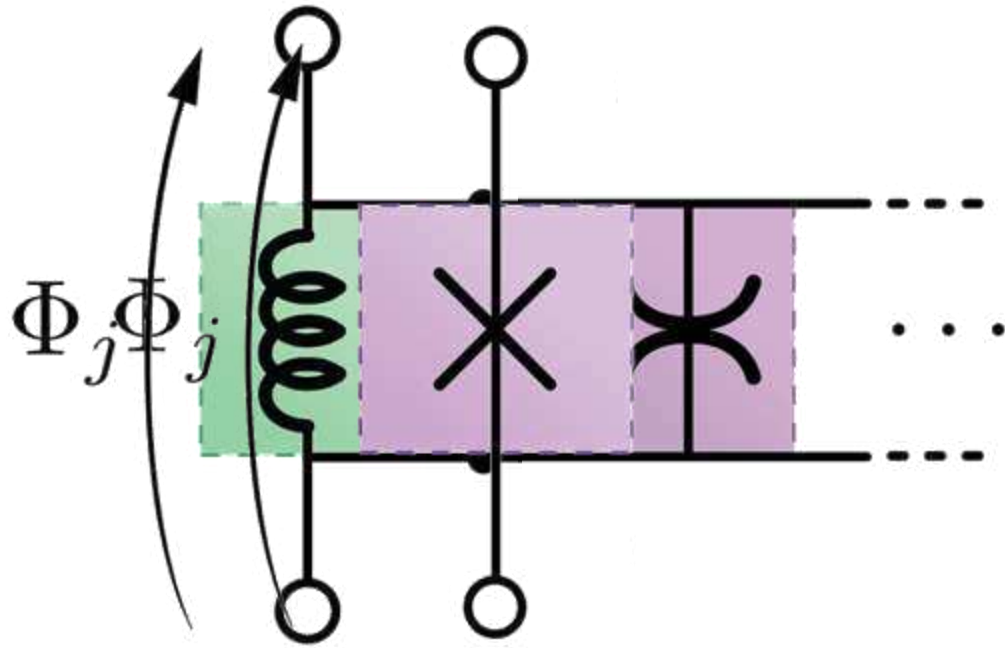




non-linear
inductor



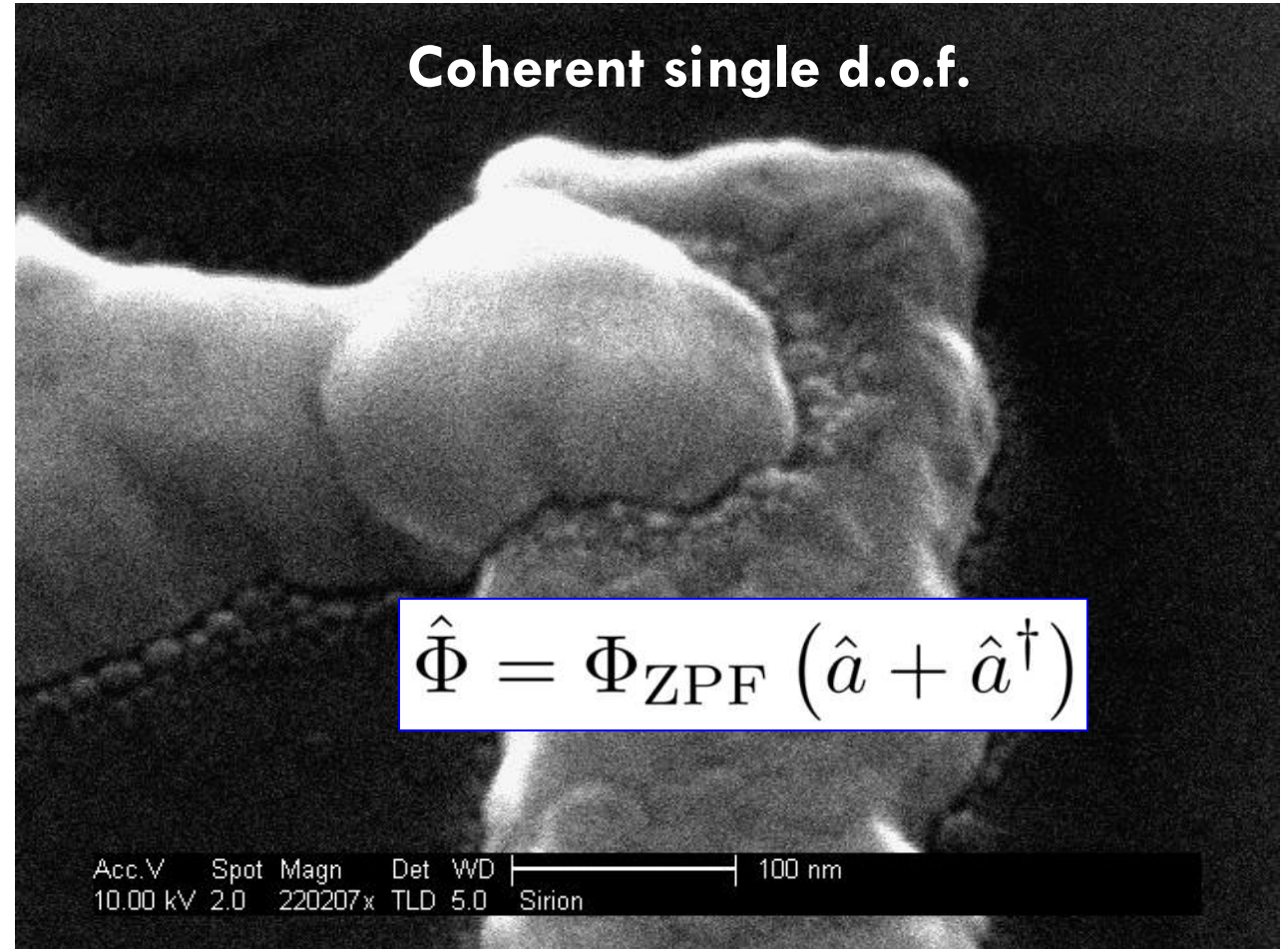
Non-linear element: Josephson tunnel junction



$$\mathcal{E}_j(\Phi_j) = E_j (1 - \cos(\Phi_j/\phi_0))$$

$$\phi_0 \equiv \hbar/2e \Phi_j + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$

$$= \frac{E_j}{2} \left(\frac{\Phi_j}{\phi_0} \right)^2 - \frac{E_j}{4!} \left(\frac{\Phi_j}{\phi_0} \right)^4 + \mathcal{O}(\Phi_j^6)$$



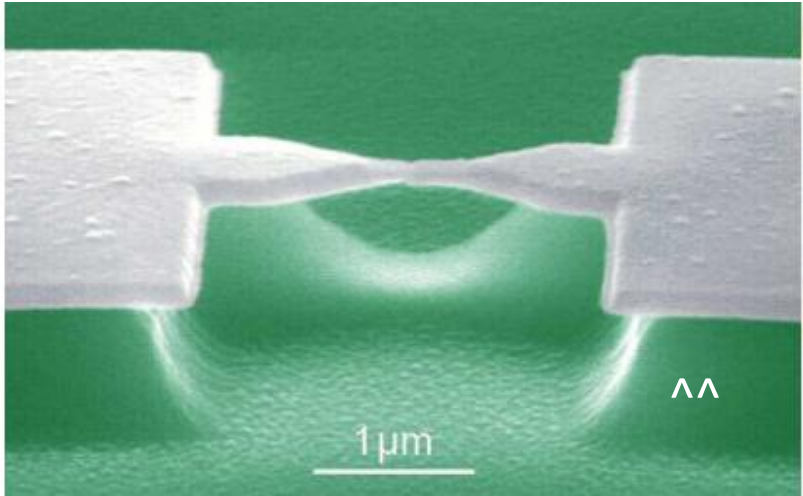
Coherent single d.o.f.

$$\hat{\Phi} = \Phi_{\text{ZPF}} (\hat{a} + \hat{a}^\dagger)$$

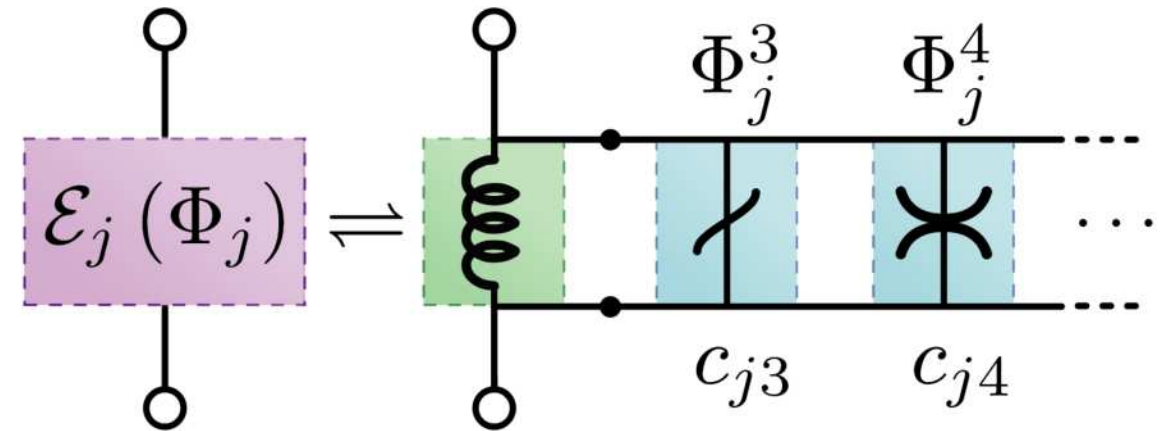
* Mineev, Sec. 4.1.1, arXiv:1902.10355

SEM image: L. Frunzio

More general non-linear and hybrid elements



$$\mathcal{E}_j(\Phi_j) = \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j) \quad *$$



weak-links
 nanobridges
 kinetic-inductance
 atomic point contacts
 semiconducting nanowires
 2degs
 ...

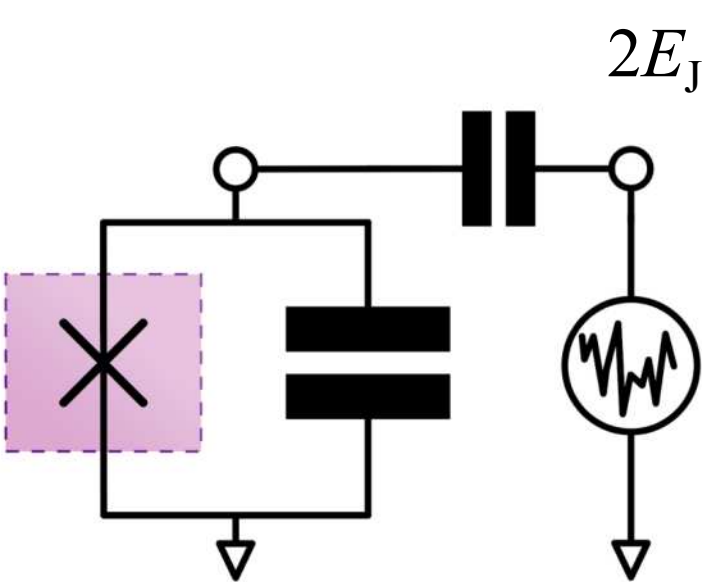
^^ image: Janvier 2015; Vijay 2010, Peltonen 2016; Mooij 2006, Abay 2014, Larsen 2015, DeLange 2015, Casparis 2016; Ho Eom 2012, Vissers 2015; Shim 2014; Shabani 2019; ...

* Mineev, Sec. 4.1.1, arXiv:1902.10355 and Mineev *et al.*, in prep (2019)

Transmon qubit

Circuit

$$\Phi_{\text{ZPF}}/\phi_0 \sim 0.1$$

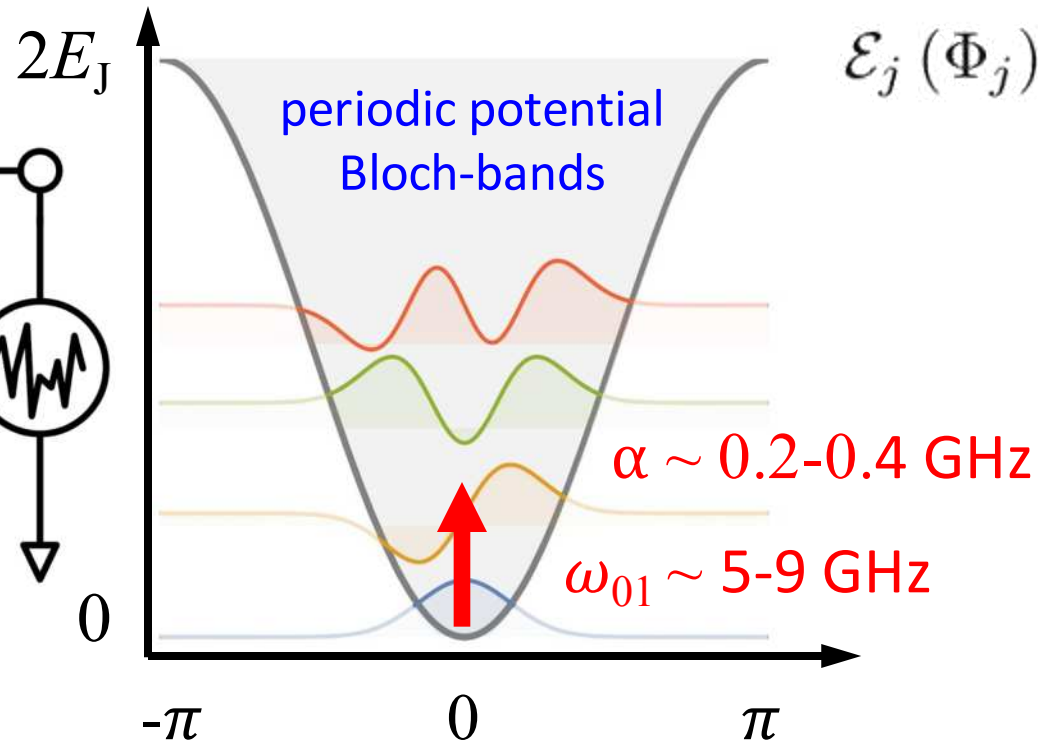


Regime reduces
charge sensitivity
Koch et al. (2007)

Blais et al. (2004)

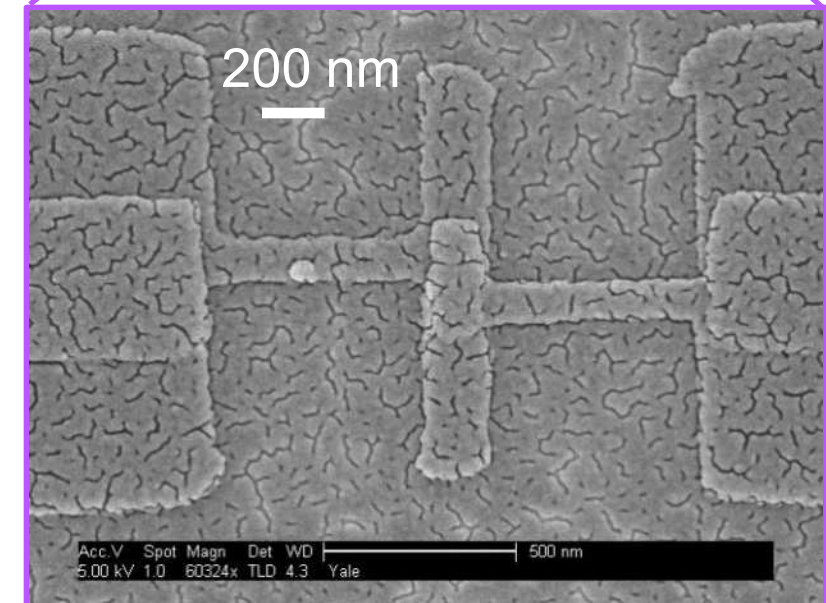
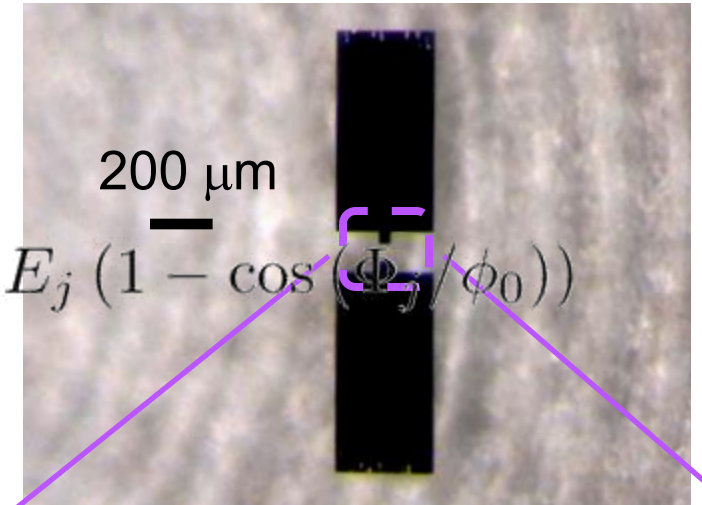
Energy spectrum

$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$



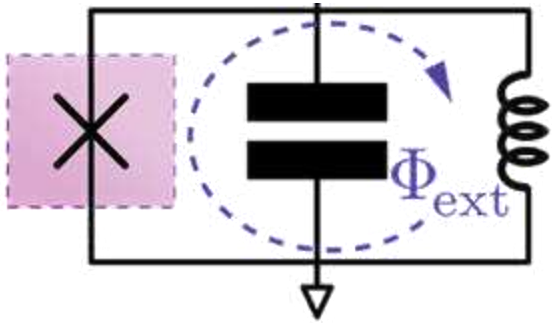
Reduced magnetic flux (Φ/ϕ_0)

Implementation

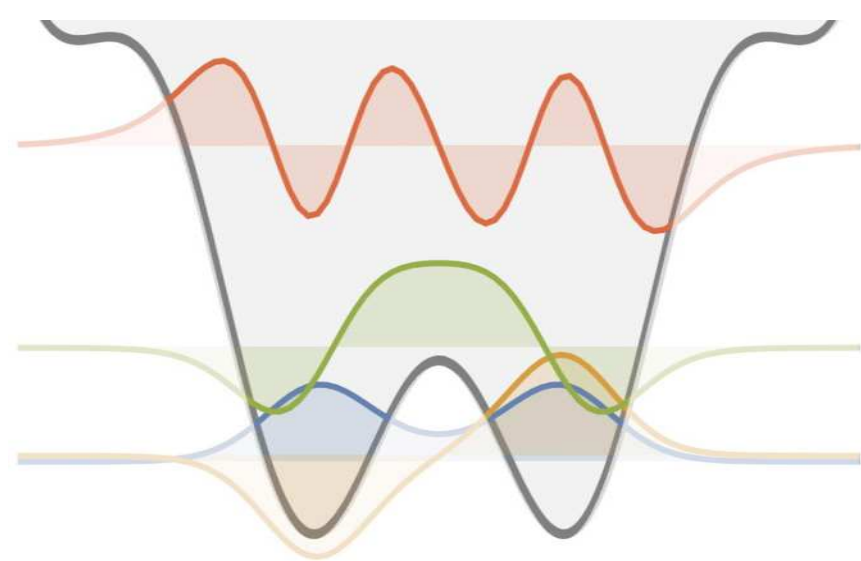


Lego land of qubits: engineer susceptibilities to ext. params.

Fluxonium

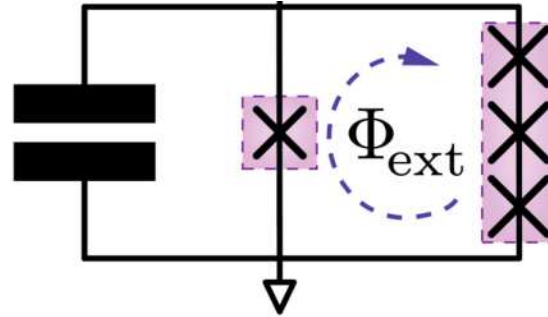


Energy vs. flux (at sweet spot)

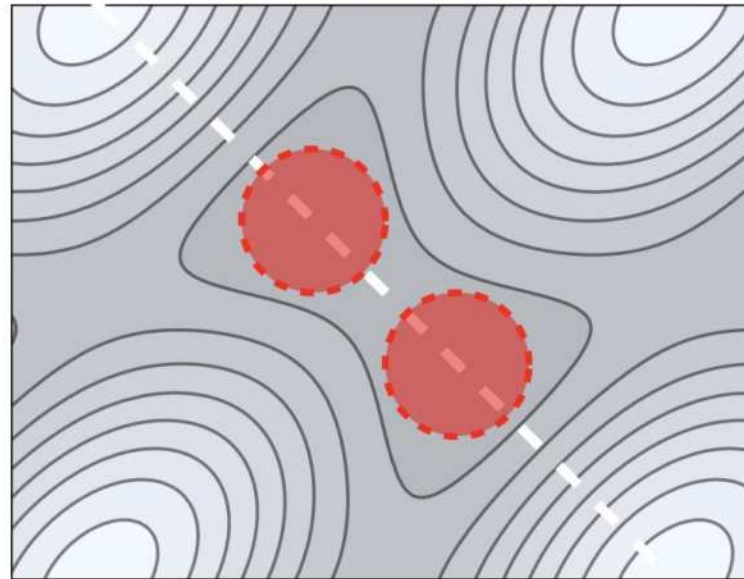


Manucharyan *et al.* (2009), Earnest *et al.* (2018), Lin *et al.* (2018), ...

(C-shunted) flux qubit



2D potential (flux degeneracy)



Orlando *et al.* (1999), Mooij *et al.* (1999), You (2006), Stern *et al.* (2014), Yan *et al.* (2016), ...

Many more

Quantrium

Phase qubit

Zero-pi qubit

Sin 2 phi qubit

Inductively-shunted
transmon

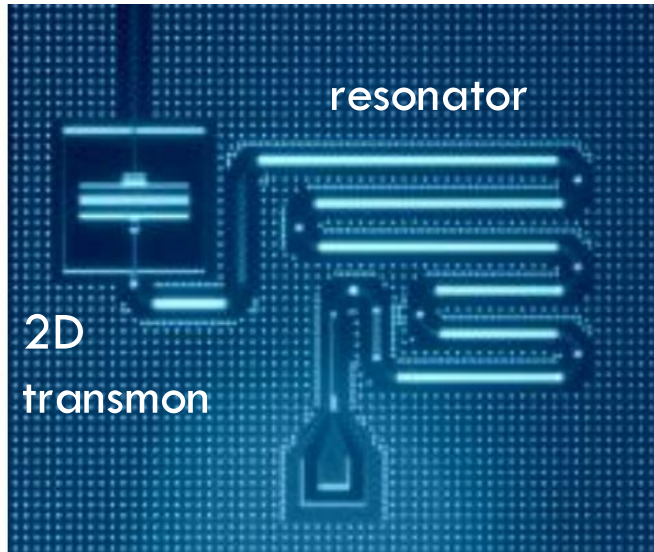
...

Versatile

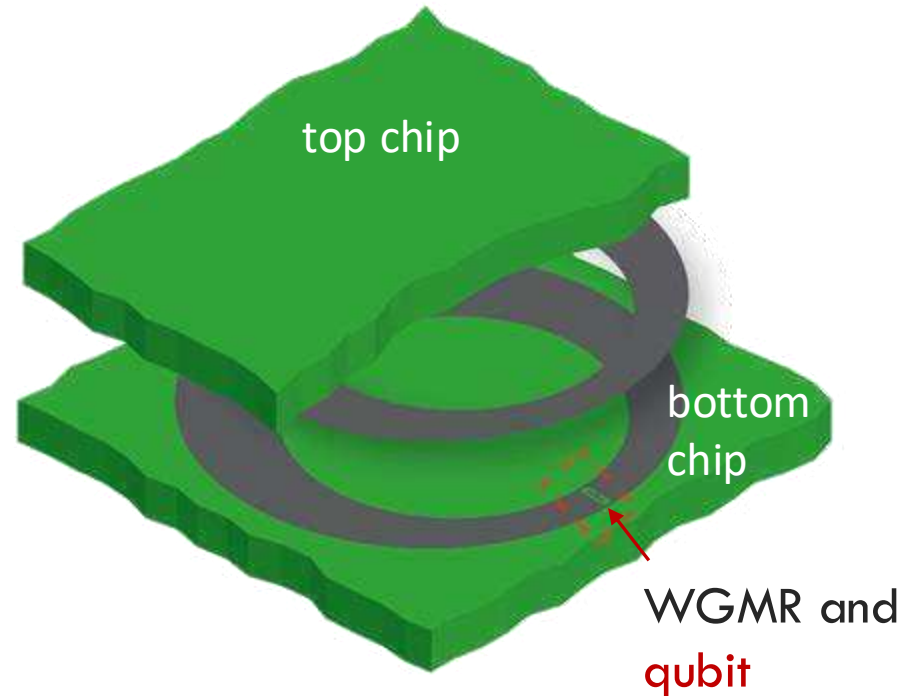
images: Minev, in prep (2019)

Hardware architectures

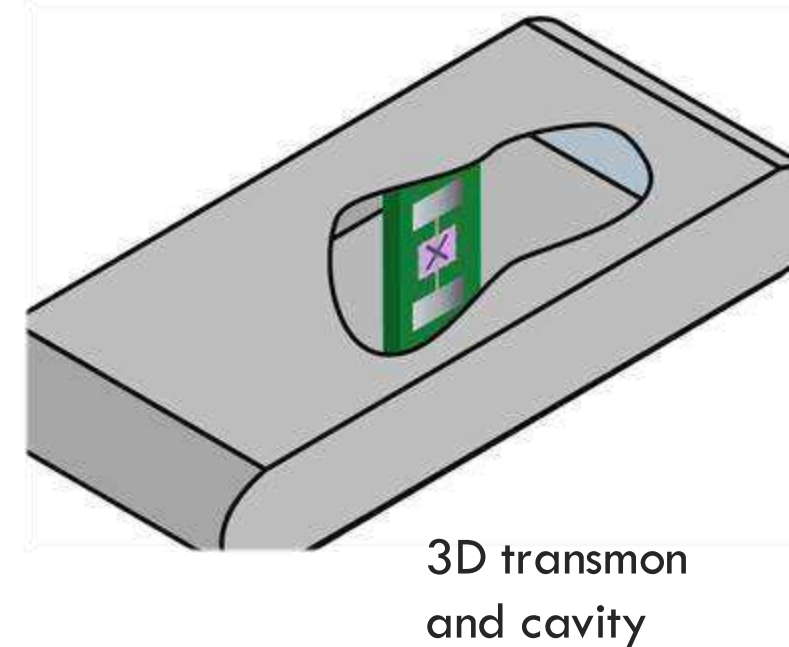
Planar (2D)



Multilayer planar (2.5D)



Three-dimensional (3D)



Blais *et al.* (2004), Wallraff *et al.* (2004),
Barends *et al.* (2014), Yan *et al.* (2014), ...

Minev *et al.* (2013, 2016),
Brecht *et al.* (2016), Rosenberg *et al.* (2017),
Yan *et al.* (2014), ...

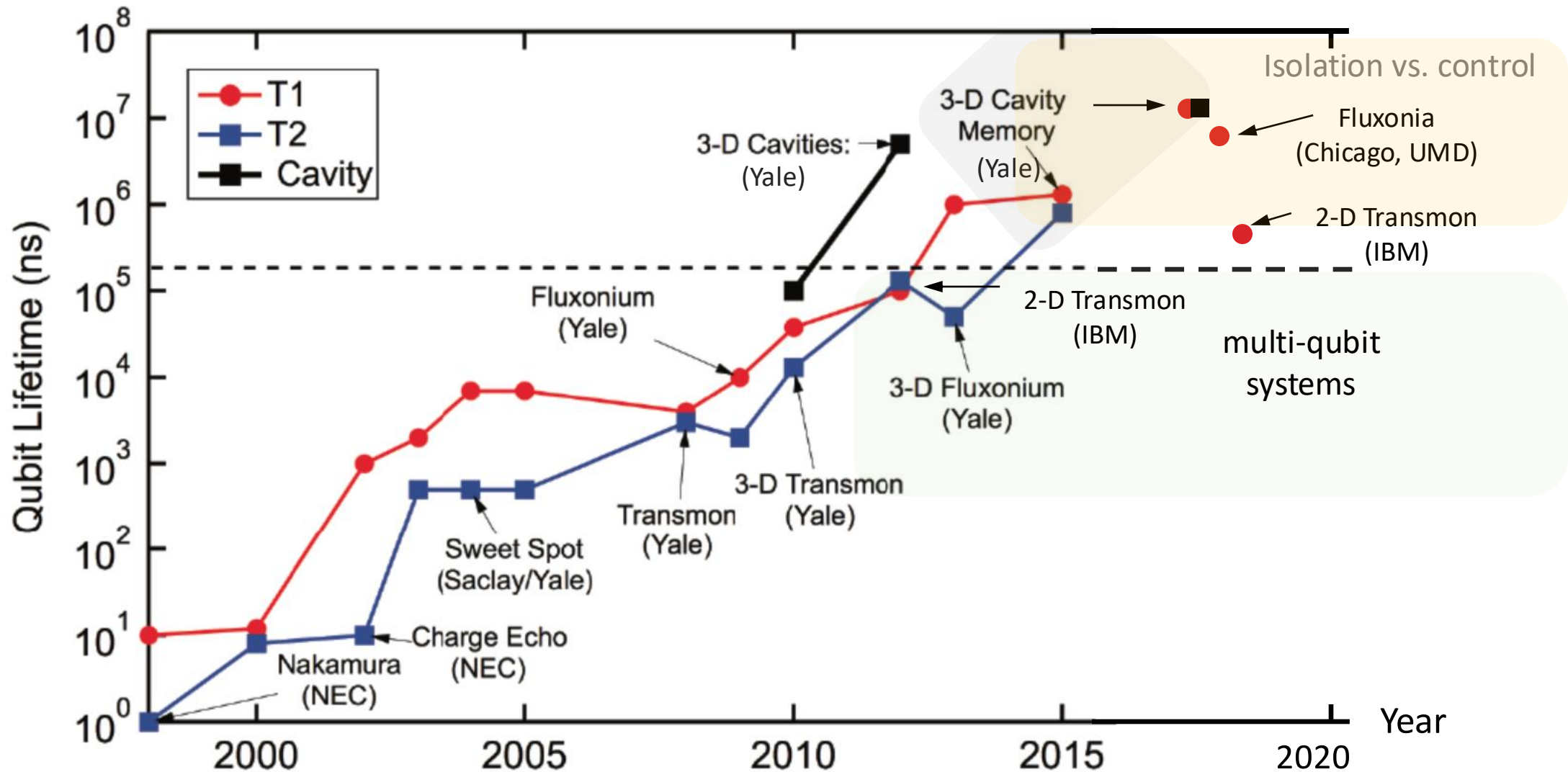
Paik *et al.* (2011), Reagor
(2016) ...

Image: Gambetta, Chow, and Steffen (2017)

Image: Minev *et al.* (2016)

Image: Minev *et al.* (2019)

Coherence in superconducting circuits



Schoelkopf's law: coherence 10x every 3 yrs. Note, for single system.

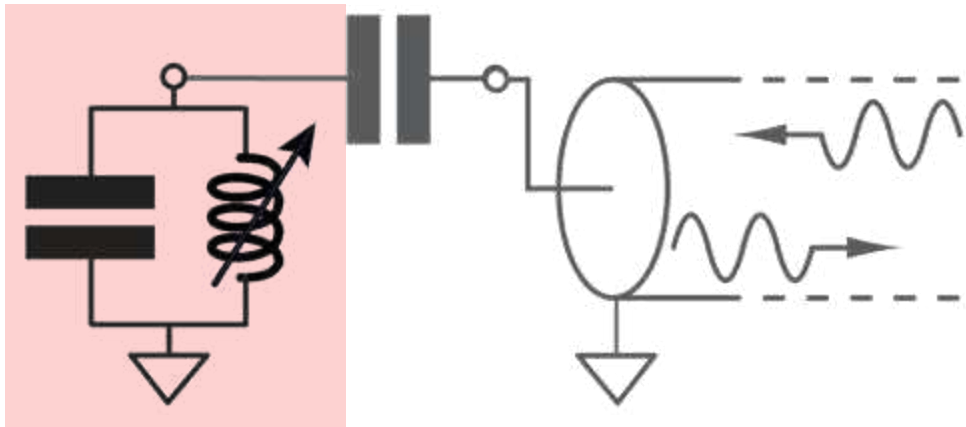
Imagereproduced Reagor (2015), an update of Devoret and Schoelkopf (2013), and updated (2019)

How to measure?

Qubit measurement with circuits

Direct monitoring

qubit



Demolition

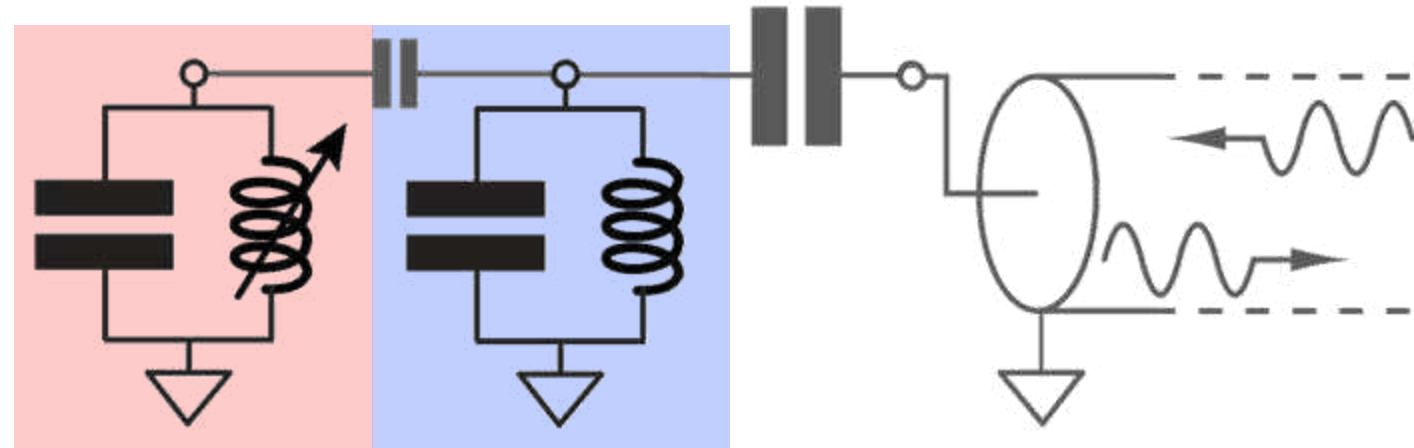


Spontaneous emission

cQED dispersive

qubit

cavity



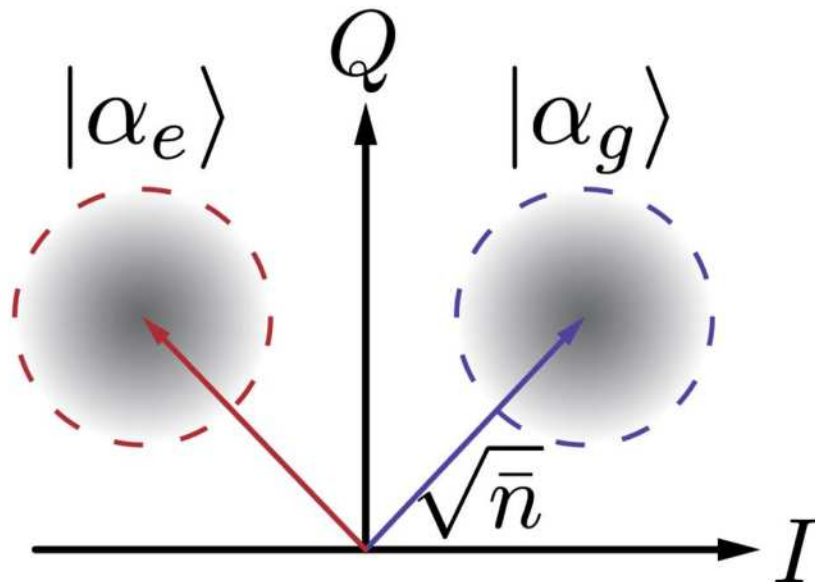
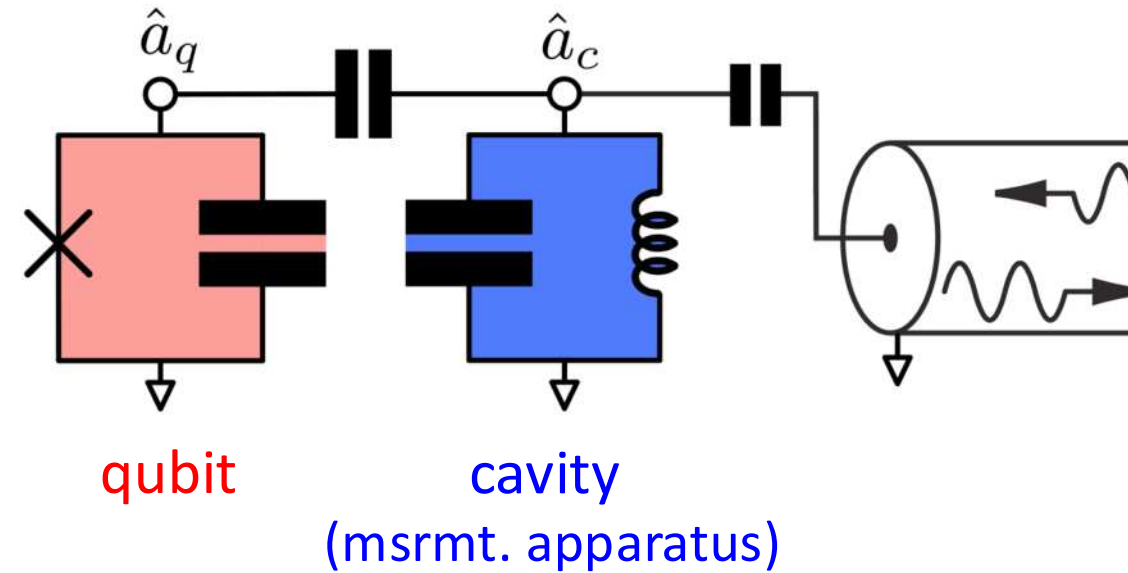
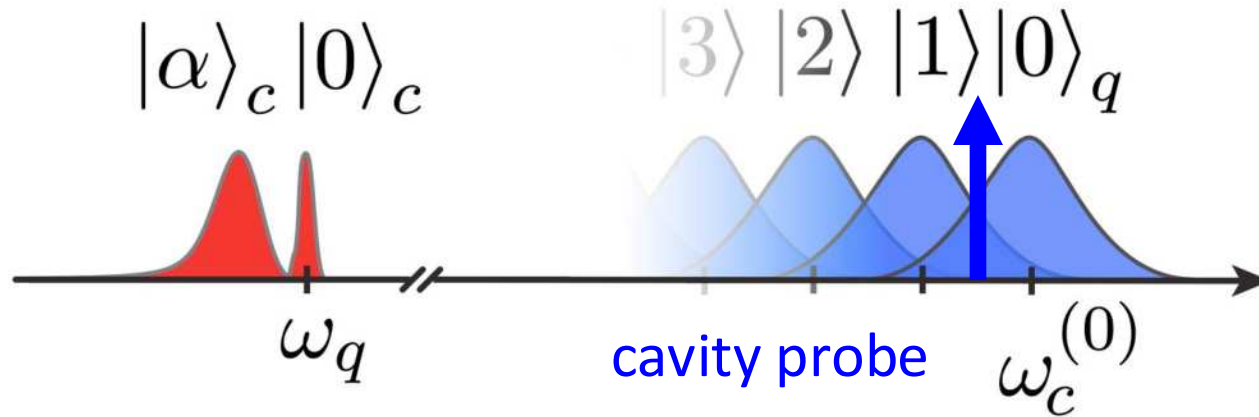
Quantum non-demolition* (QND)



Inhibited spontaneous emission

cQED dispersive readout: measuring Z

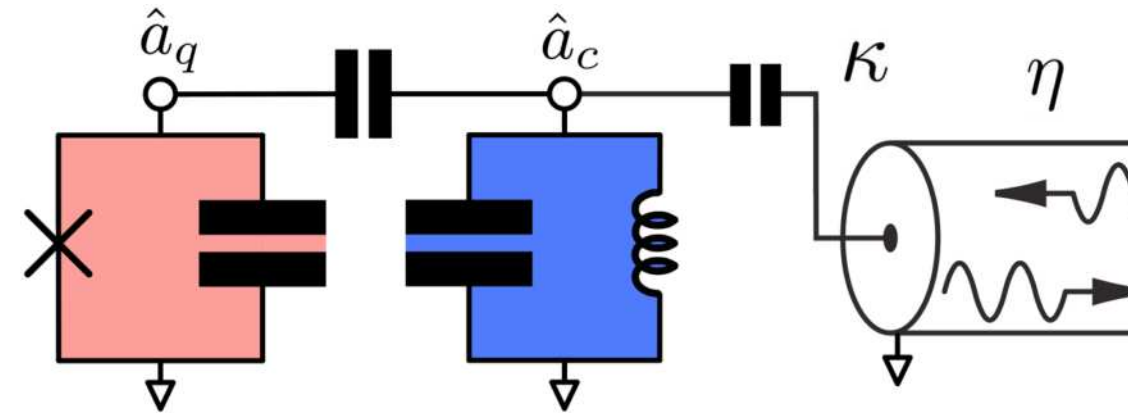
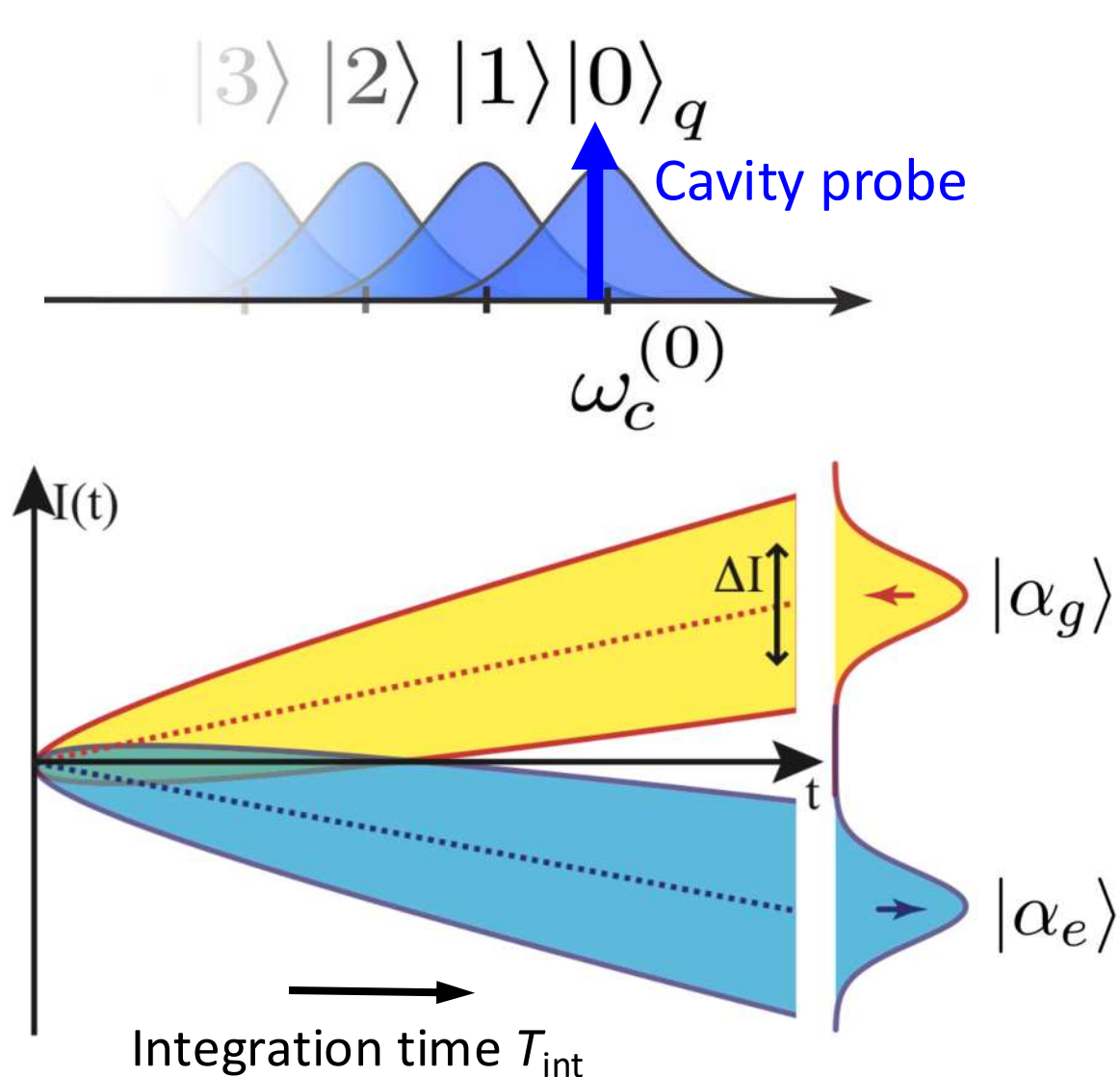
Response spectrum



$$\hat{H}_{\text{int}} = \hbar\chi \hat{a}_q^\dagger \hat{a}_q \hat{a}_c^\dagger \hat{a}_c$$

Pointer states of measurement apparatus (cavity) correspond to the qubit in the ground (g) and excited (e) states

cQED dispersive readout: measuring Z



$$\text{SNR} = \frac{1}{2} \eta \kappa T_{\text{int}} |\alpha_e - \alpha_g|^2$$

distinguishability

$$\eta_{\text{disc}} = \frac{1}{2} \text{erfc} \left[-\sqrt{\frac{\text{SNR}}{2}} \right]$$

efficient, fast measurements

Tradeoff: sensitivity vs. noise-susceptibility

Purcell effect

$$T_1$$

$$T_1^{\text{Purcell}} = \frac{C_q}{\text{Re}[Y(\omega_q)]}$$

Esteve *et al.*, PRB (1986)
Neeley *et al.*, PRB (2008)
Reed, Thesis (2013)
...

Photon-shot-noise
dephasing

$$T$$

$$\Gamma_{\text{th}} = \frac{4\chi^2 n_{\text{th}}^{\text{eff}}}{\kappa_{\text{tot}}} (n_{\text{th}}^{\text{eff}} + 1) \quad \text{large } \kappa_{\text{tot}}/\chi$$

Gambetta *et al.*, PRA (2006)
Schuster, Thesis (2007)
Rigetti *et al.*, PRB (2012)
...

T_1 vs. $\bar{n}$

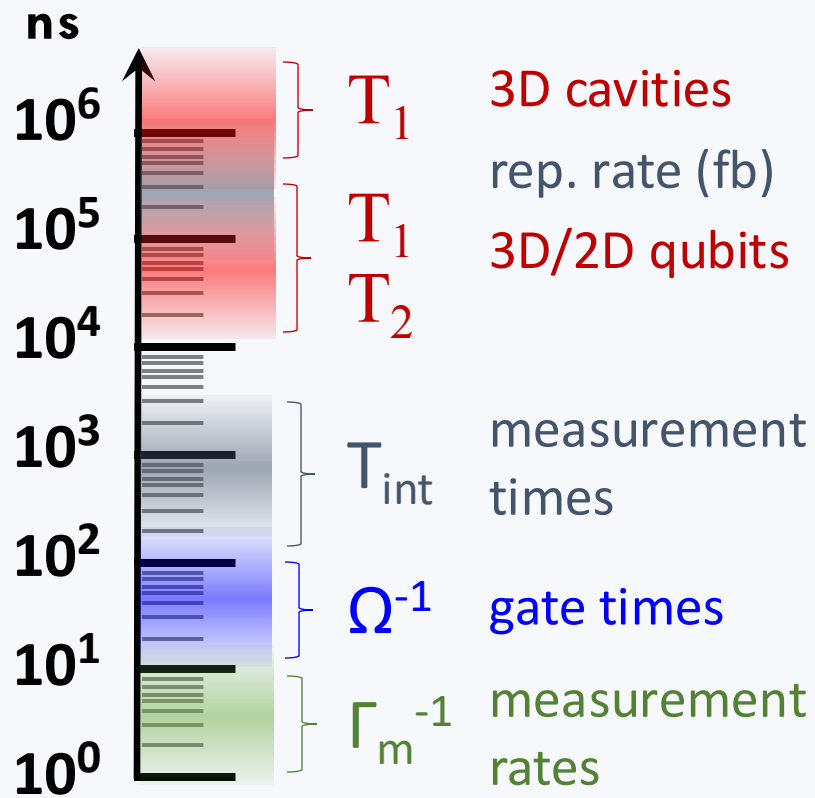
readout, QND

?

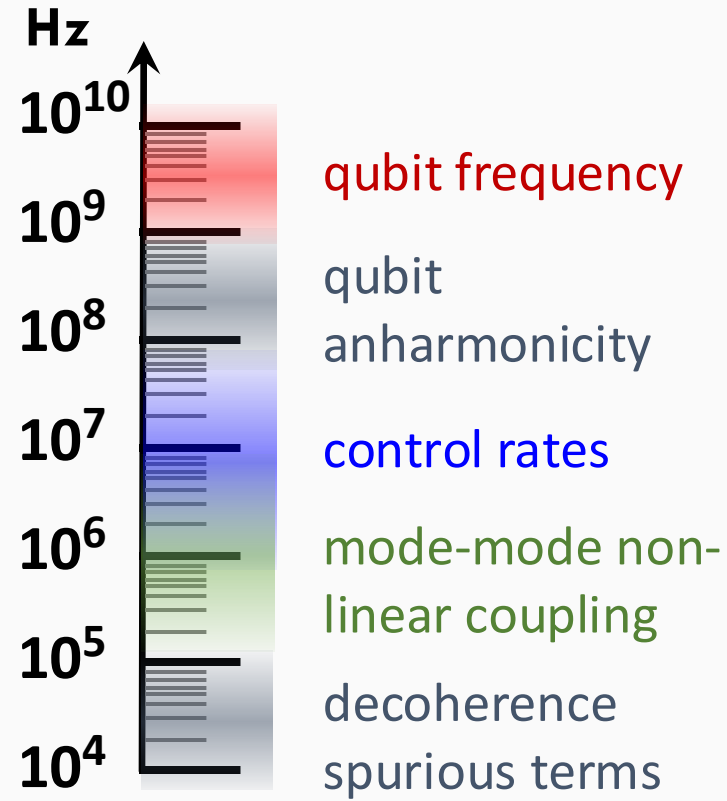
Boissonneault *et al.*, PRA (2009)
Slichter *et al.*, PRL (2012)
Shantanu, MLS (2015)
Sank *et al.*, PRL (2016)
...

High separation of timescale b/w noise and control

System timescales

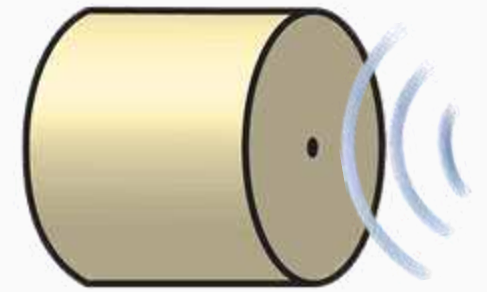


Frequency bands



High efficiency detection

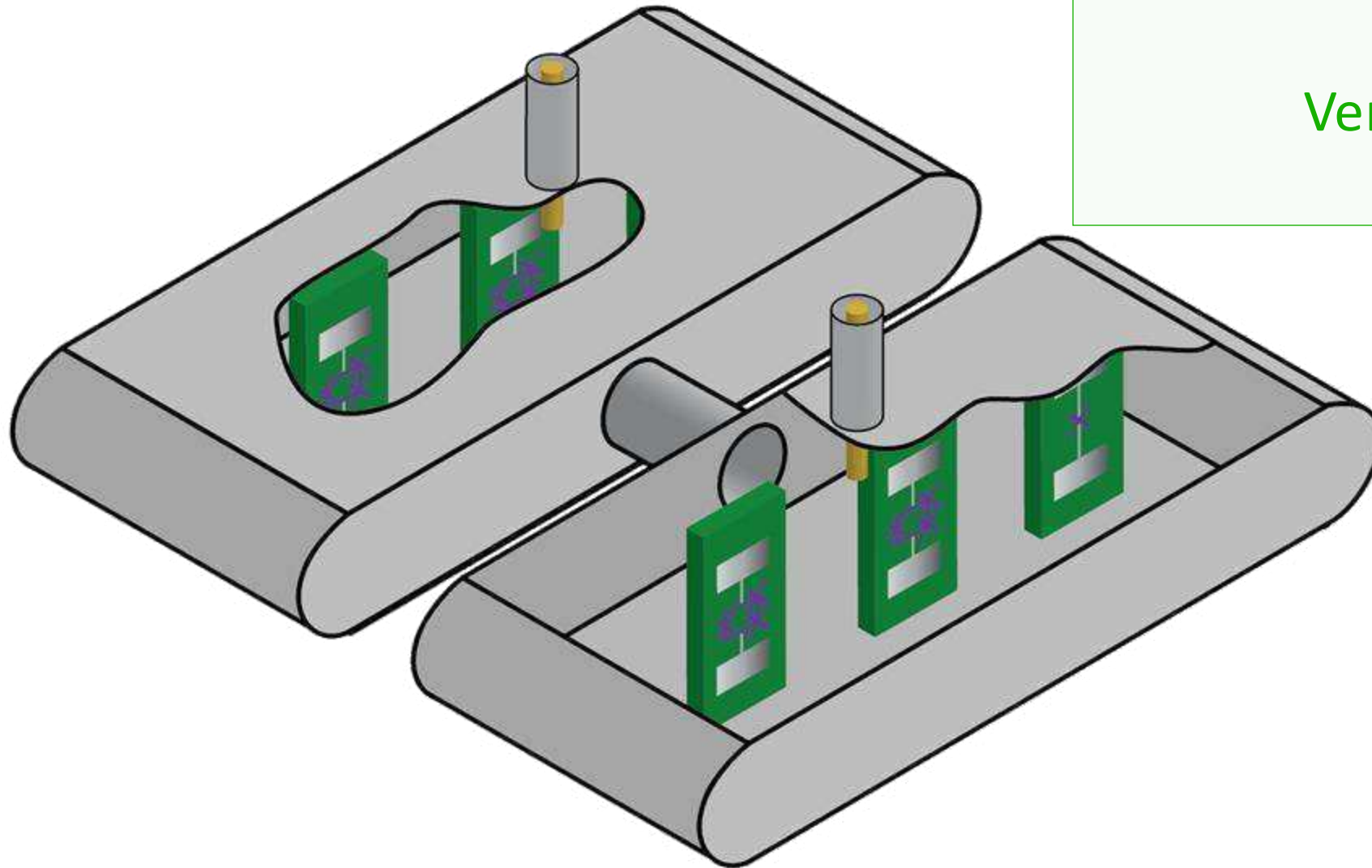
$$\eta \sim 0.1-0.5$$



(using quantum-limited amplifiers)

How to design circuits with versatility and precision?

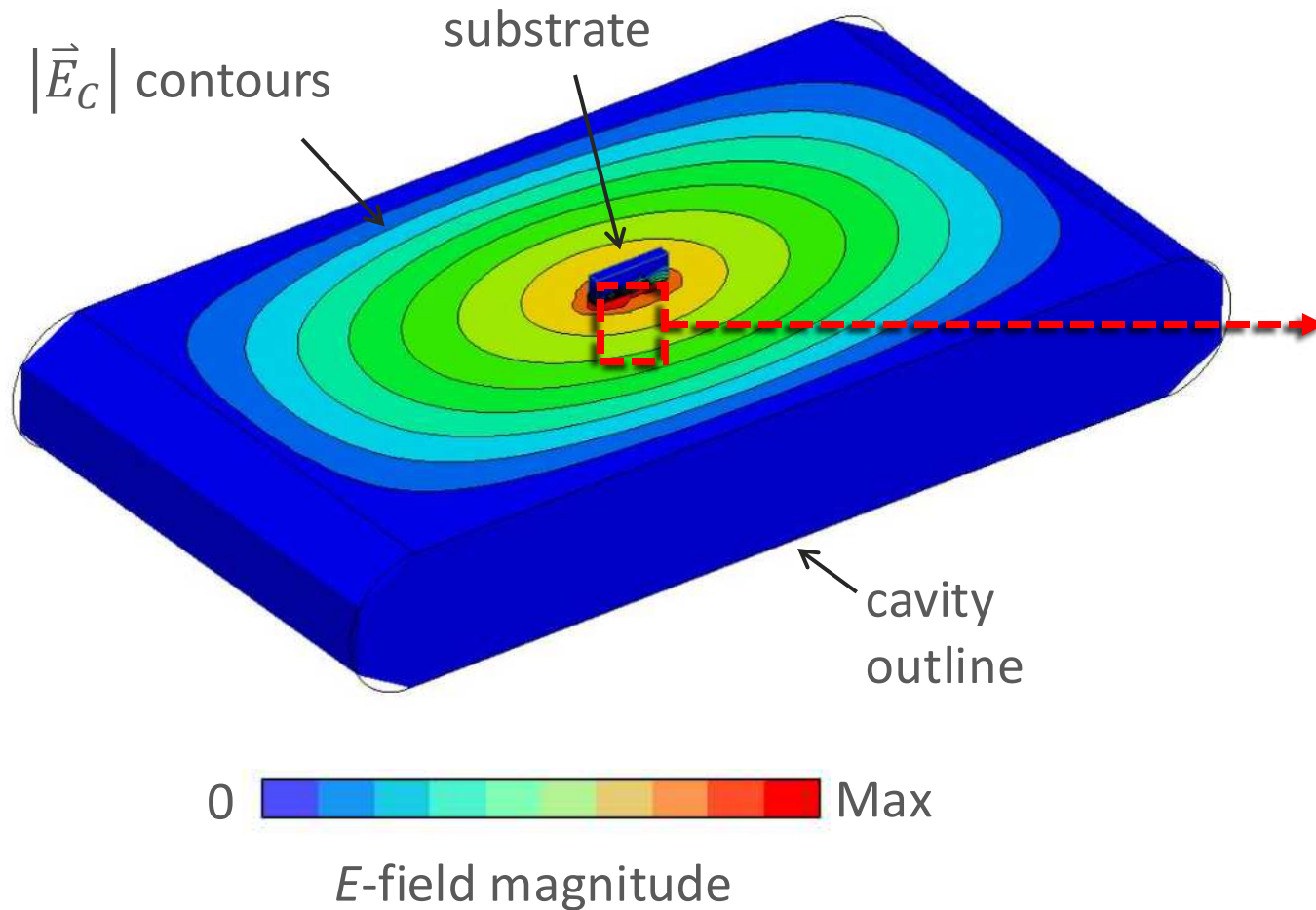
Minev, arXiv:1902.10355 (2018), Minev *et al.* in prep. (2019)



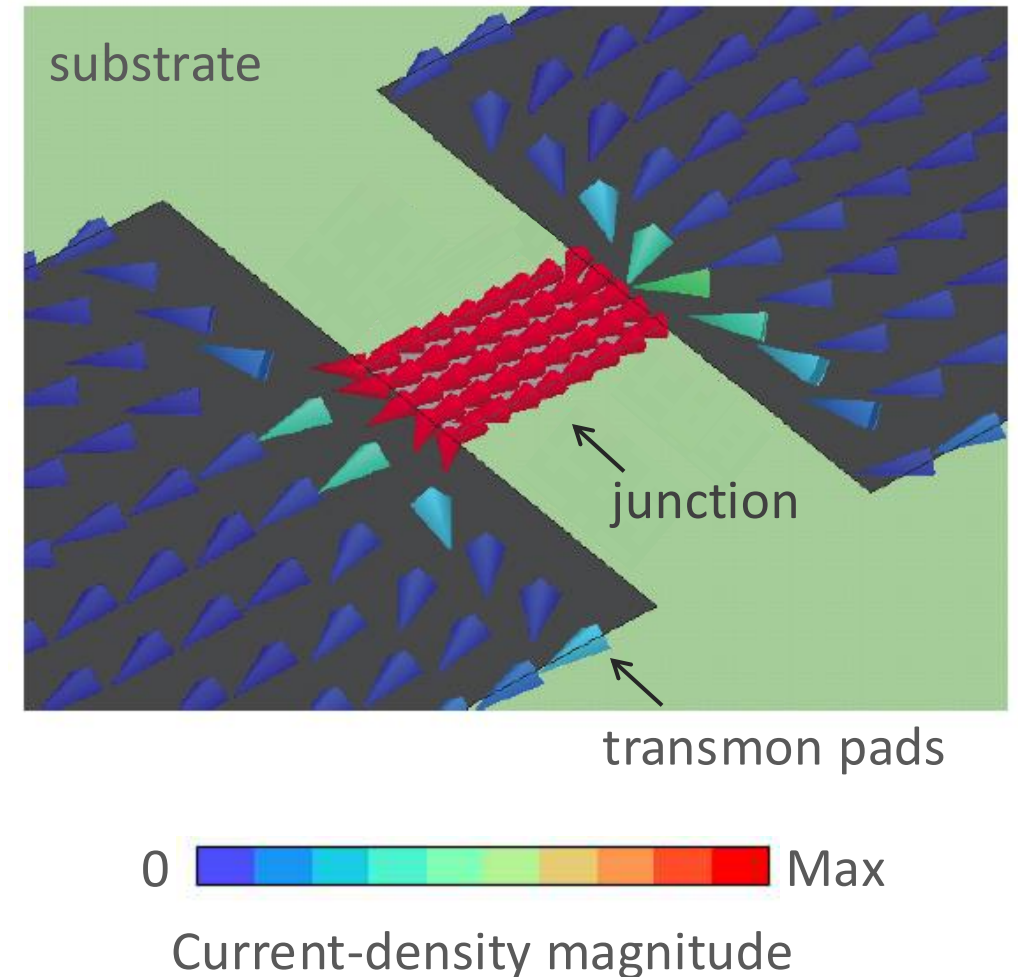
Versatile

$\mathcal{H}_{\text{lin}}$ eigen modes

Cavity mode (7.0 GHz)

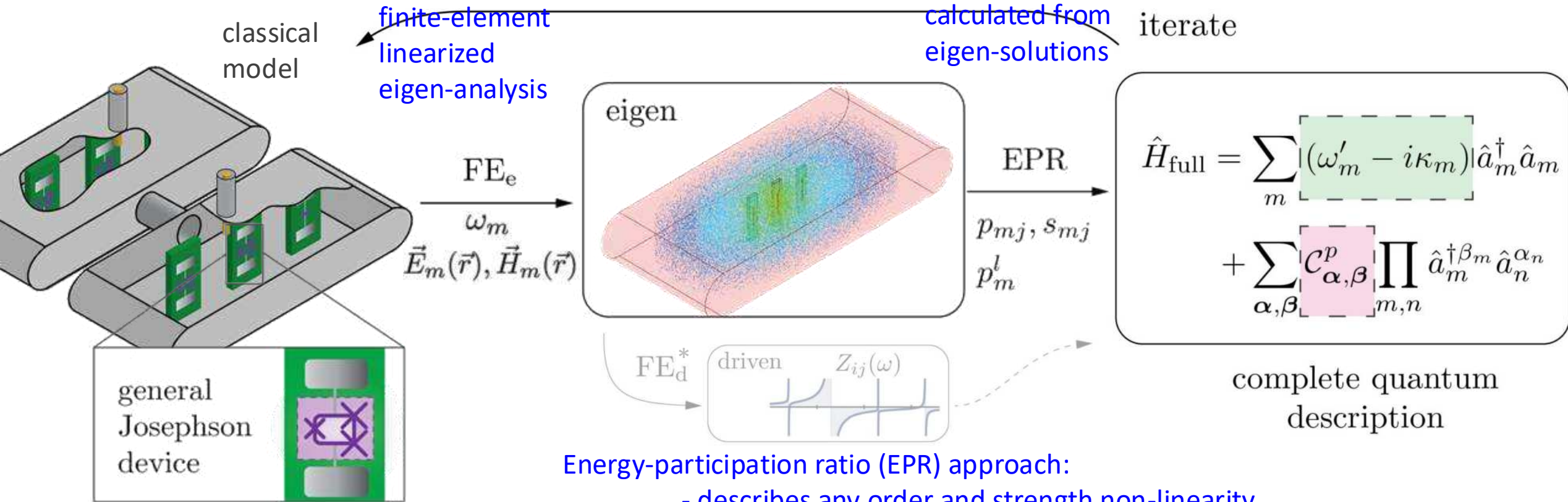


Qubit mode (linearized, 5 GHz)



Overview of energy participation approach

Minev, Ch. 4, arXiv:1902.10355 (2018), Minev *et al.* in prep. (2019)



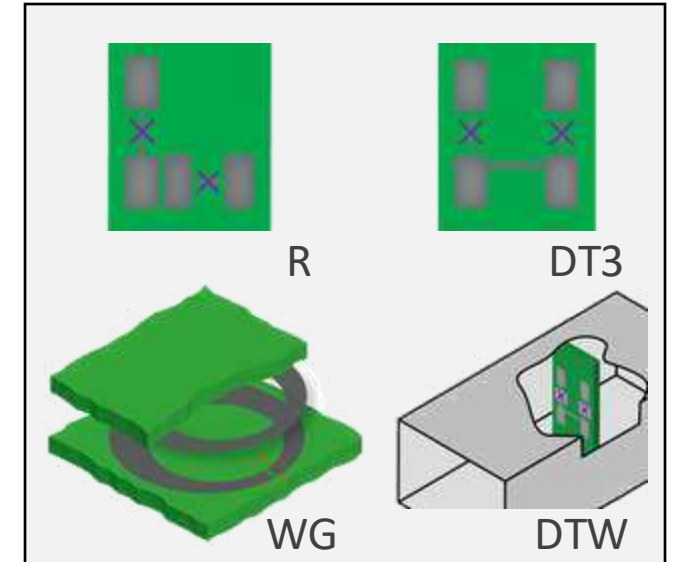
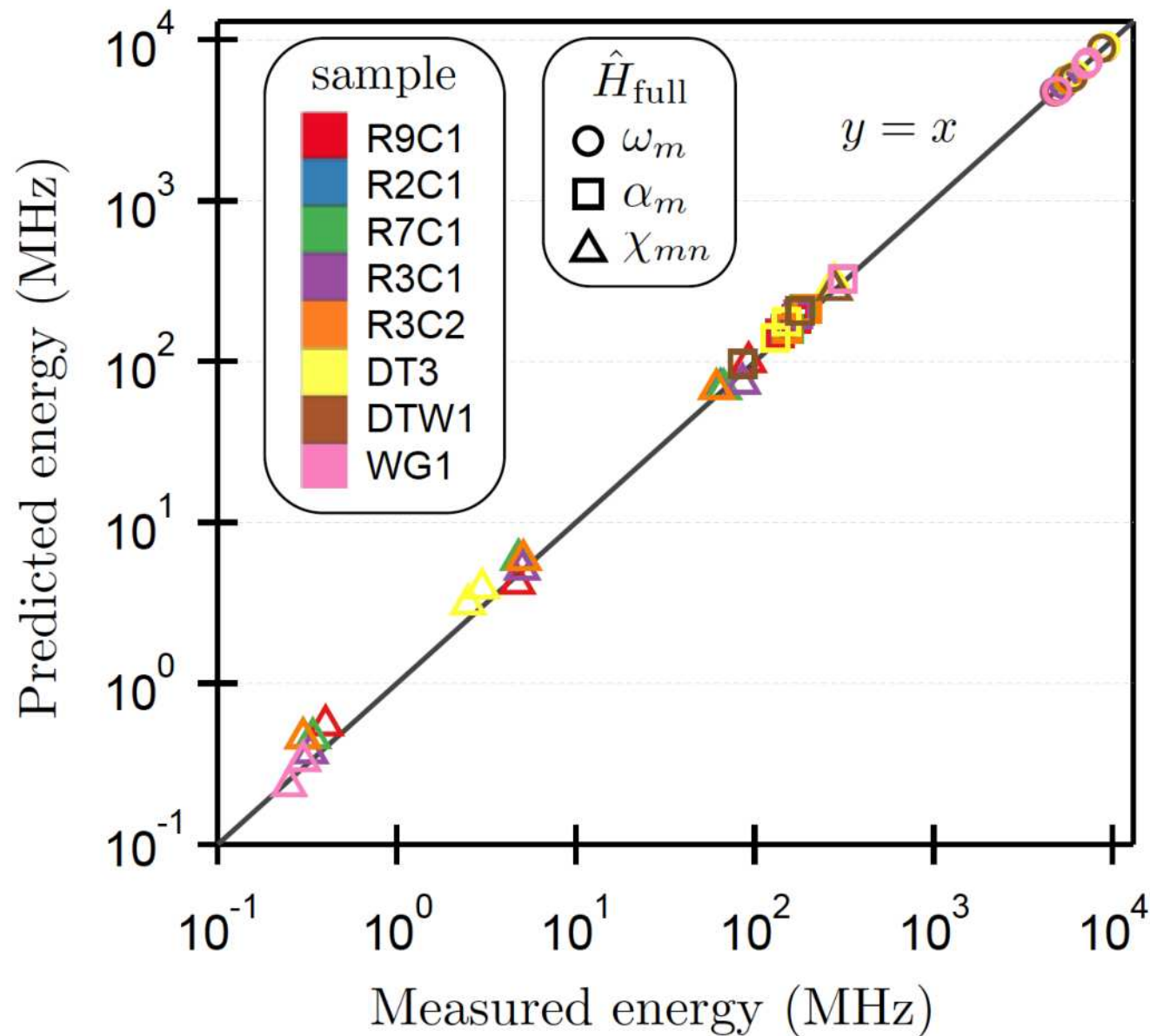
Energy-participation ratio (EPR) approach:

- describes any order and strength non-linearity
- describes arbitrary (composite) non-linear inductive devices
- first-principle derivation
- zero approximations (aside from truncation of modes)
- fully automated in python (github.com/zlatko-minev)

Practical limits: Fock and mode basis truncation due to computing power

* Nigg, Paik, *et al.*, PRL (2012),
Bourassa *et al.* (2012),
Solgun *et al.* (2014, 2015, 2017), ...

Theory vs. experiment: agreement over 5 orders of magnitude

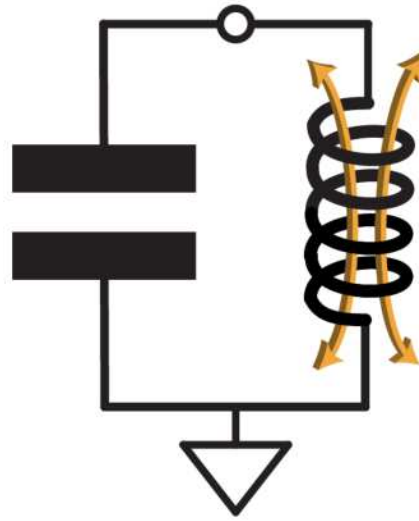


R: Minev *et al.* (2018)

WG: Minev *et al.* (2013, 2016)

DT3, DTW: Minev *et al.* (2019)

Quantum sensing with superconducting quantum circuits

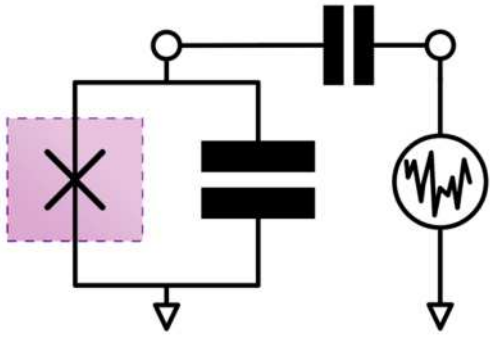


(an incomplete) survey

Sensing with circuits: survey

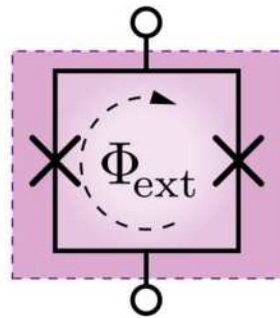
Charge

transmon

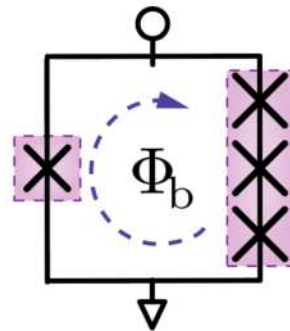


Flux

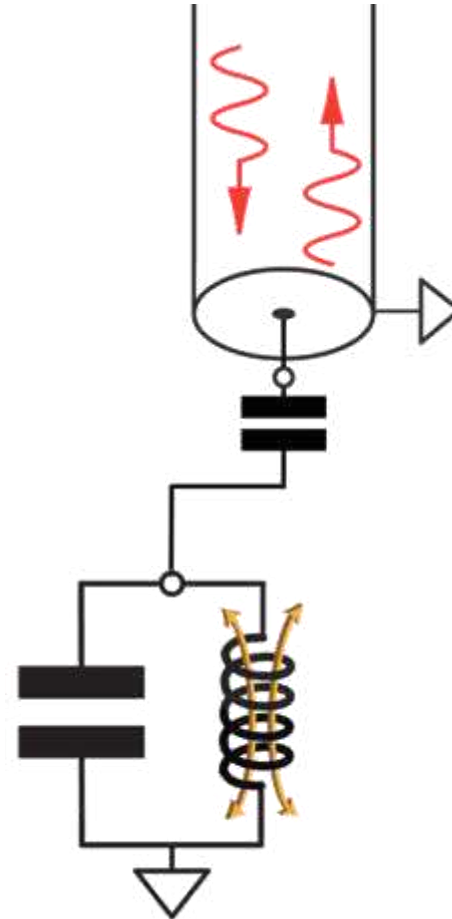
SQUID



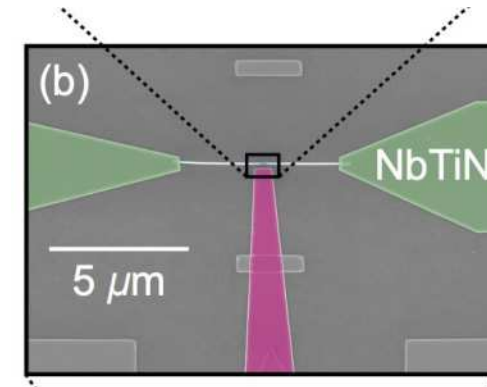
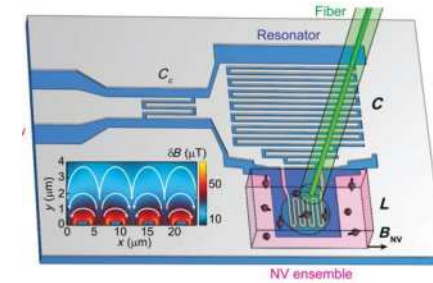
flux qubit



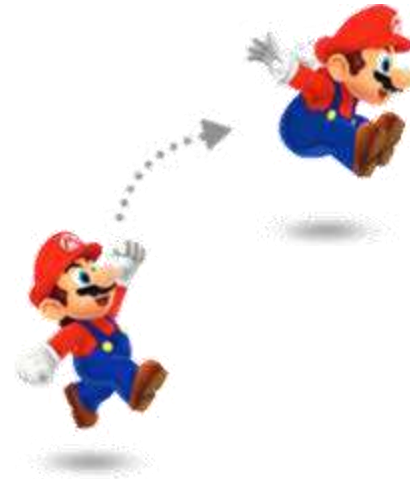
Frequency/ adaptive



Hybrid



Photons/ feedback



SQUID magnetometer

Sensor characteristics

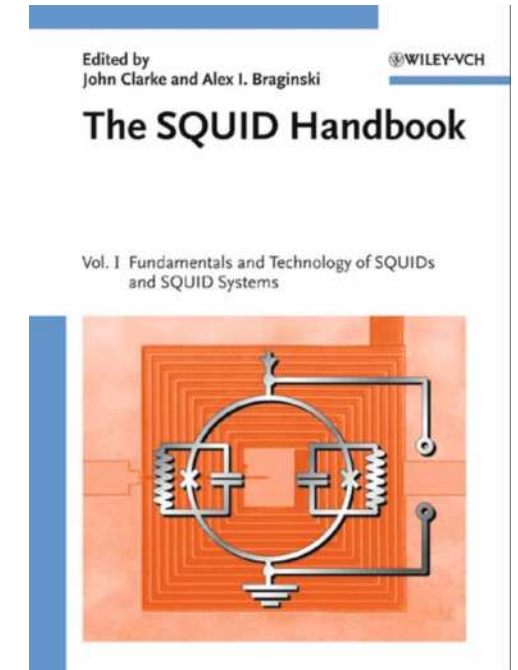
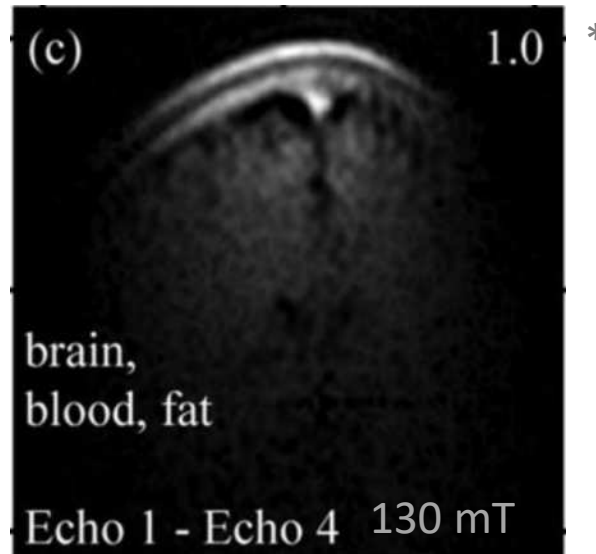
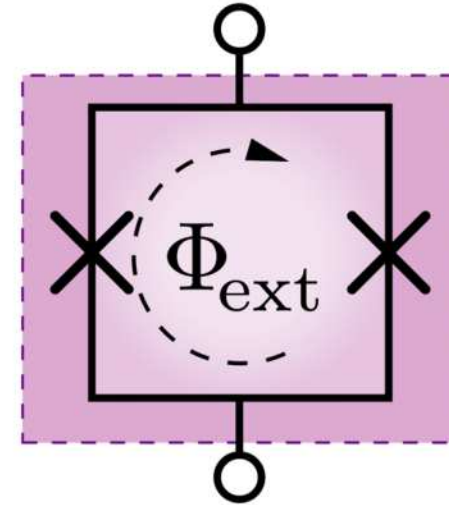
responds to: magnetic field
frequency: dc-GHz

min field: $\sim$ aT

max sensitivity: $0.01\text{-}10 \text{ aT}/\sqrt{\text{Hz}}$

size: can be submicron

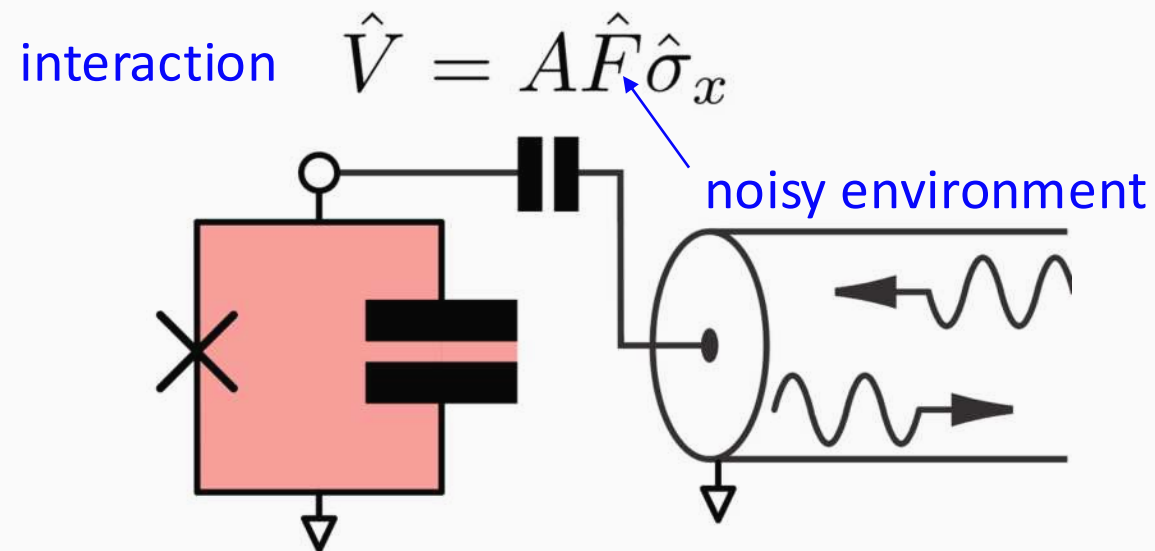
status: commercial



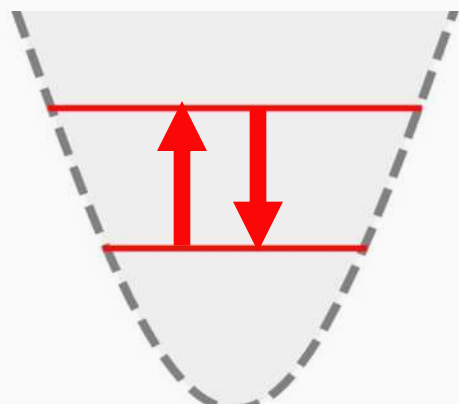
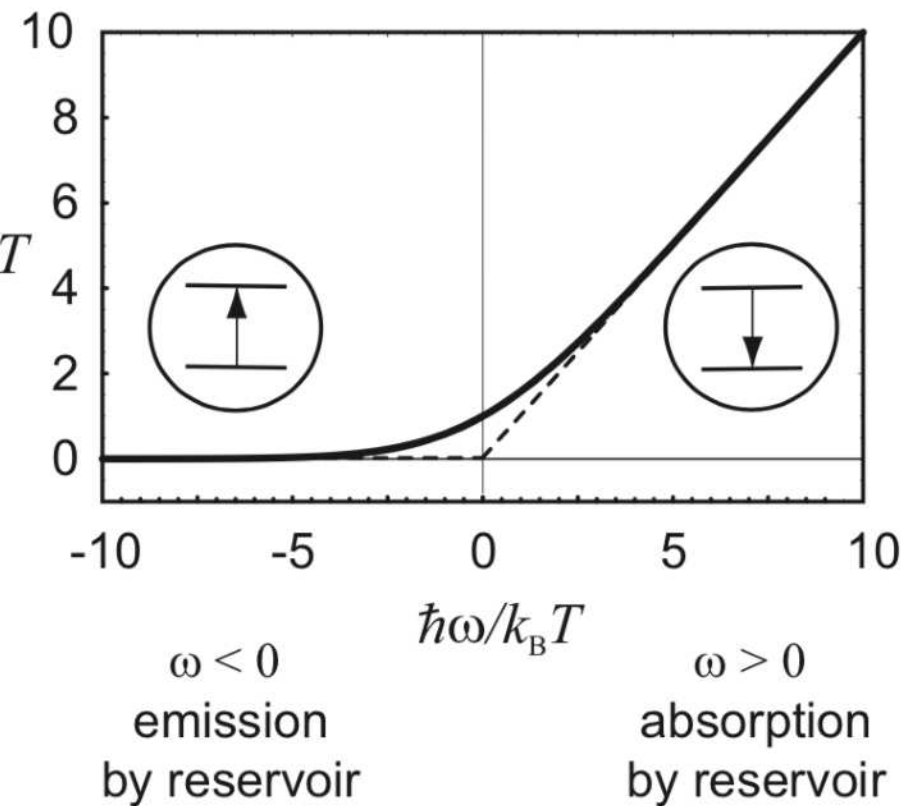
* image: Inglis (2013)

Jaklevic *et al.* (1965); Clarke and Braginski (2004); Fagaly (2006)

Qubit as a spectrometer (relaxometry, thermometry, etc.)



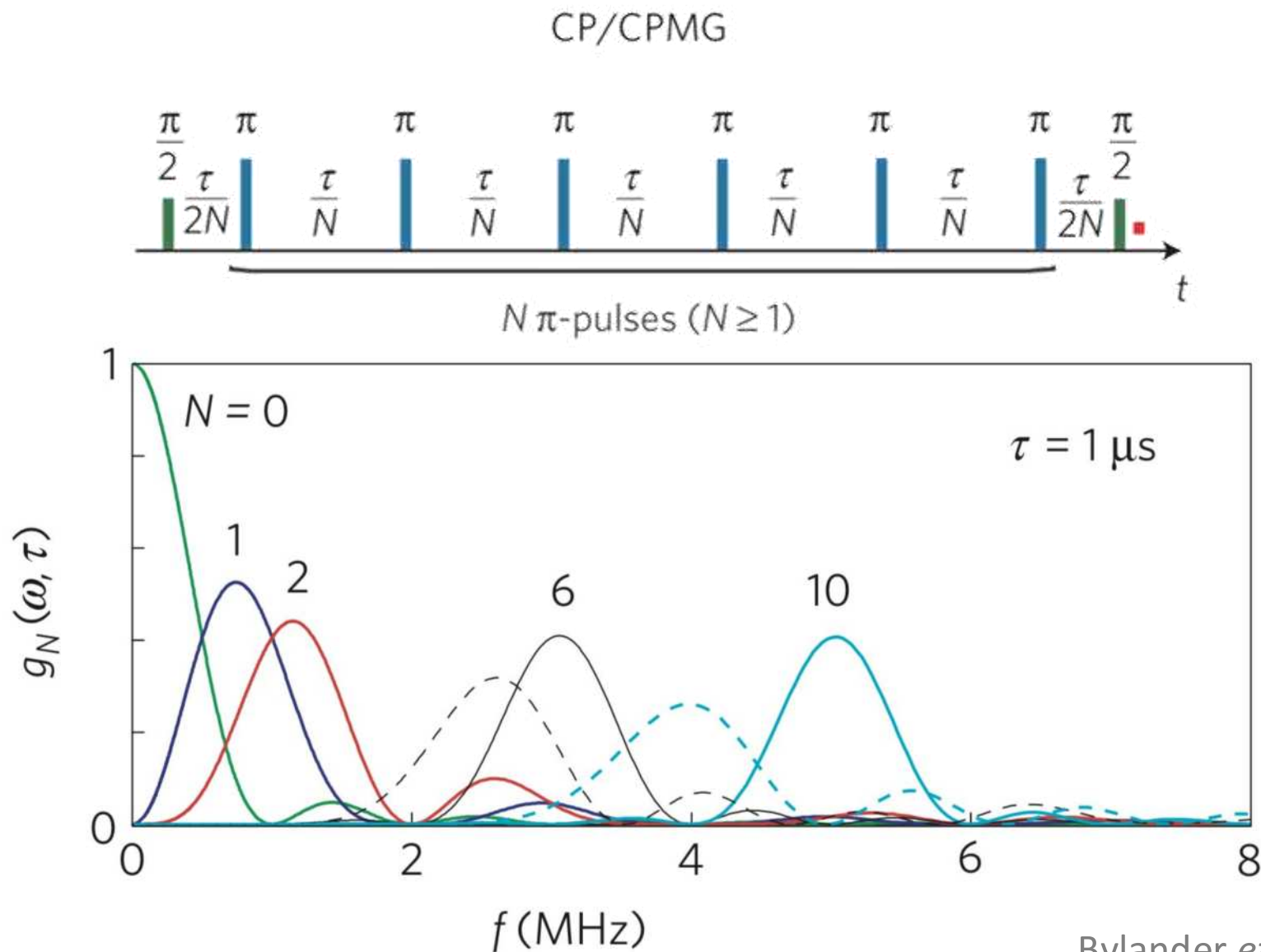
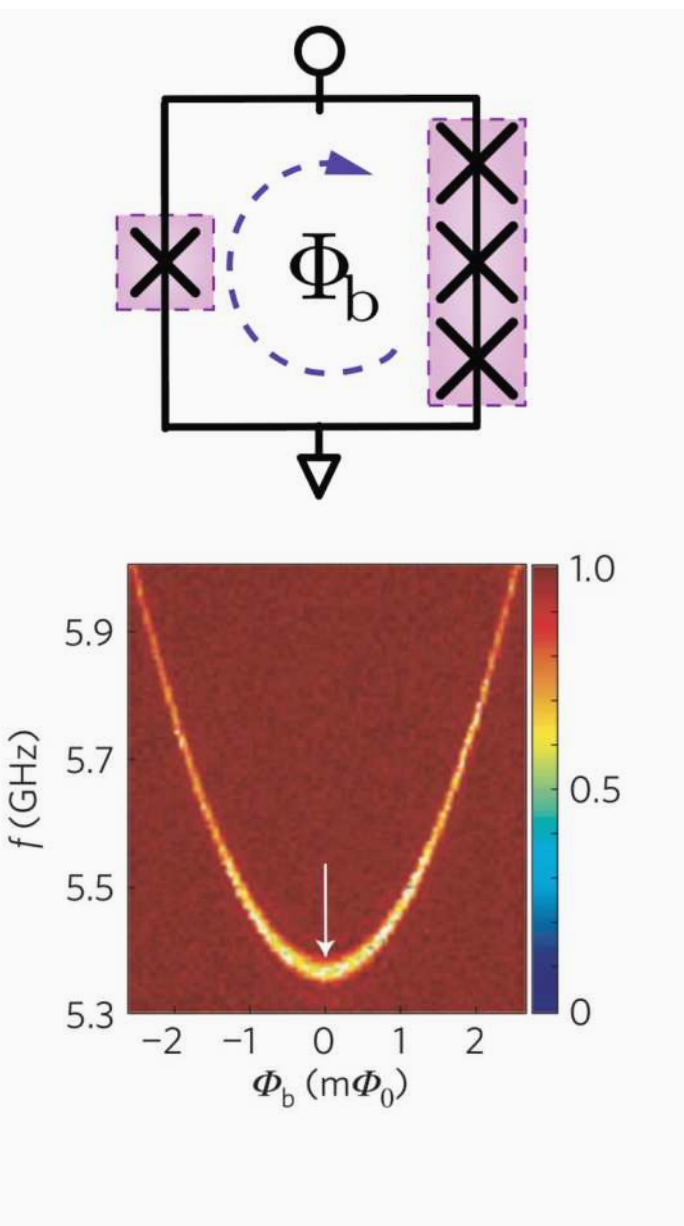
$$S_{VV}[\omega]/2Rk_B T$$



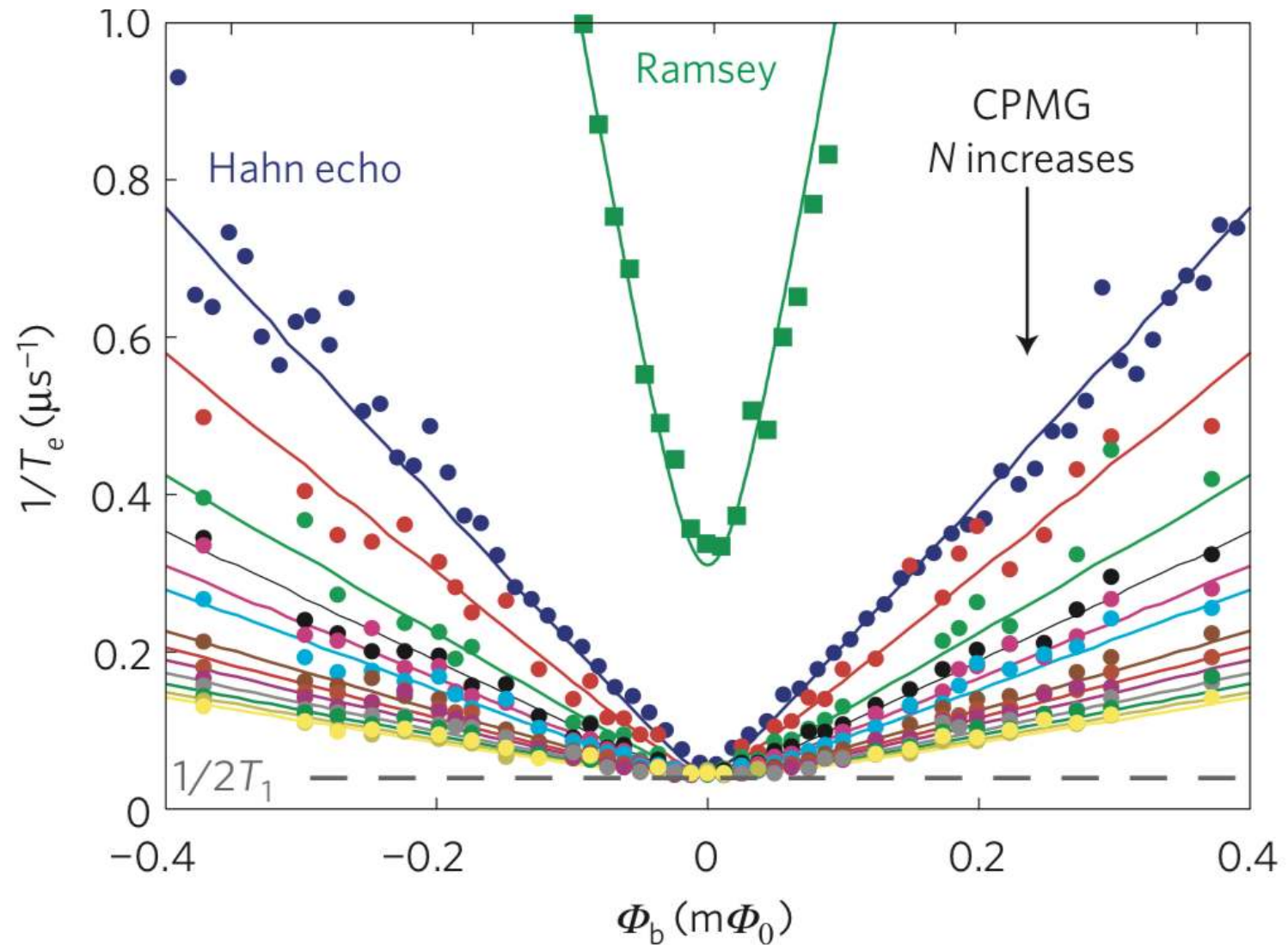
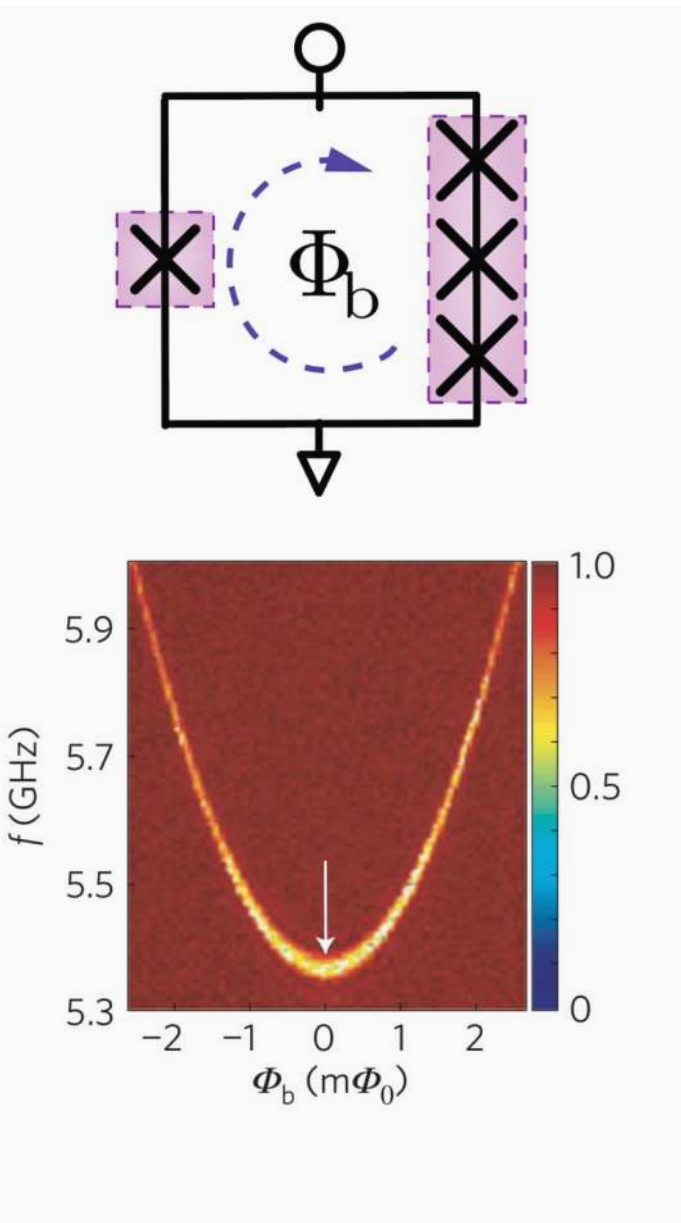
$$\Gamma_{\uparrow} = \frac{A^2}{\hbar^2} S_{FF}[-\omega_{01}]$$

$$\Gamma_{\downarrow} = \frac{A^2}{\hbar^2} S_{FF}[+\omega_{01}].$$

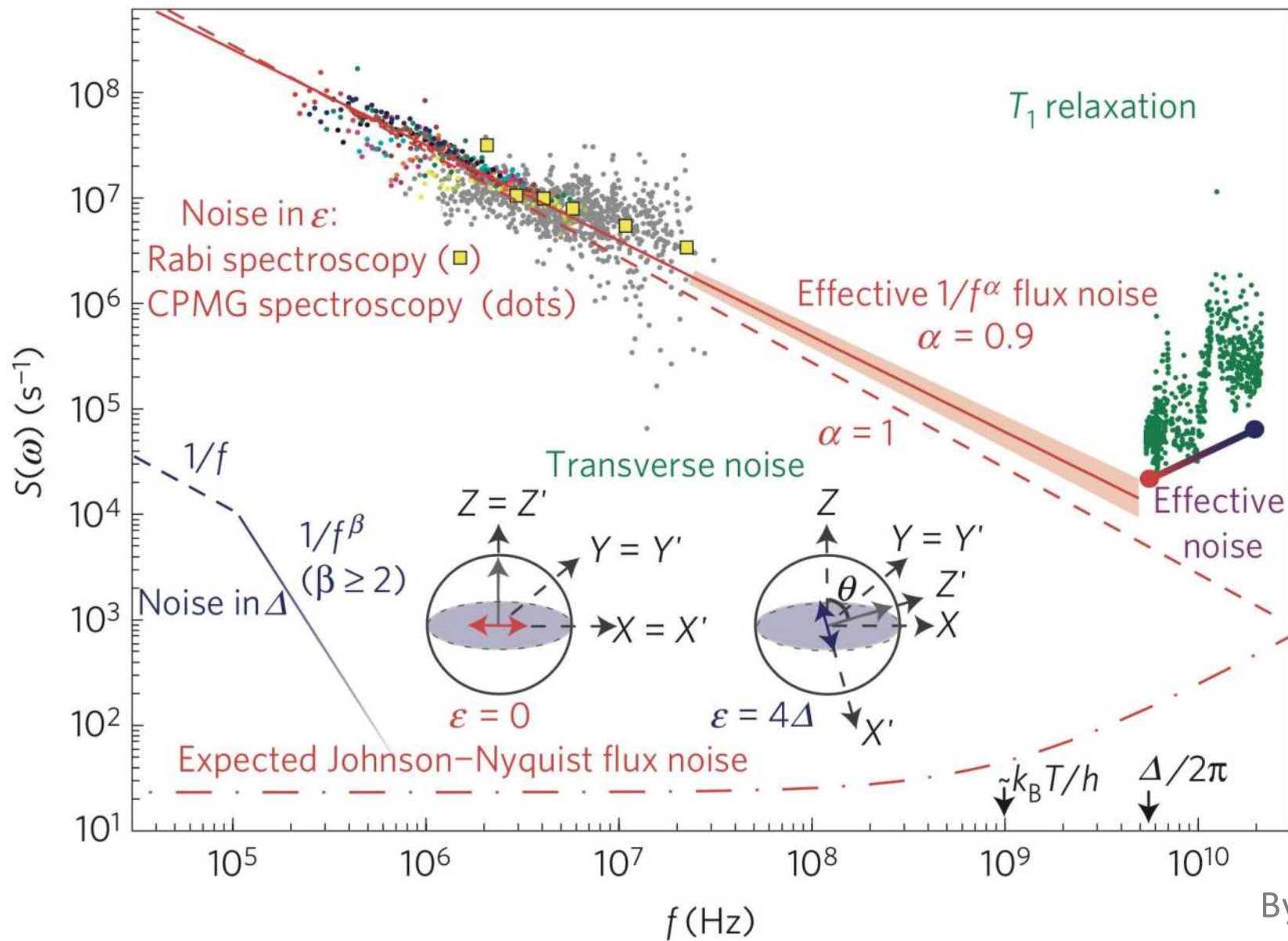
Flux noise sensing with flux qubits (T_2 relaxometry)



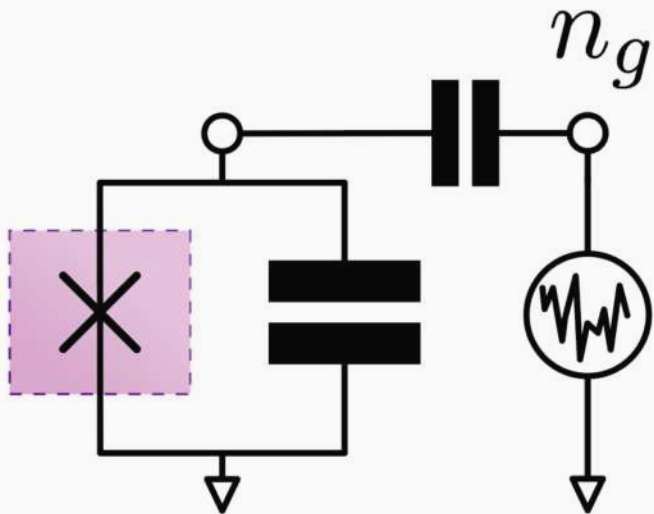
Flux noise sensing with flux qubits (T_2 relaxometry)



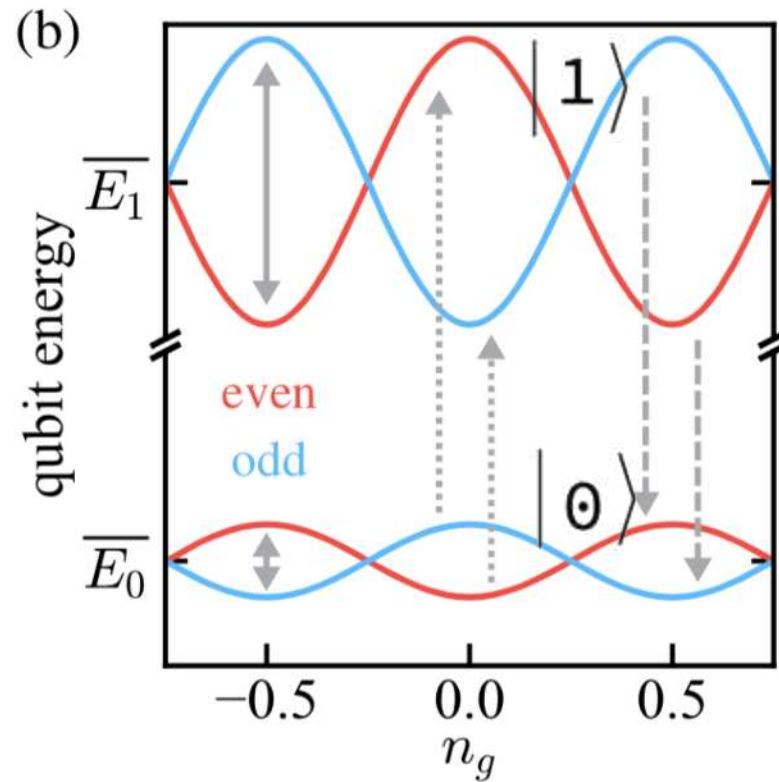
Flux noise sensing with flux qubits (T_1, T_2 relaxometry)



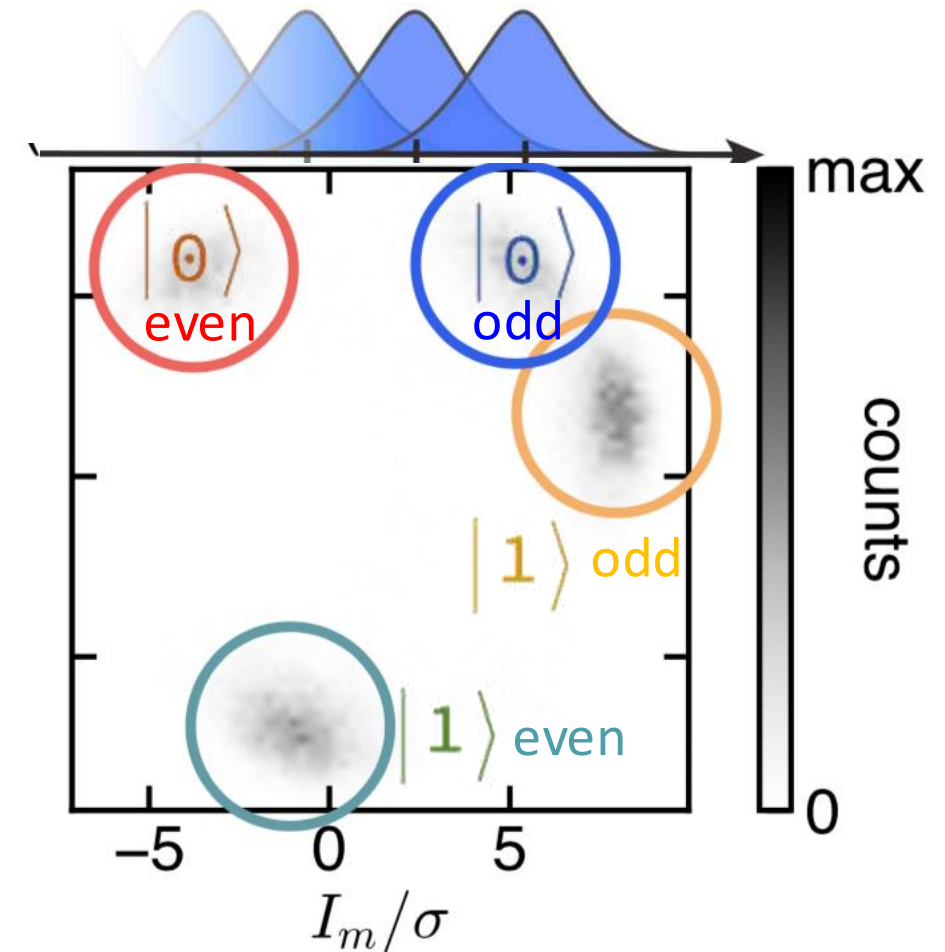
Charge sensing with OCS transmon



Energy susceptibility

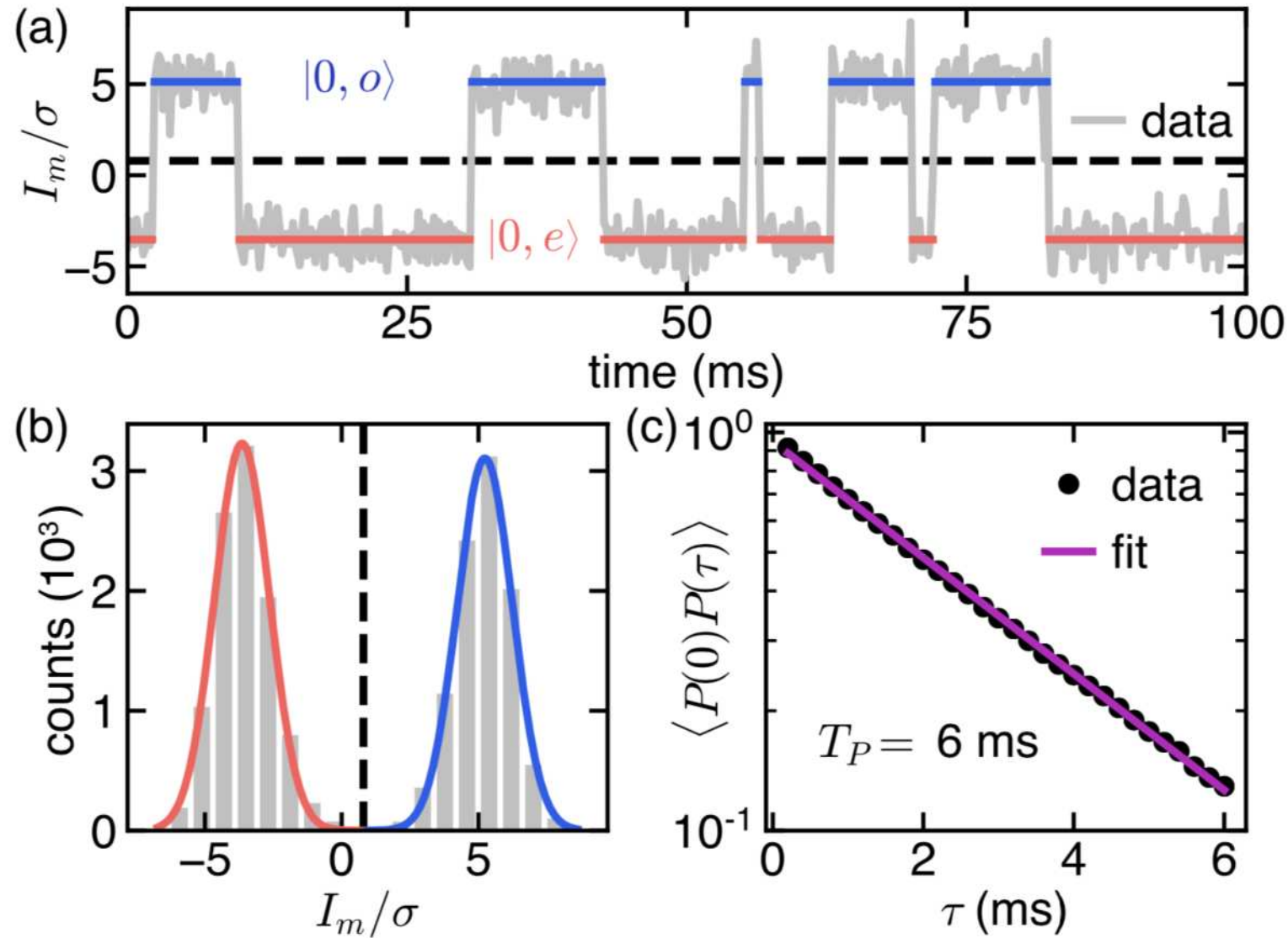


Cavity response

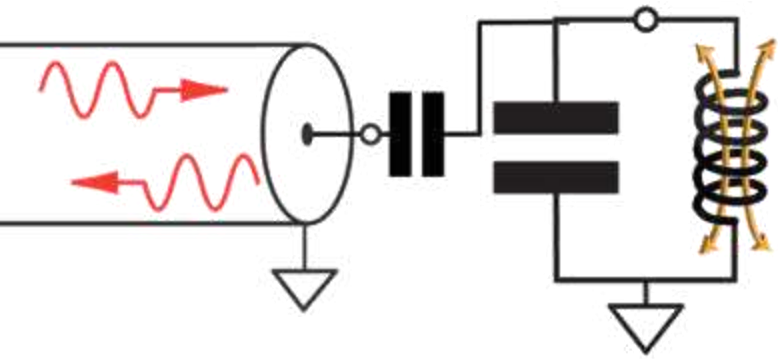


Charge sensing with OCS transmon

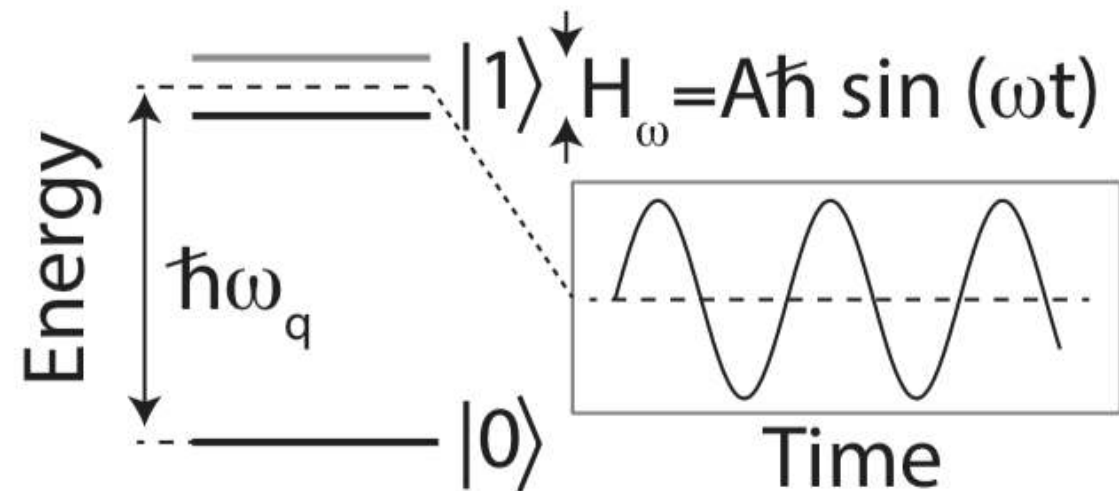
Quantum jumps of quasiparticle parity in real-time



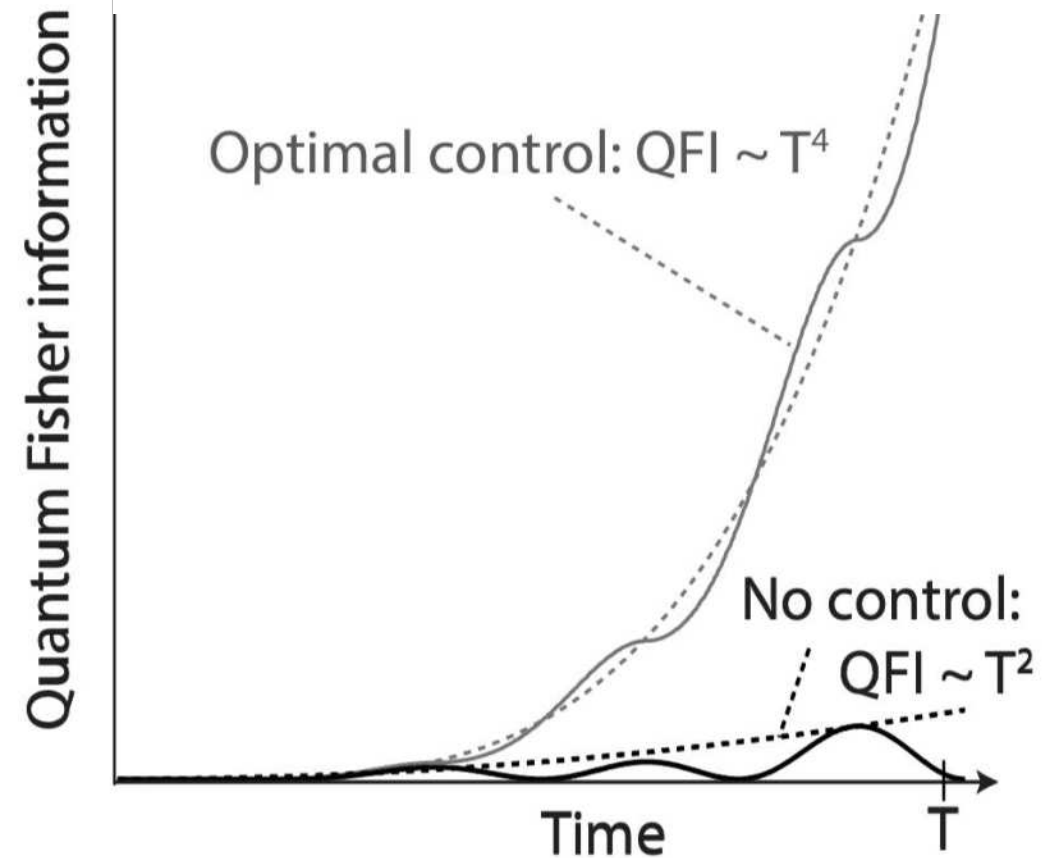
Adaptive sensing: frequency estimation



Goal: Estimate drive freq.



Improved the QFI by factor of 740.
First observation of this T^4 scaling.
Absolute fundamental limit.



Hybrid systems and sensors

NV / spins

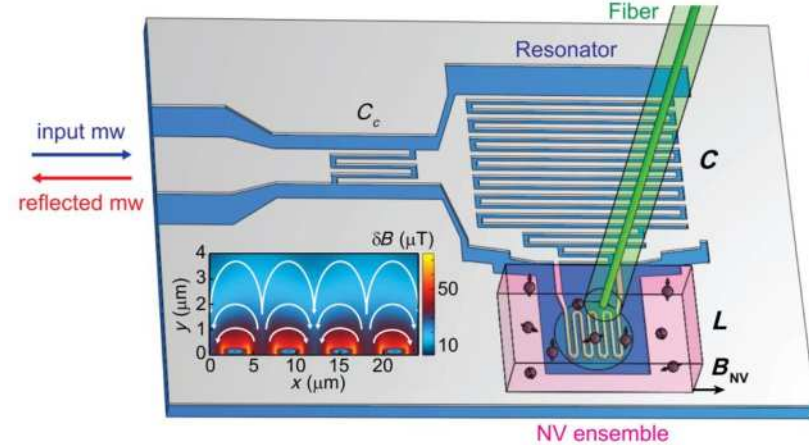


image: Grezes (2014), see also Bienfait (2015)

Quantum magnonics

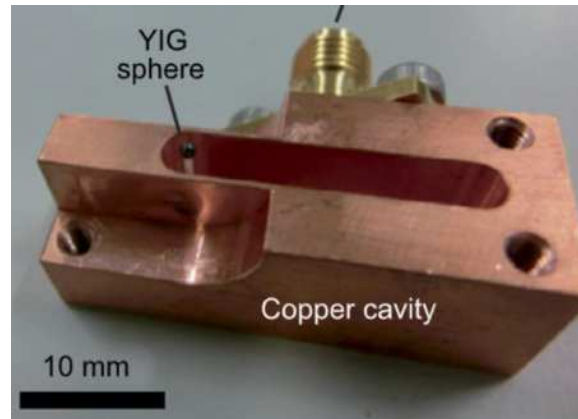


image: Tabuchi (2016)

Squeezed-vacuum bath

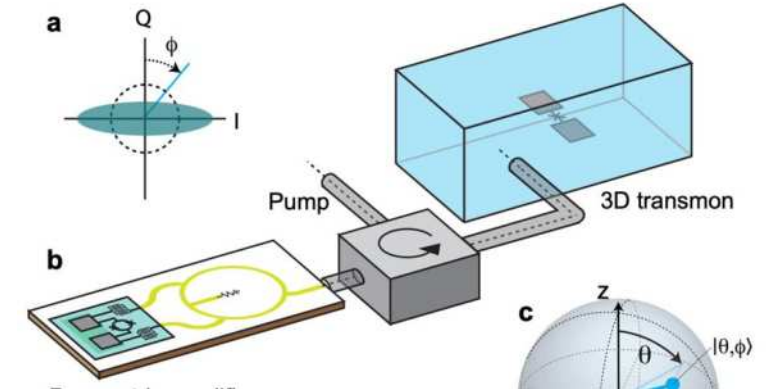


image: Murch (2018)

ABS (break jj)

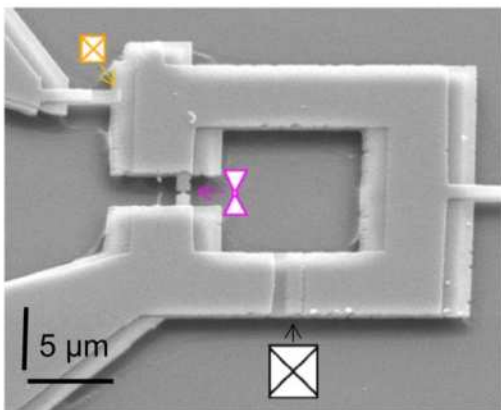


image: Bretheau (2013)

Semiconductor nanowire

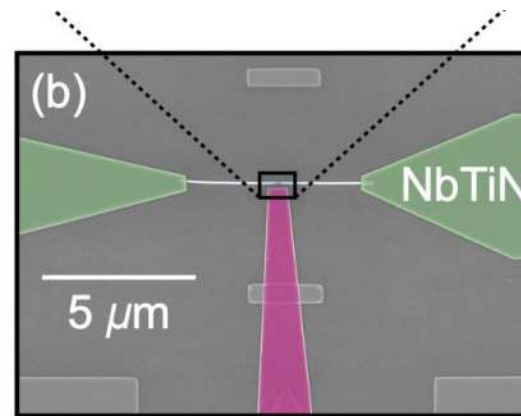


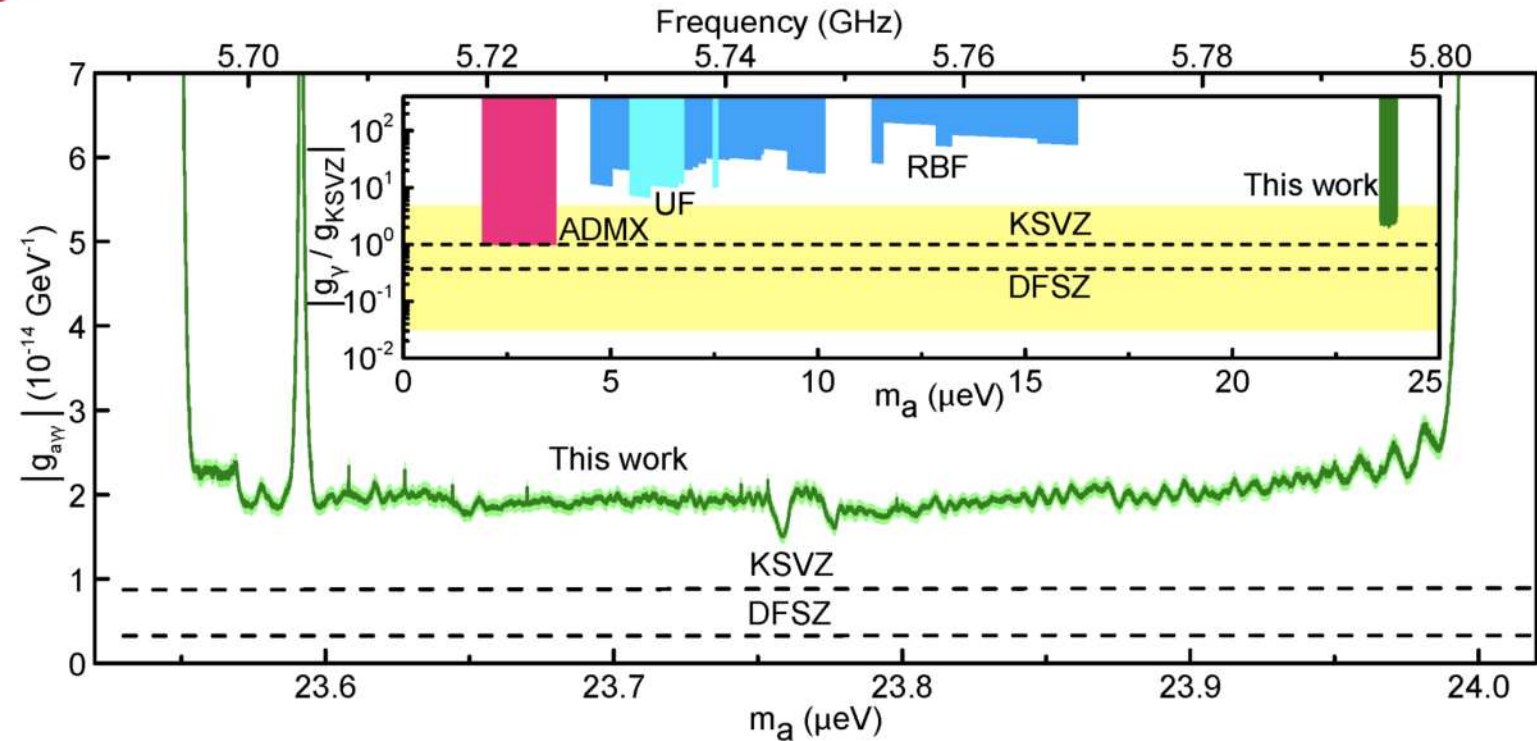
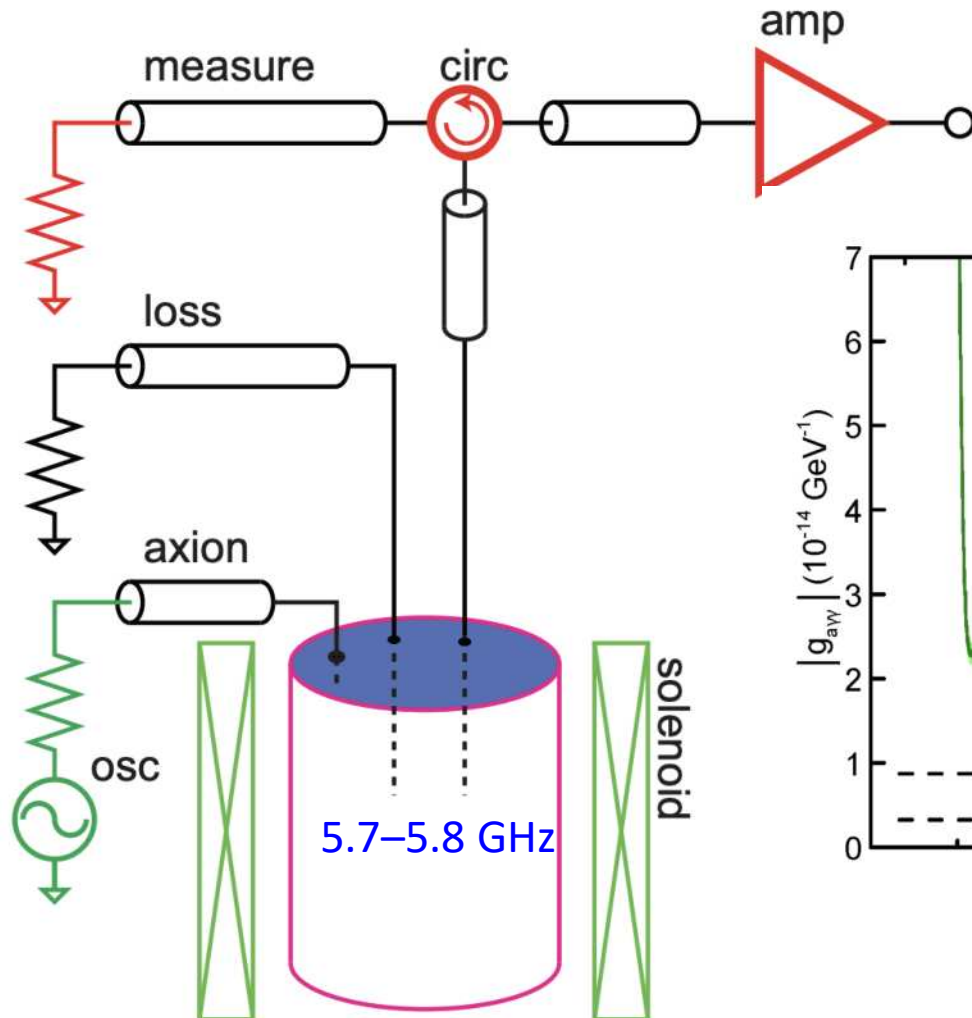
image: Hays (2018)

and many more

...

Axion search: detecting photos

a microwave cavity search for cold dark matter (CDM) axions with masses above $20\ \mu\text{eV}$

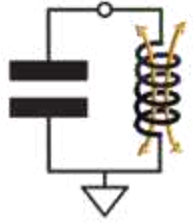


Zheng et al. (2016), Brubaker et al. (2017)

Detecting photos

Case study

Advancing Quantum Sensing via Quantum Control with



Good at measurements

Demonstrate highest efficiency to
measure single atomic de-excitation

$$\eta \approx 1$$

Good at real-time control

Demonstrate ns-latency feedback
and conditional measurements

Versatile

Demonstrate an atom experiment, not natural to circuits, that
cannot be done with real atoms



To catch and reverse a quantum jump mid-flight



Nature 570, 200 (2019)

Thesis: arXiv:1902.10355



Nintendo Corp.

Zlatko Minev*

Department of Applied Physics, Yale University

S.O. Mundhada

S. Shankar

P. Reinhold

R. Gutiérrez-Jáuregui⁺

R.J. Schoelkopf

M. Mirrahimi[‡]

H.J. Carmichael⁺

M.H. Devoret

* Present: IBM Research ⁺ U. Auckland, New Zealand [‡] INRIA Paris, France



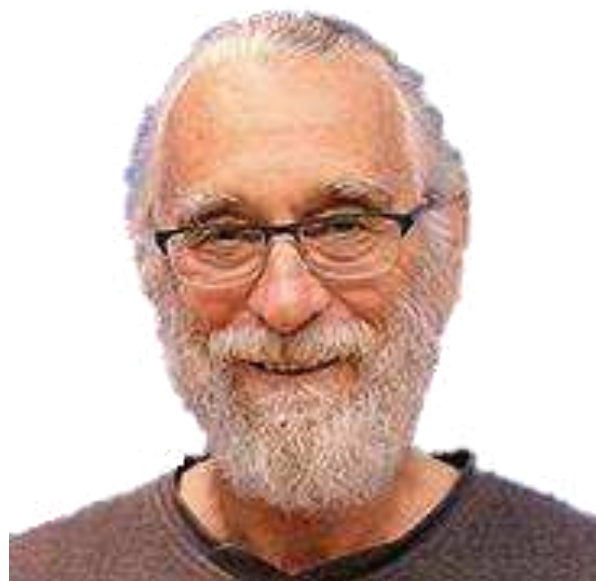
Yale Institute for Nanoscience
and Quantum Engineering

Is it possible to sense if there is an
advance warning signal
that a quantum jump is about to occur?

measurements

versatility

real-time control



H.J. Carmichael

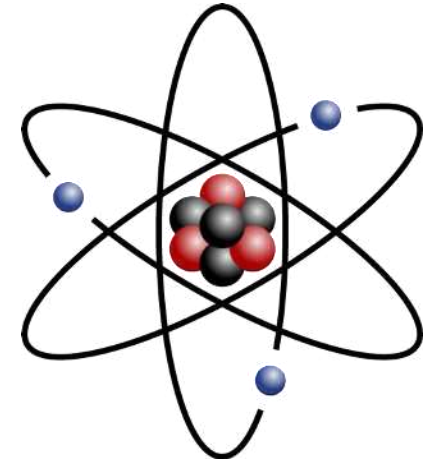


Outline

Jumps in a quantum system

original observation

quantum trajectory prediction

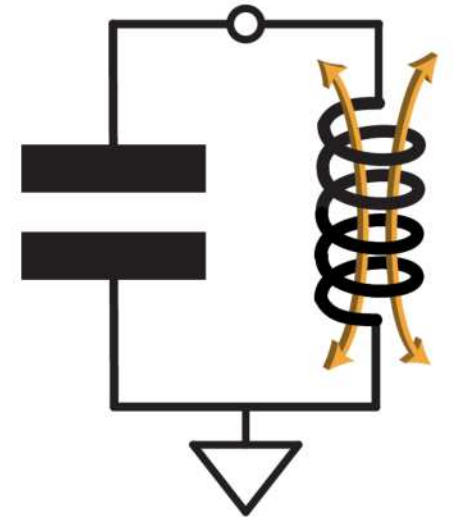


Art: Indoleces

Circuit quantum electrodynamics realization

experimental apparatus

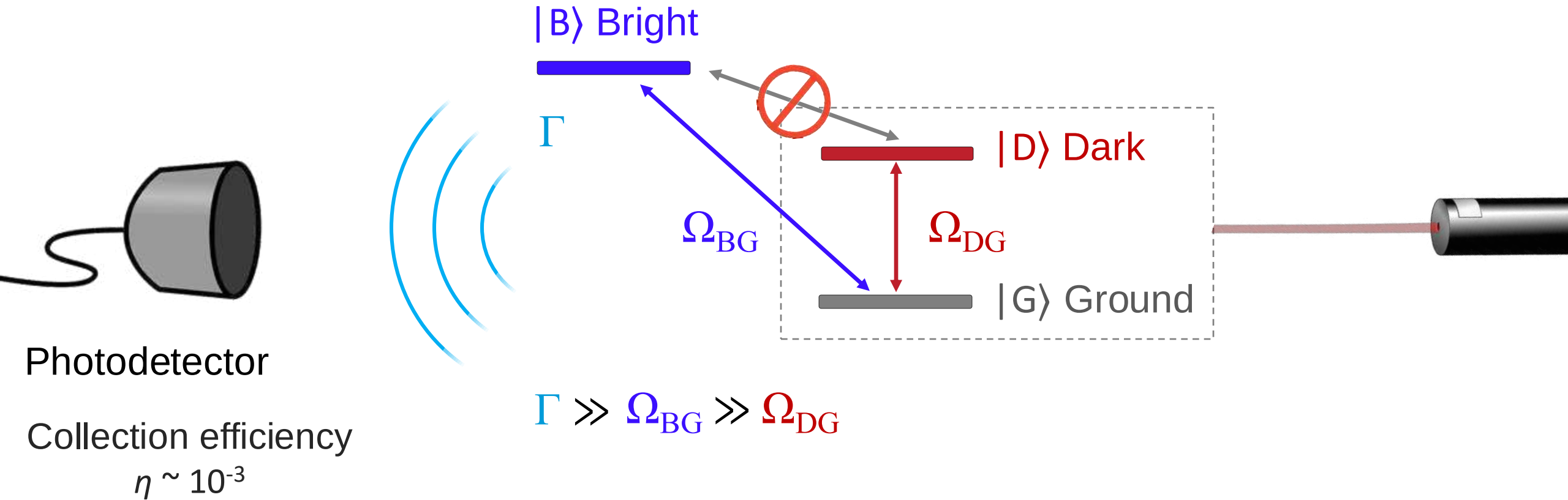
results on jump coherence and determinism



Quantum jumps of an *individual* atom



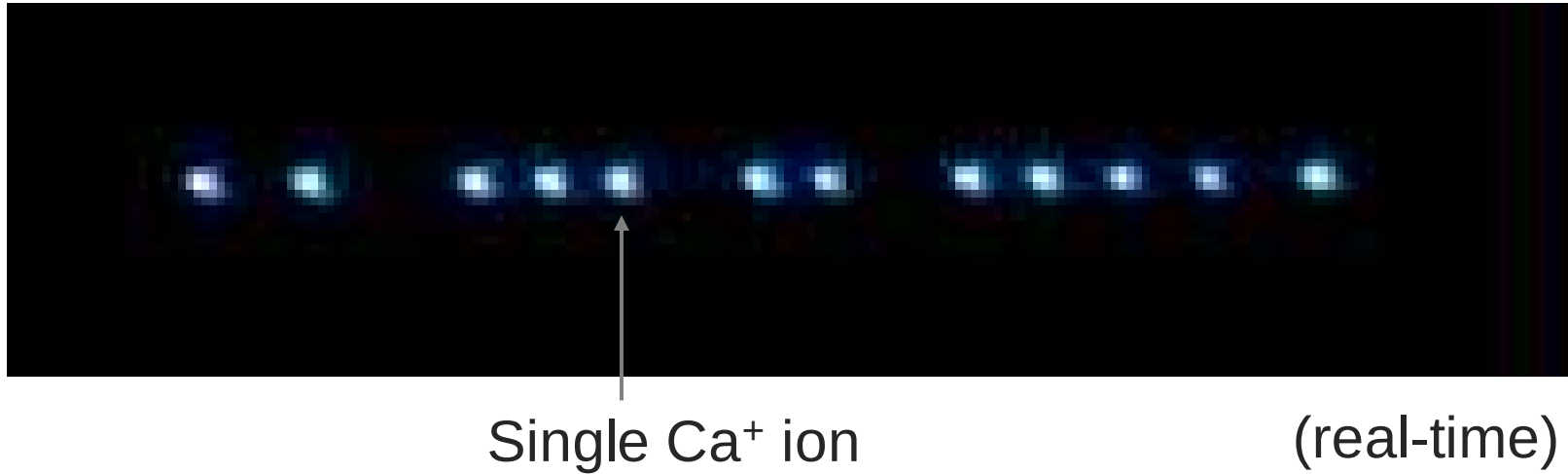
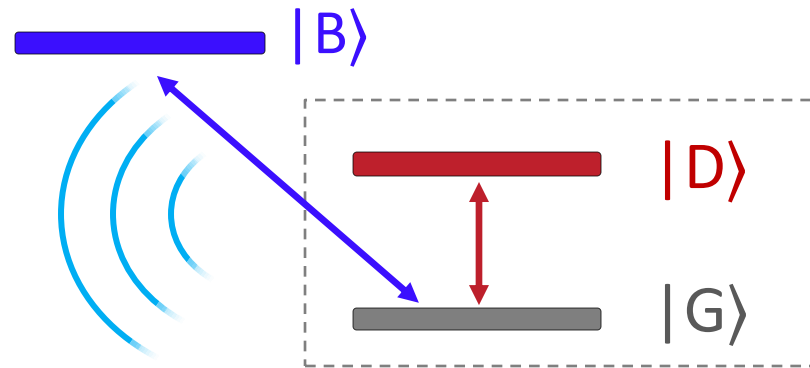
Observation of jumps in an *individual* atom



Exp: Berquist *et al.*, PRL (1986); Sauter *et al.*, PRL (1986); Nagourney *et al.*, PRL (1986)

Thy: Cook & Kimble, PRL (1985); Cohen-Tannoudji & Dalibard, EPL (1986), ...

Quantum jumps of *individual* ions



Principle of quantum jumps observation

Photodetector

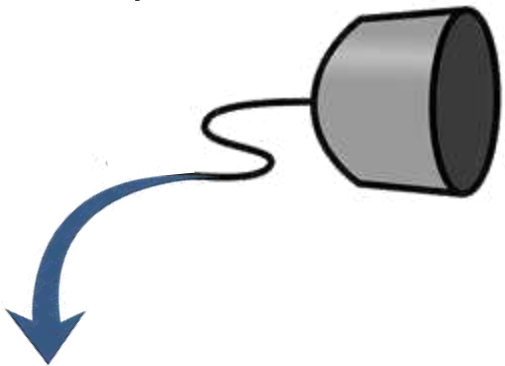
$$\eta \sim 10^{-3}$$

$$\Gamma \gg \Omega_{BG} \gg \Omega_{DG}$$

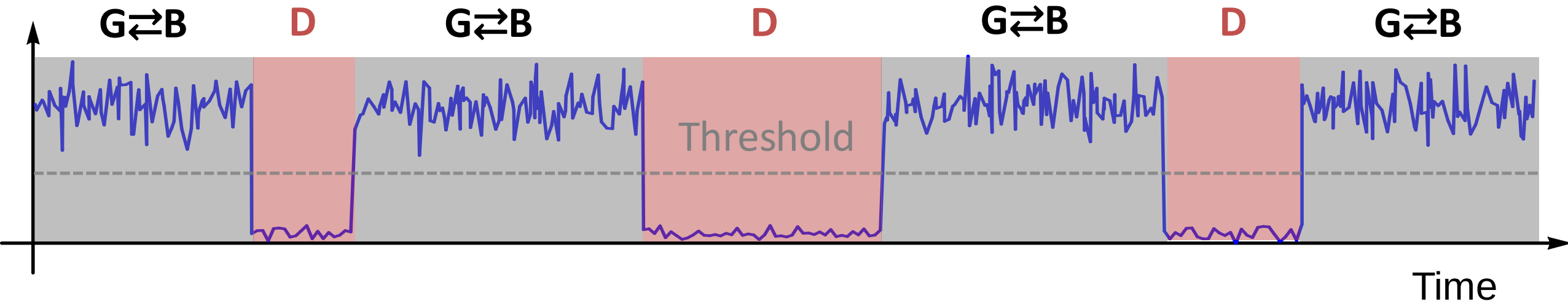
$|B\rangle$ Bright

$|D\rangle$ Dark

$|G\rangle$ Ground



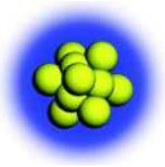
Measurement record (fluorescence)



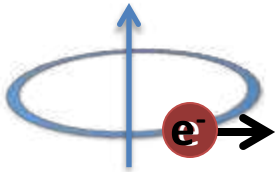
Observations of quantum jumps since 1986

Trapped massive particles

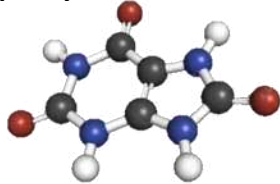
Single *ion* [1-3] (1986)



Single *electron* [4] (1992)
($1e^-$ cyclotron oscillator)

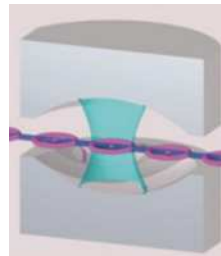


Single *molecule* [5] (1995)
(large polyatomic molecule)

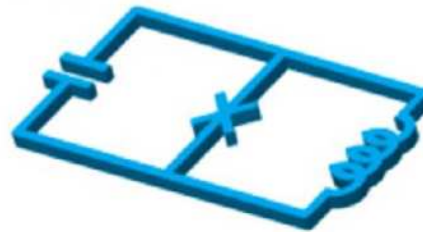


Light quanta

Microwave cavity
(*CQED*) [6] (2007)

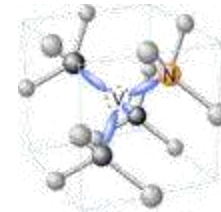


Superconducting qubit
& cavity (*cQED*) [9-11] (2011)

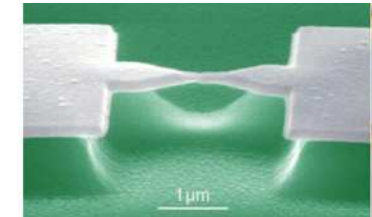


Solid state and mesoscopic

Nitrogen–vacancy center
in diamond [7,8] (2010)



Atomic-point contact
(Andreev states) [12] (2015)



And others!

[1] PRL 56, 2797 (1986)

[2] PRL 57, 1696 (1986)

[3] PRL 57, 1699 (1986)

[4] PRL 83, 1287 (1999)

[5] Nature 373, 132 (1995)

[6] Nature 446, 297 (2007)

[7] Science 329, 542 (2010)

[8] Nature 477, 574 (2011)

[9] PRL 106, 110502 (2011)

[10] Science 339, 178 (2013)

[11] Nature 511, 444 (2014)

[12] Science 349, 1199 (2015)

“If all this damned *quantum jumping* were really to stay,
I should be *sorry* I ever got involved with quantum theory.”

- E. Schrödinger

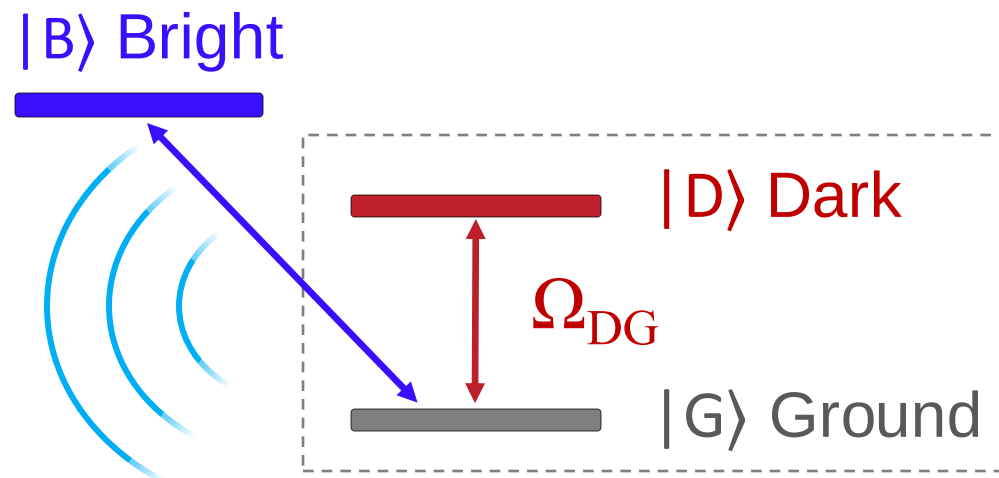


The case of the fully efficient observer

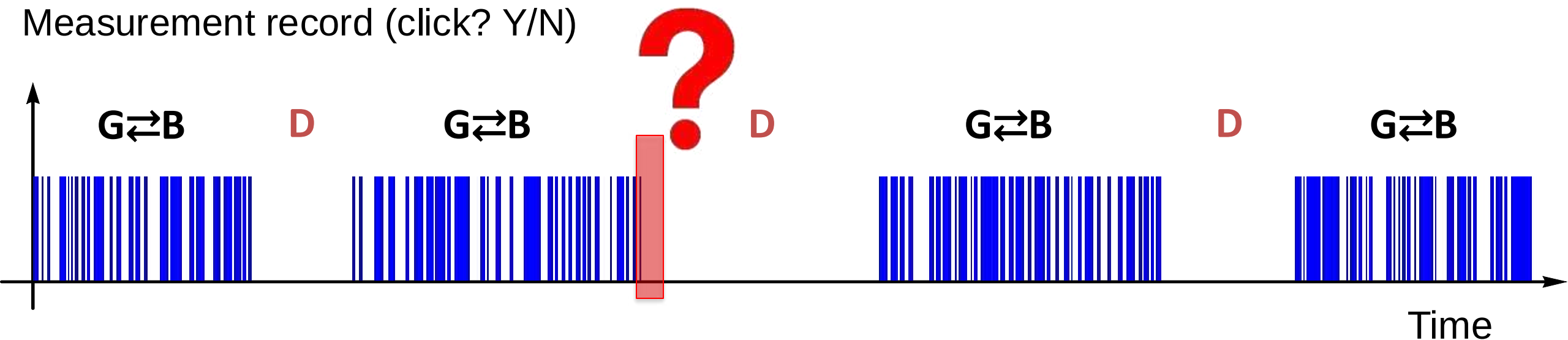
Hypothetical efficiency $\eta = 1$

$$BW_{\text{det}} \gg \frac{\Omega_{\text{BG}}^2}{\Gamma} \gg \Omega_{\text{DG}}$$

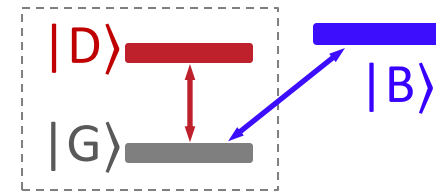
$$\Gamma \gg \Omega_{\text{BG}} \gg \Omega_{\text{DG}}$$



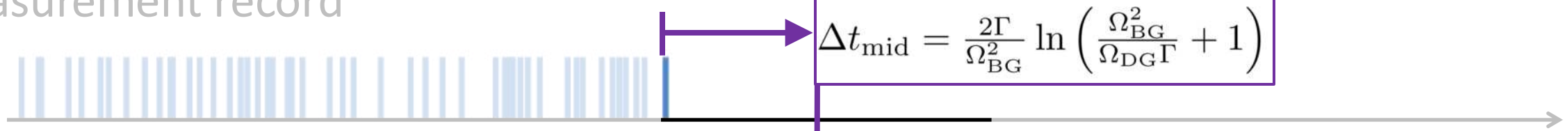
Measurement record (click? Y/N)



No-click trajectory

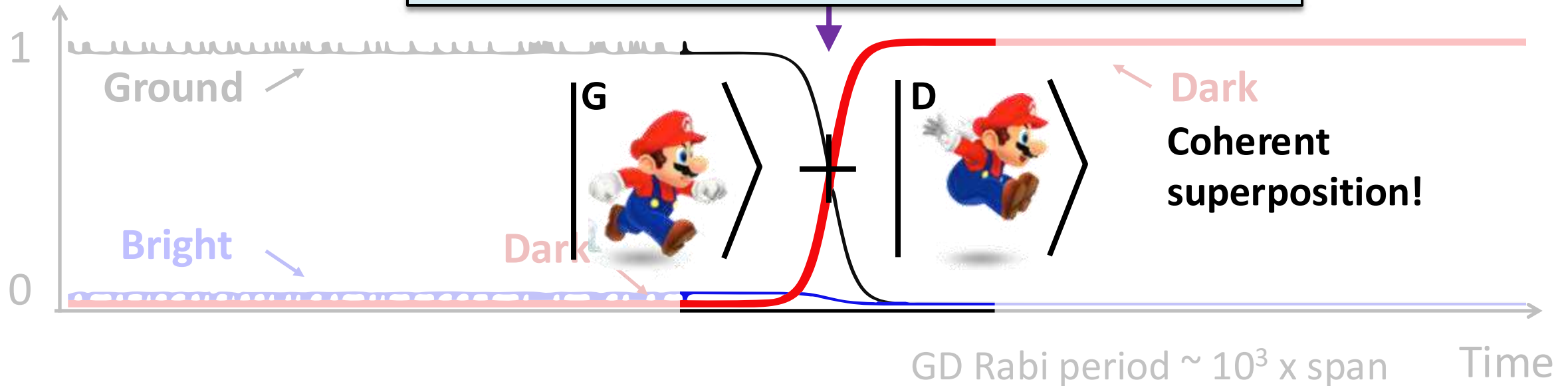


Measurement record



Inferred population

Requires not missing a **SINGLE** click!



Carmichael, SUSSP71 (2015); Ruskov et al., PRB (2007); Related: Porra & Putterman, PRA (1987); Mabuchi & Zoller PRL (1996); Plenio & Knight, RMP (1998); Katz et al., Science (2006) & PRL (2008)

Principle of our cQED experiment

Related quantum circuit experiments on trajectories of 2-level atoms:

Hacohen-Gourgy *et al.*, PRL (2018)

Cottet *et al.*, PNAS (2017)

Hacohen-Gourgy *et al.*, Nature (2016)

Slichter *et al.*, NJP (2016)

Naghiloo *et al.*, Nat. Comm. (2016)

Campagne-Ibarcq *et al.*, PRL (2016)

Weber *et al.*, Nature (2014)

de Lange *et al.*, PRL (2014)

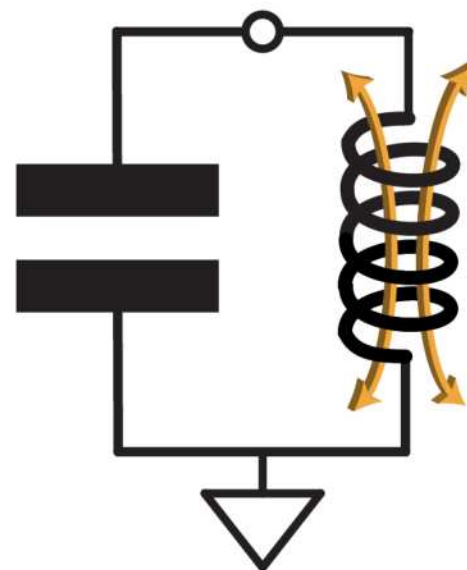
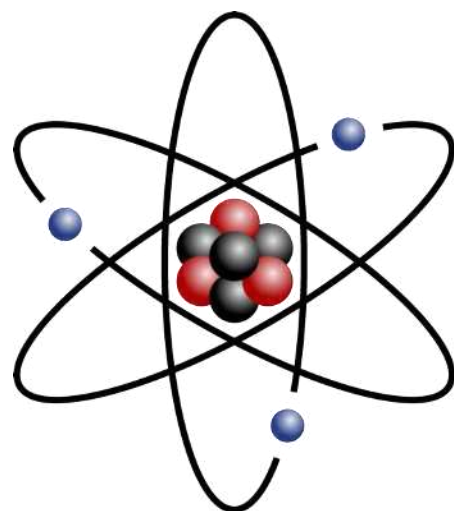
Murch *et al.*, Nature (2013)

Campagne-Ibarcq *et al.*, PRX (2013)

Katz *et al.*, PRL (2008)

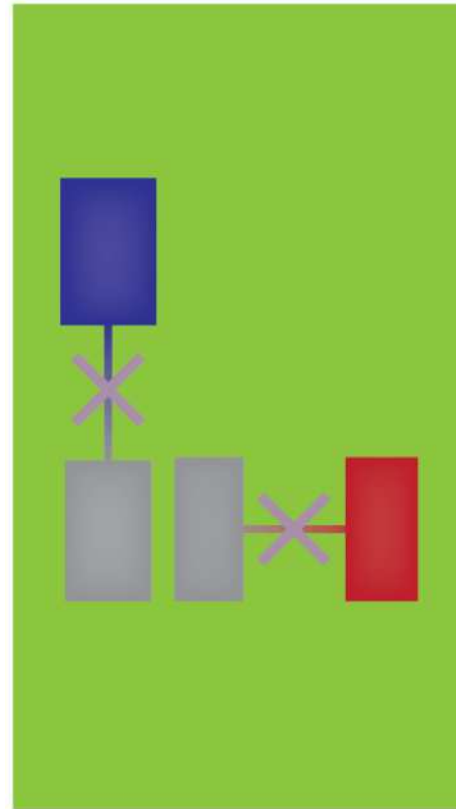
Katz *et al.*, Science (2006)

...

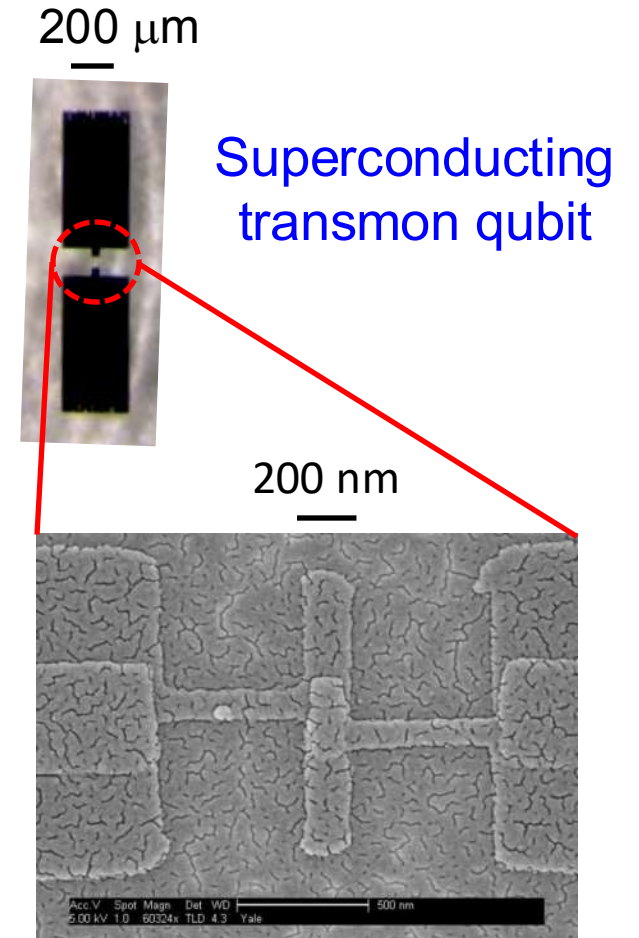


Quantum superconducting circuit architecture

Schematic representation of chip

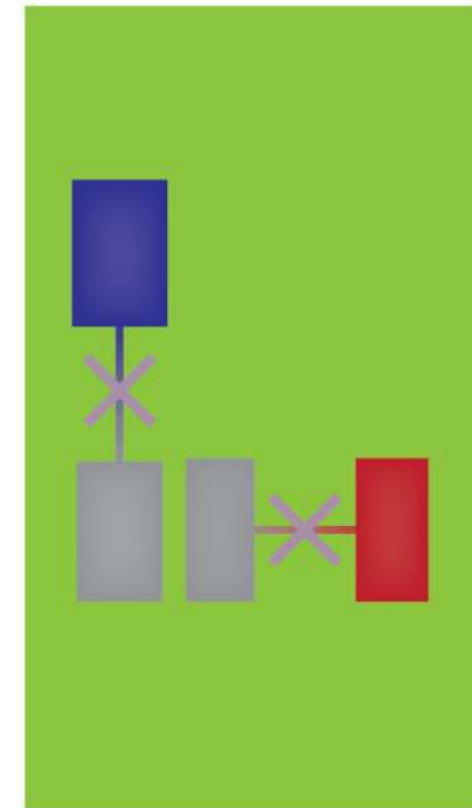
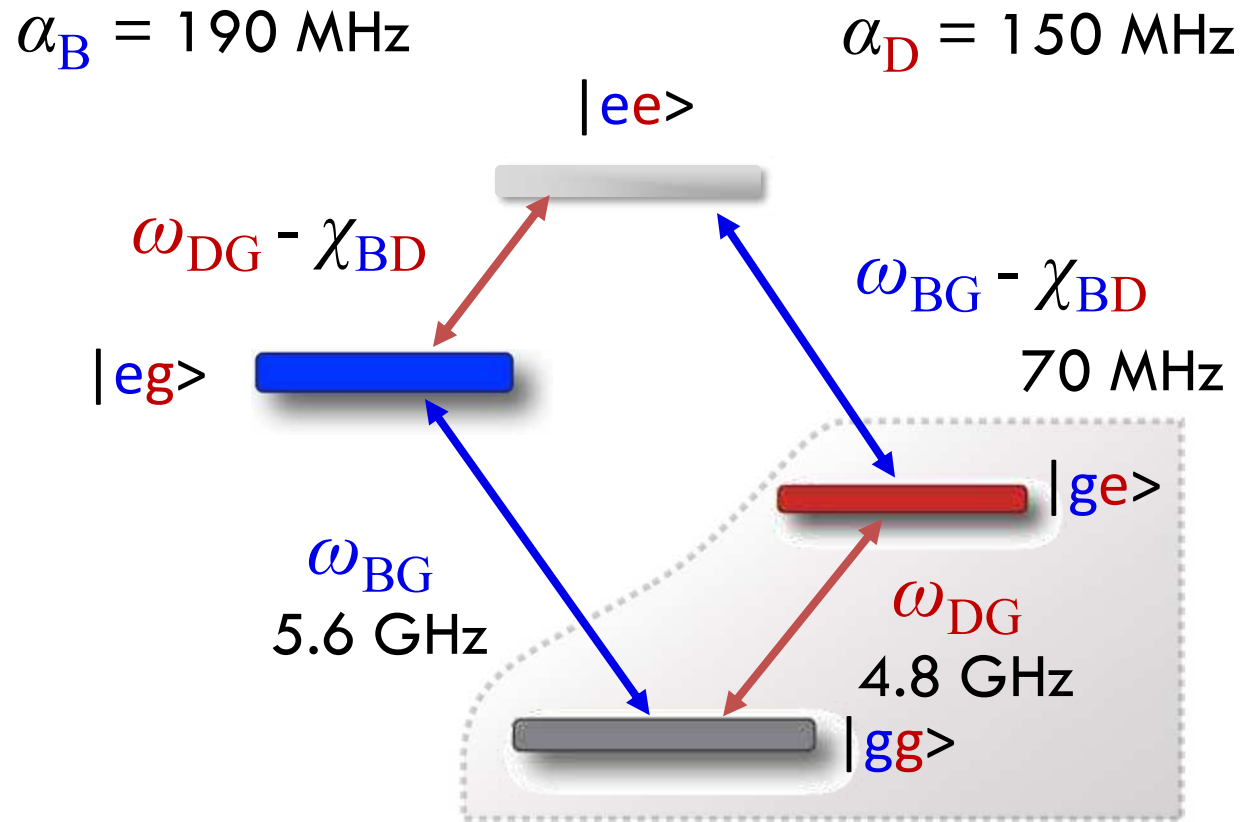


Substrate
(not to scale)



cQED implementation of V-system

Schematic representation of chip



Substrate
(not to scale)

Related work: Gambetta *et al.*, PRL (2011)

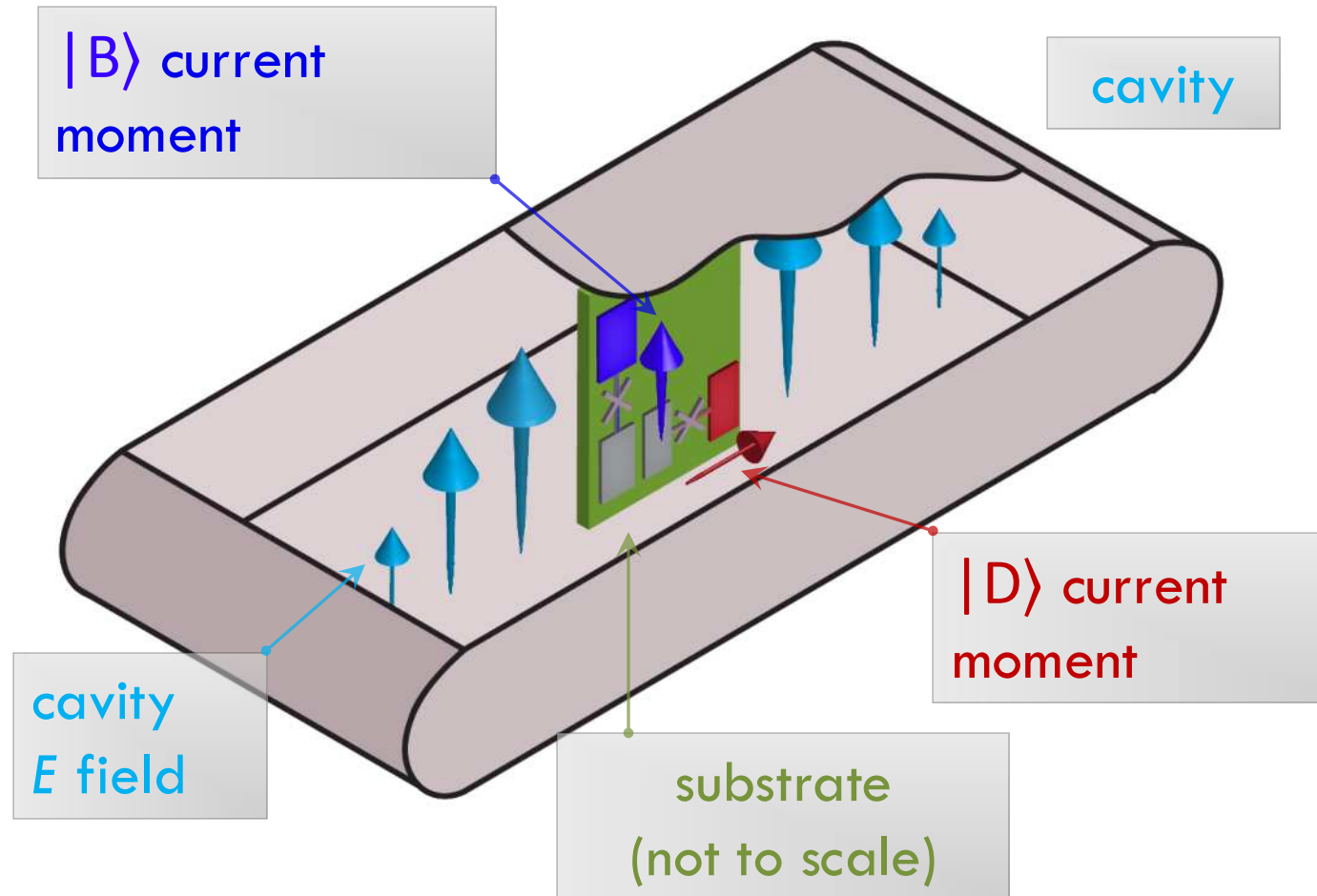
Srinivasan *et al.*, PRL (2011)

Dumur *et al.*, PRB (2015)

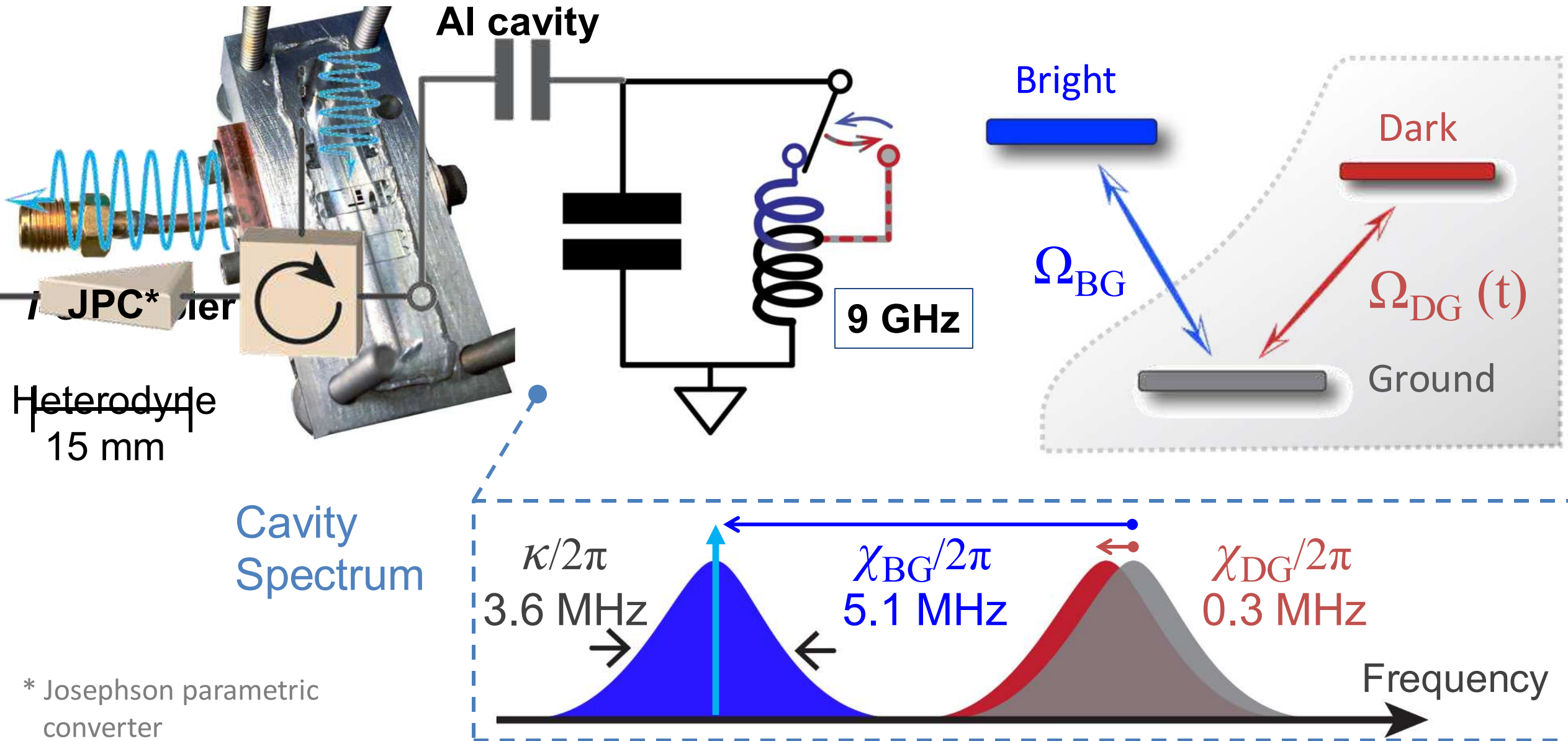
Zhang *et al.*, Nature JQI (2017) ...

cQED implementation of V-system

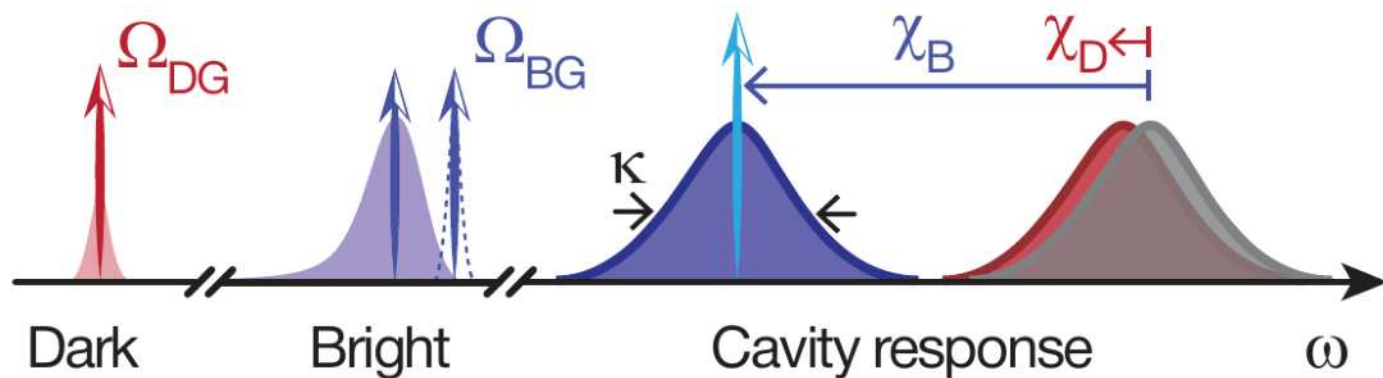
Schematic representation of design



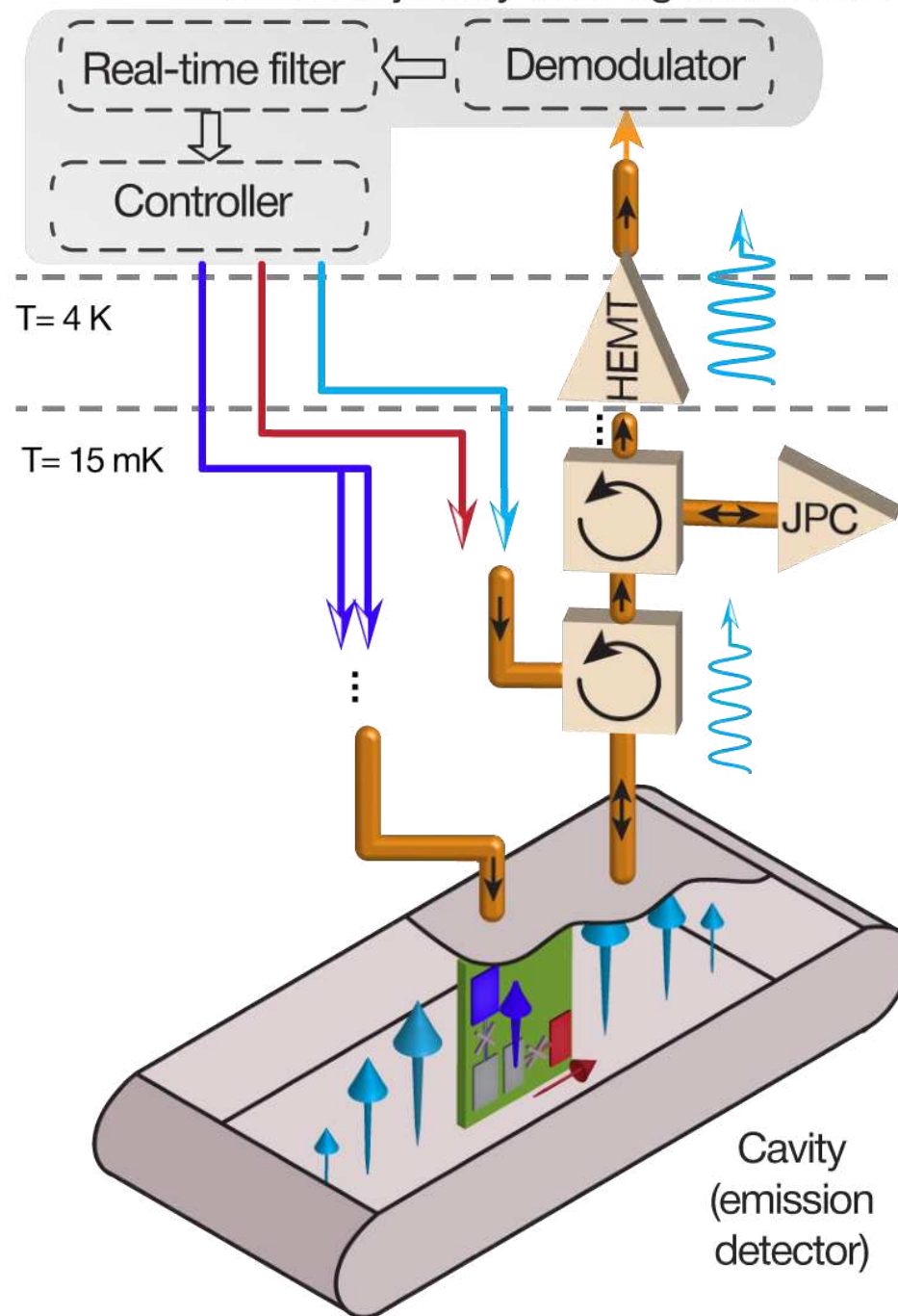
Measuring B / non-B with 90% efficiency



Frequency landscape

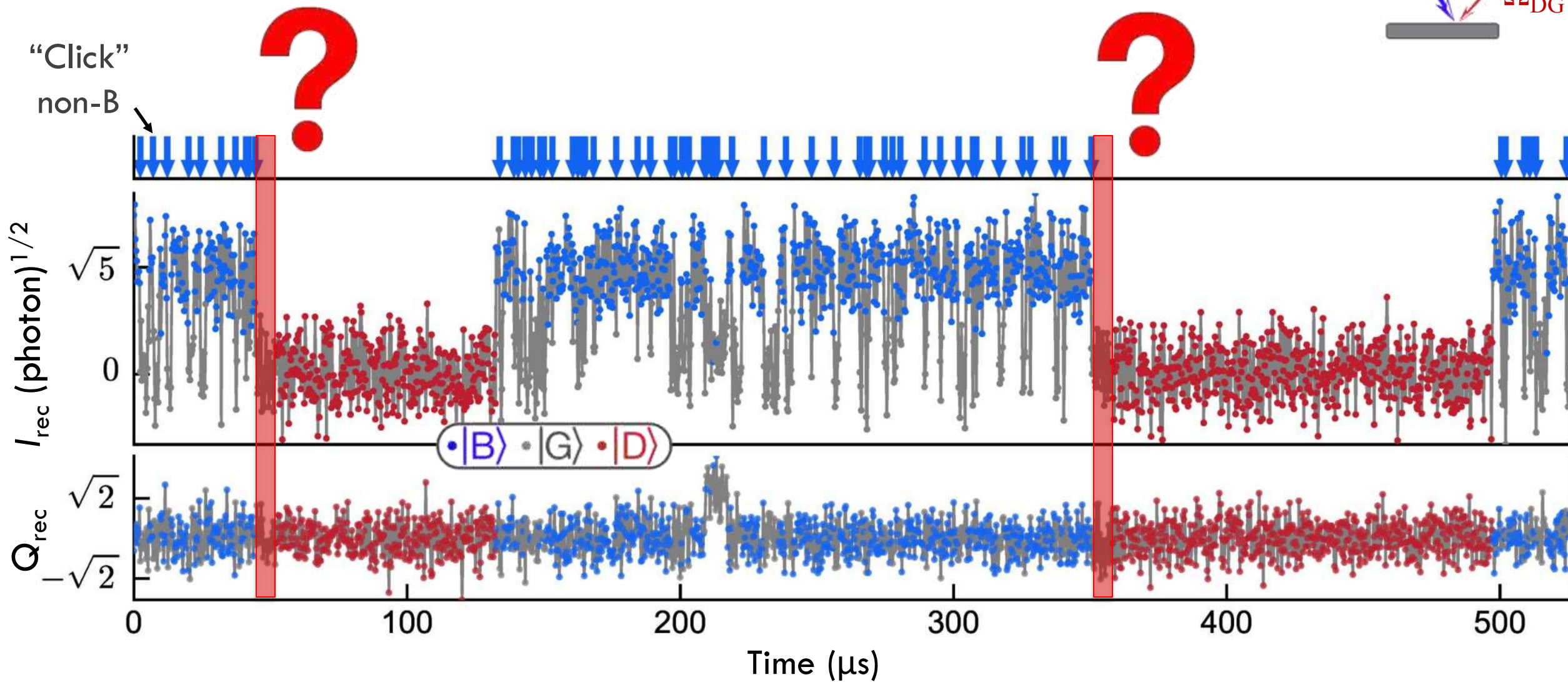
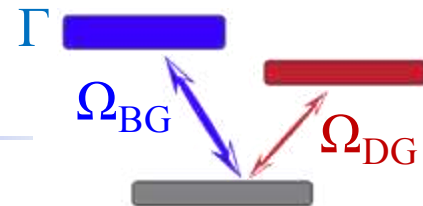


T= 300 K FPGA trajectory tracking and control



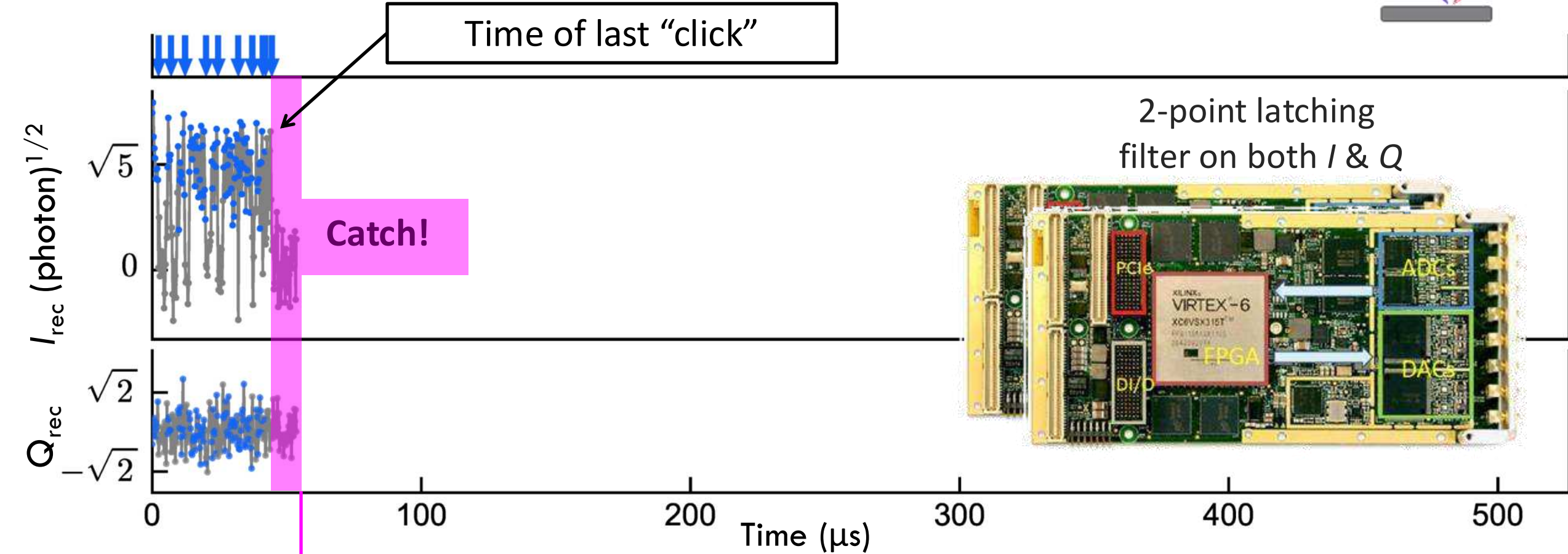
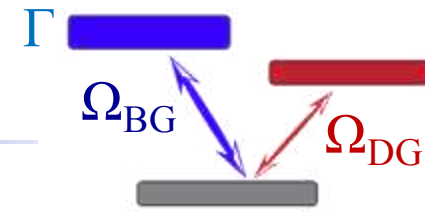
Unconditioned quantum jumps

Inferred state & click times



Real-time detection of a quantum jump

Catch protocol



Msmt. rate
 $\Gamma_m^{-1} \sim 8 \text{ ns}$

$\ll$

Integration time
 $2.6 \times 10^2 \text{ ns}$

$\ll$

GB jump timescale
 10^3 ns

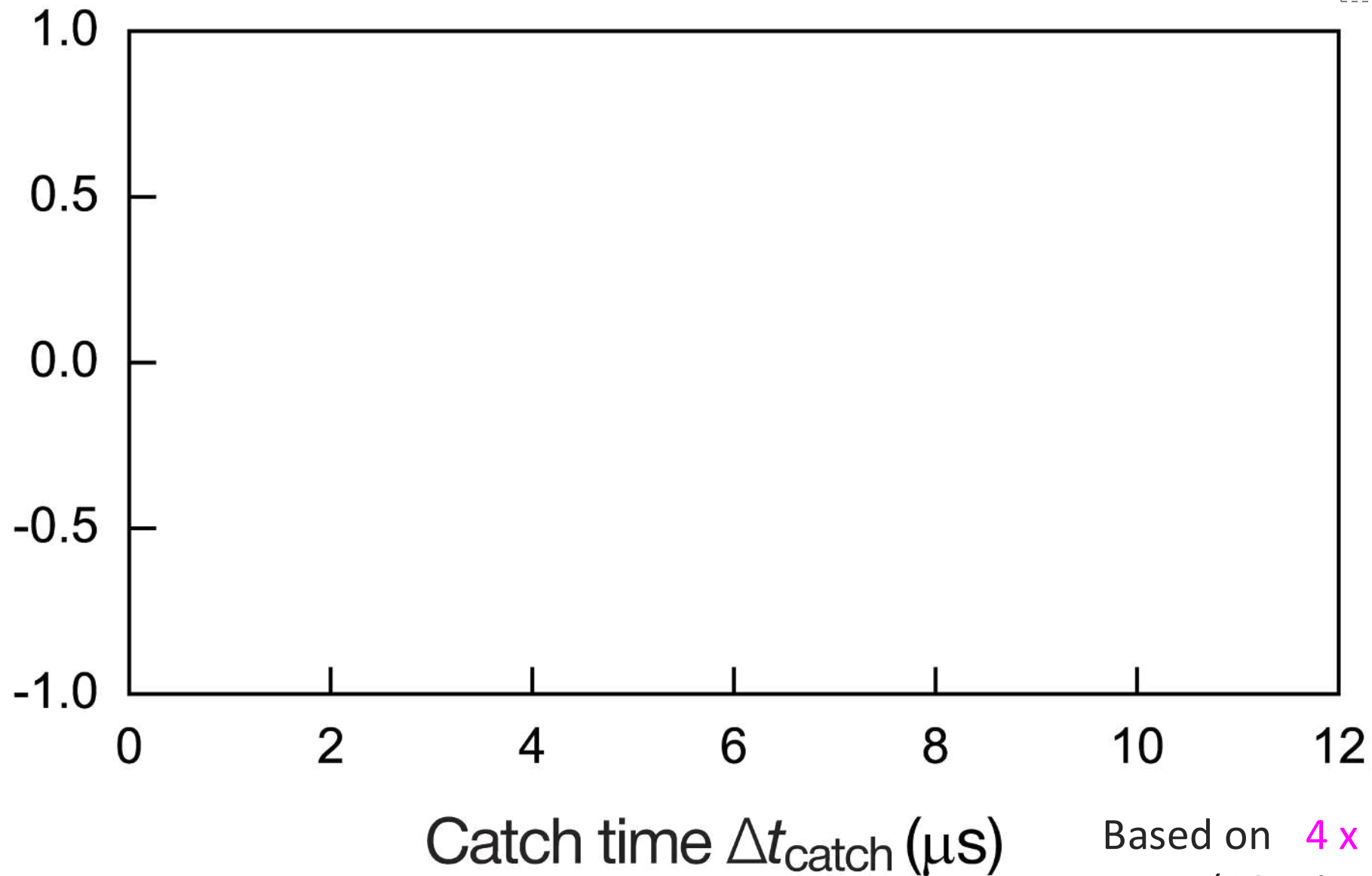
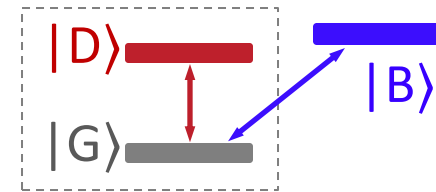
$\ll$

t_{mid}

$\ll$

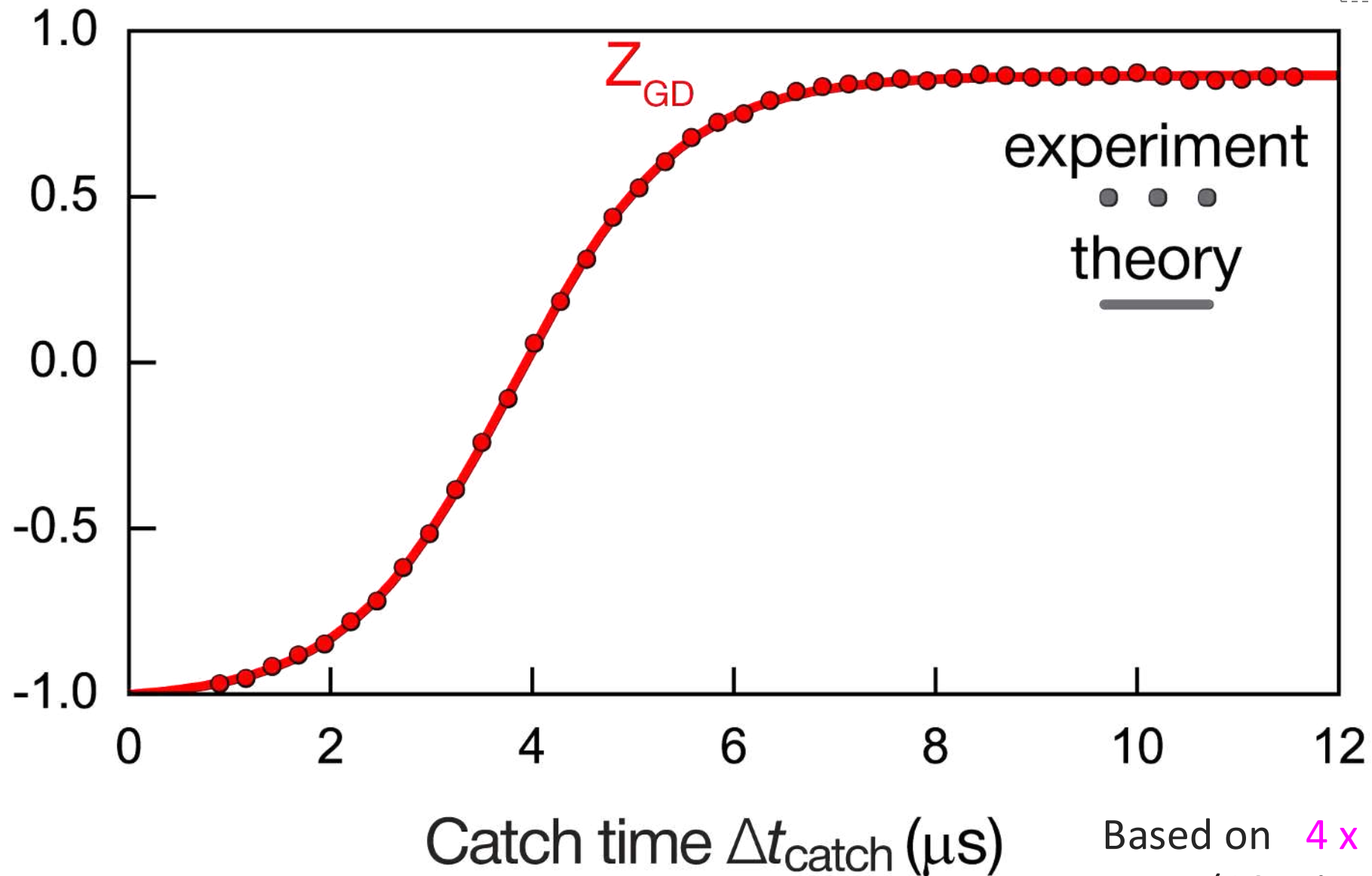
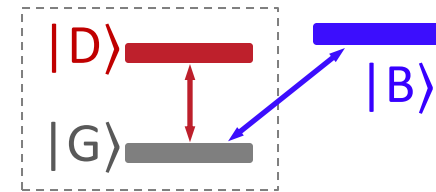
GD jump timescale
 $\sim 10^5 \text{ ns}$

Catching the quantum jump mid-flight



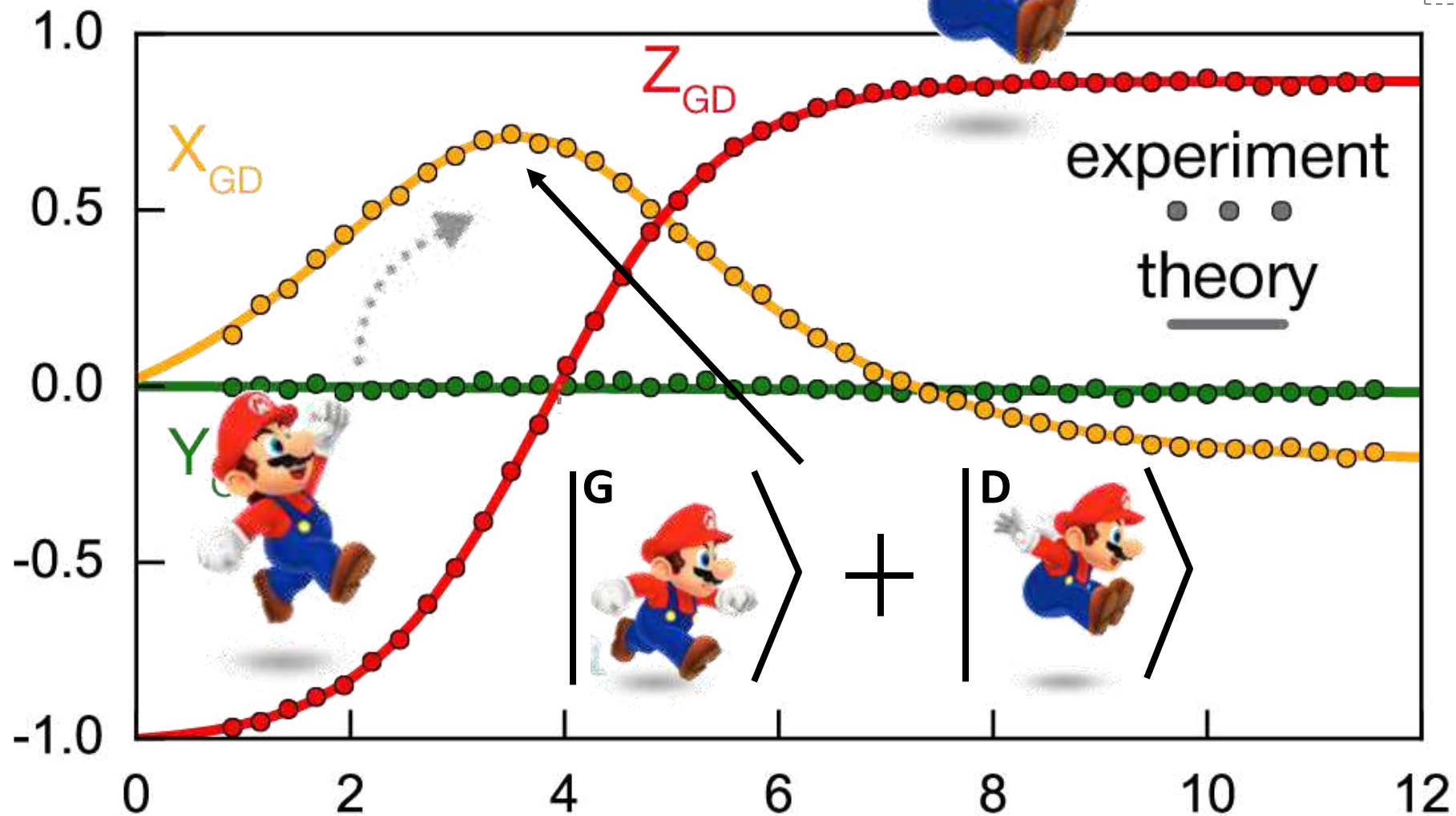
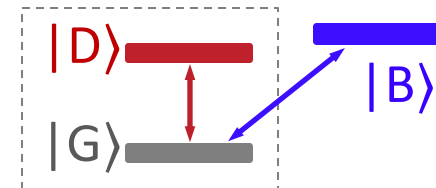
Based on 4×10^6 catch events
(10 min. tracking time)

Catching the quantum jump mid-flight



Based on 4×10^6 catch events
(10 min. tracking time)

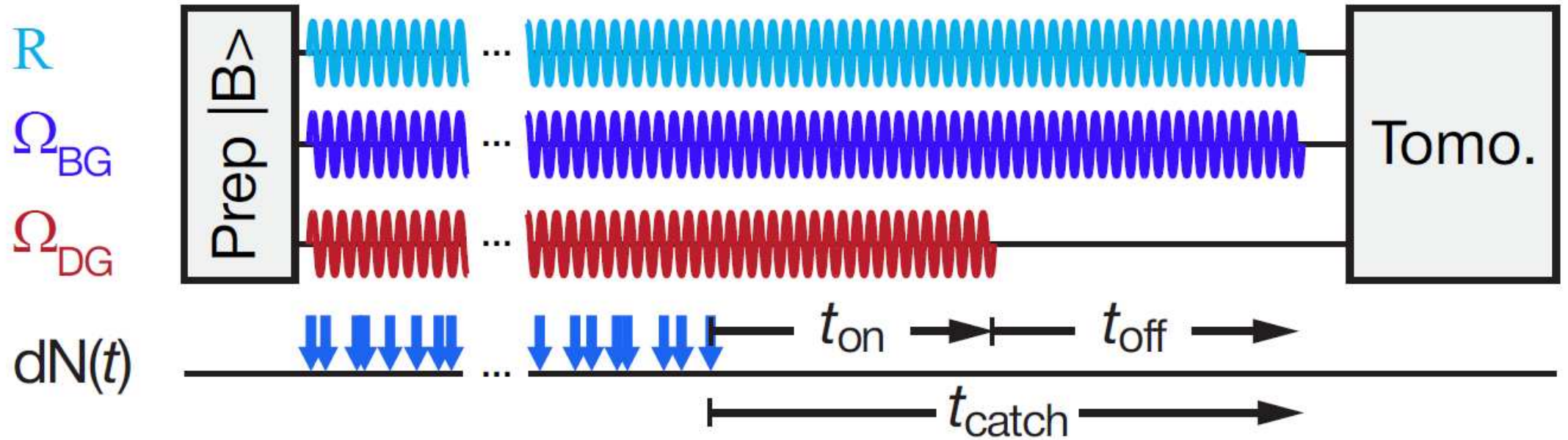
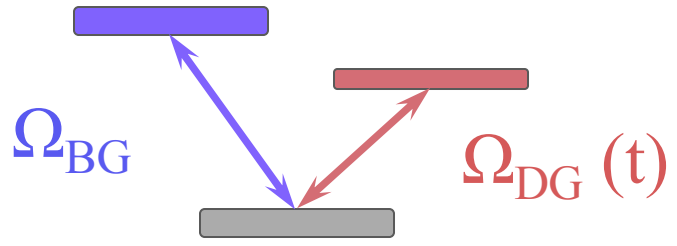
Catching the quantum jump and flight



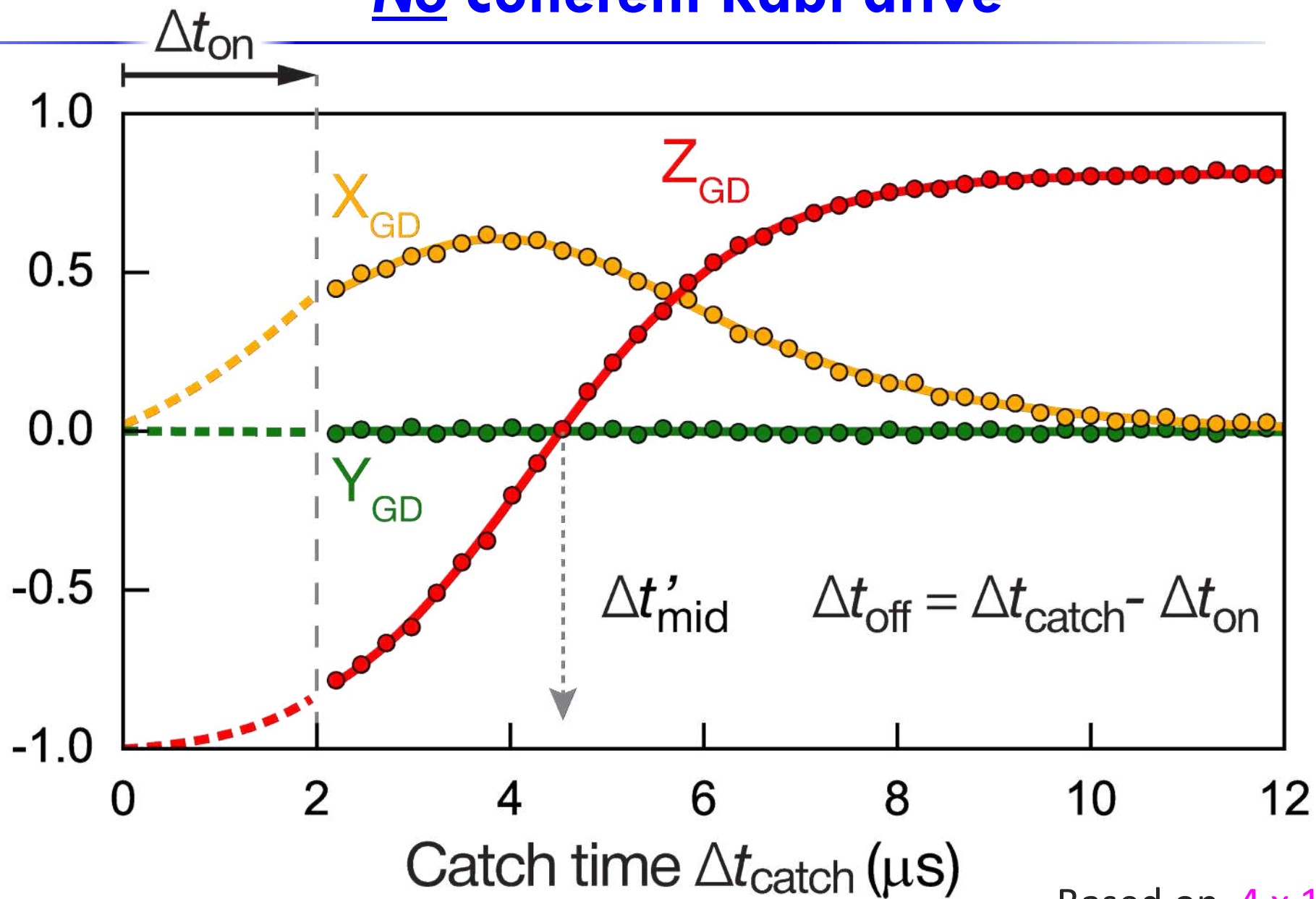
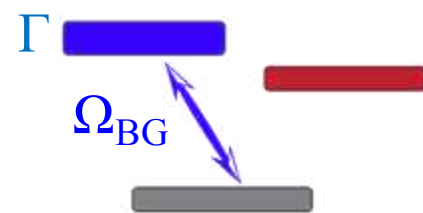
Catch time Δt_{catch} (μs)

Based on 4×10^6 catch events
(10 min. tracking time)

Catch in *absence* of Dark drive

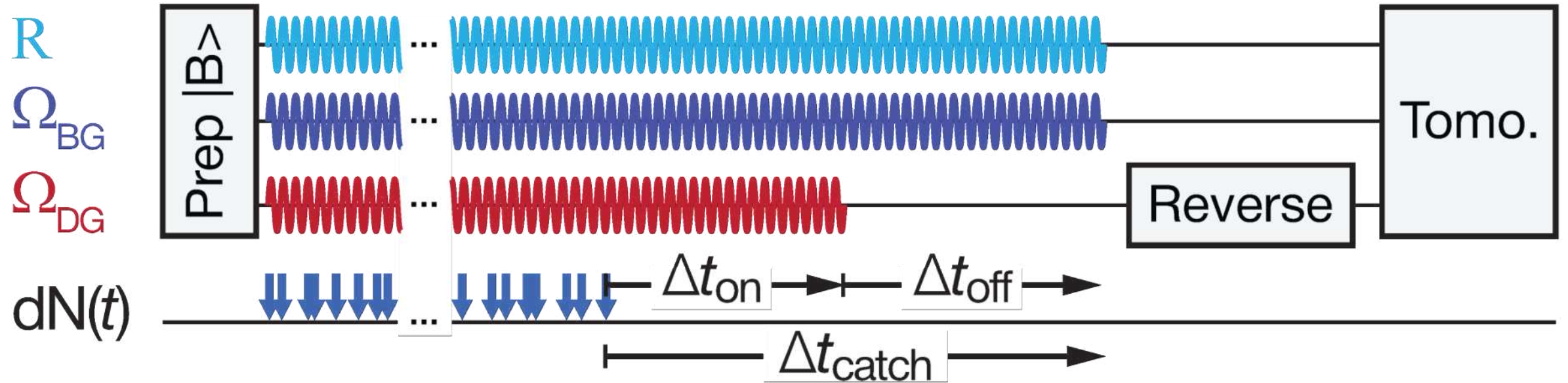


No coherent Rabi drive



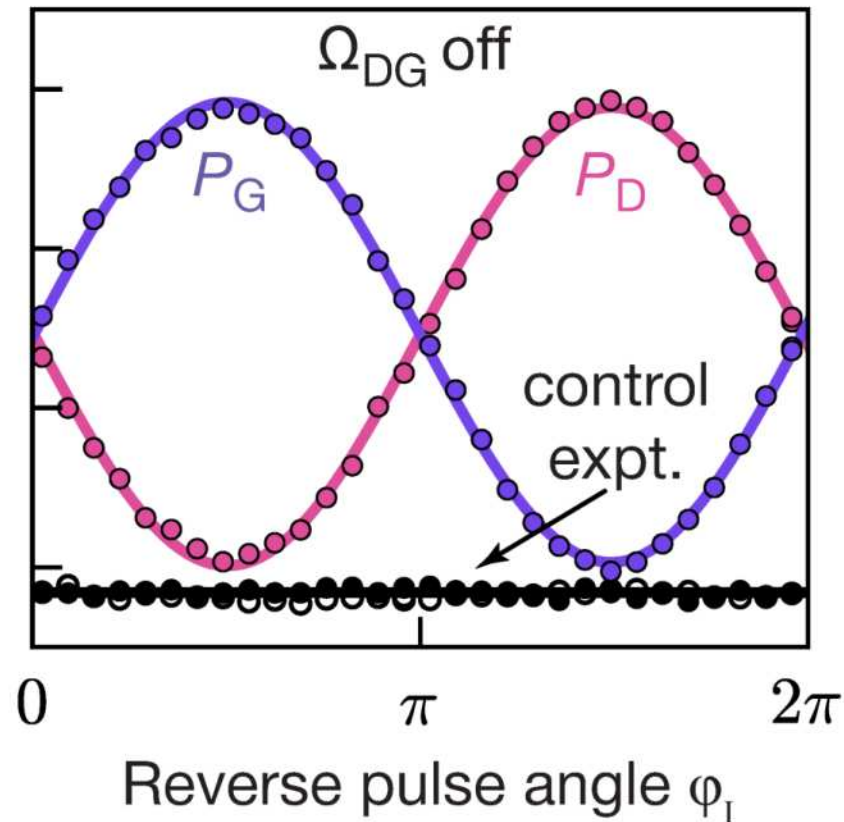
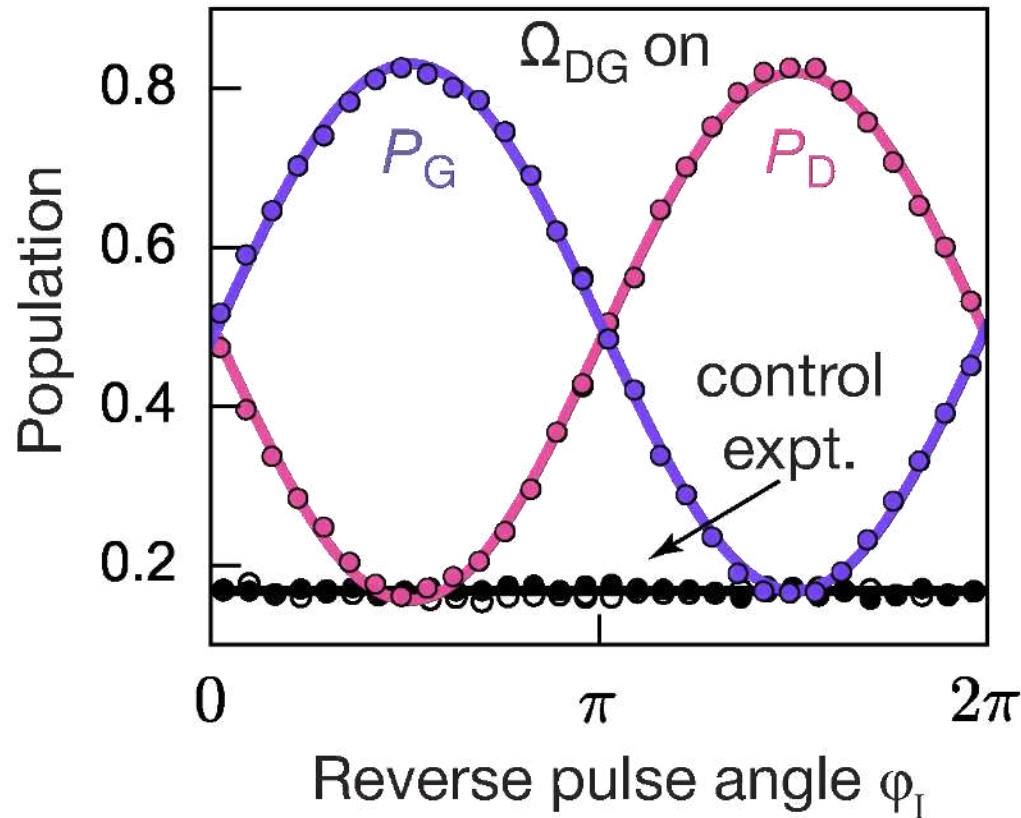
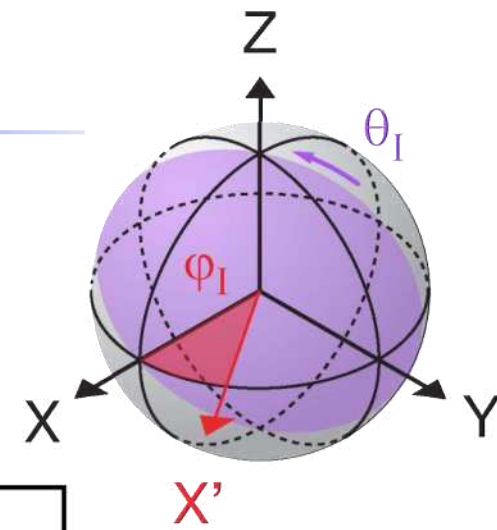
Based on 4×10^6 catch events

Reversing the quantum jump mid-flight

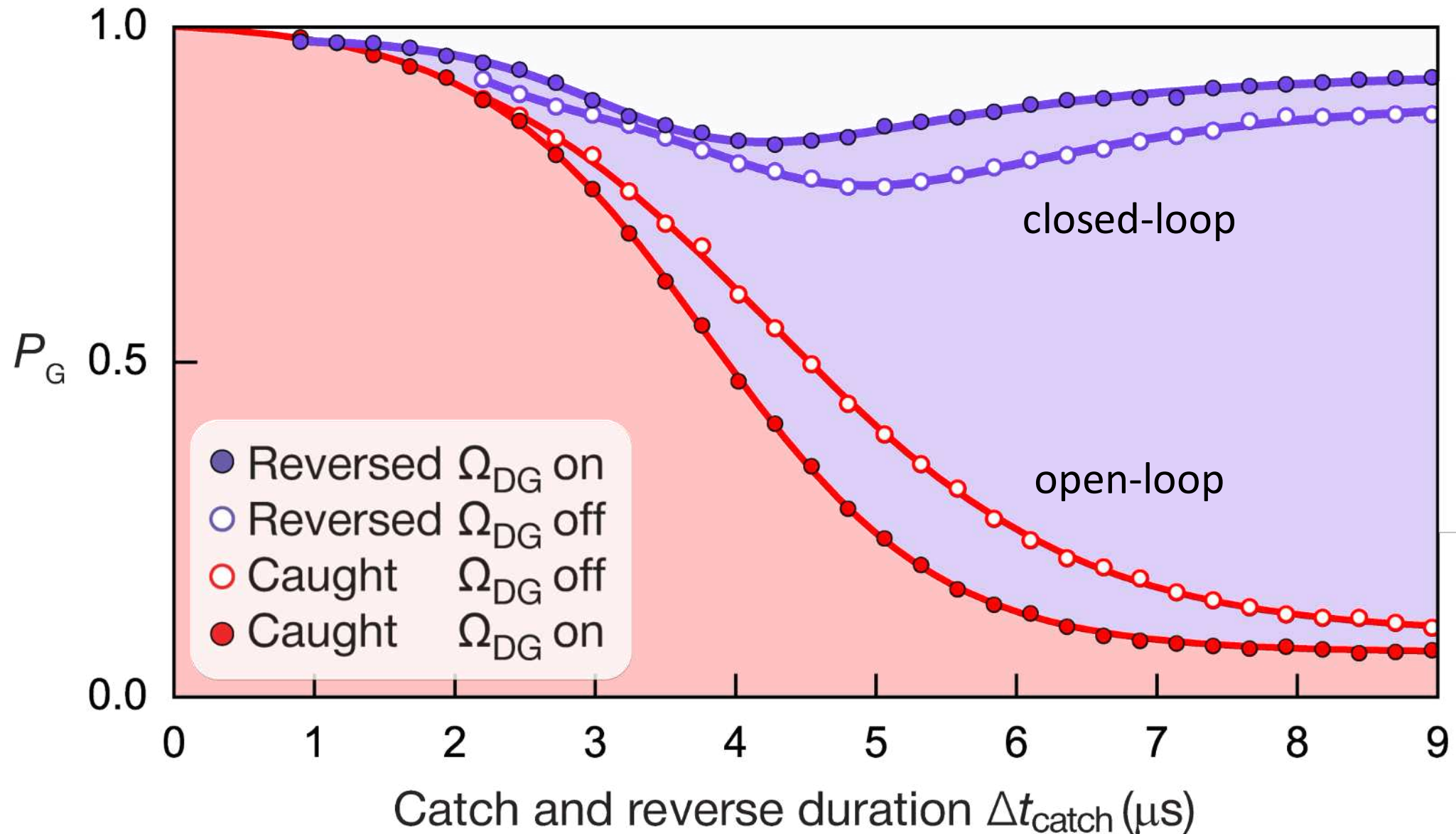


Experimental reversal of jump

No click until $\Delta t_{\text{mid}} =$ advance warning of individual jump

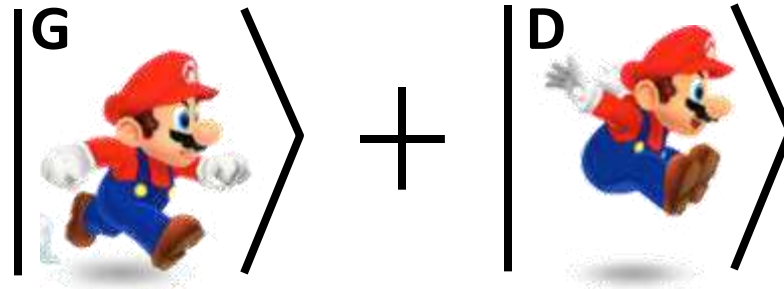


Catching and reversing the quantum jump mid-flight



Conclusions & Future directions

Observed jump “internal coherence” using fast electronics and feed-forward control



Caught individual jumps in mid-flight and reversed them

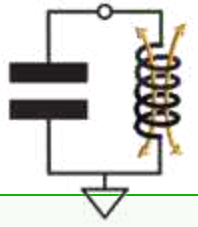
Quantum trajectory efficiently explains details of exp. results

Nature 570, 200 (2019) Mineev, Ph.D. Thesis (2018)

Continuous syndrome measurement for autonomous quantum error correction

Follow ups: [arXiv:1808.00726](#), [arXiv:1810.03225](#), ...

Advancing Quantum Sensing via Quantum Control with



Good at measurements

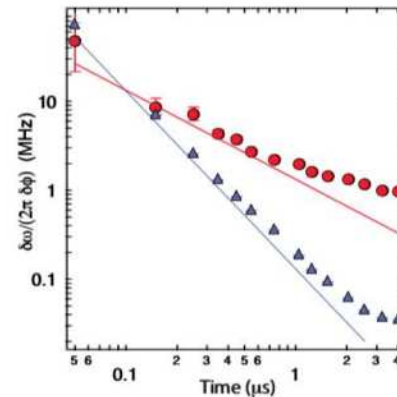
Time-resolved, quantum-limited measurements
near-unit quantum efficiency η
engineer measurements on-demand: X, Y, Z, ...
engineer bath: squeezed vacuum, interaction, ...

Good at real-time control

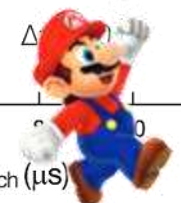
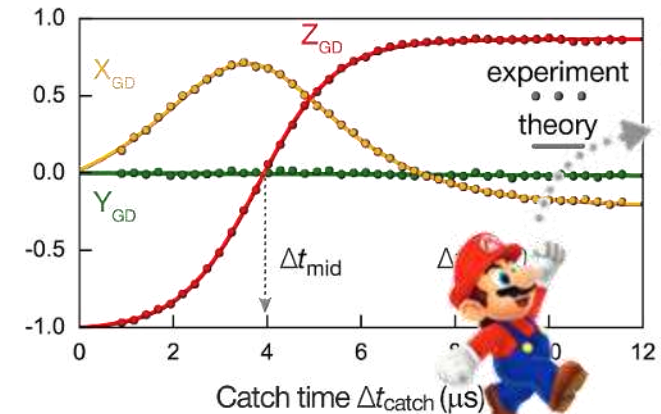
Nanosecond-latency control
feedforward, feedback
adaptive, engineerable
Off-the-shelf stability and precision of control

Versatile

1st observation of fundamental T^4
improved quantum fisher x740



90% quantum detection efficiency



@zlatko_minev



zlatko-minev.com

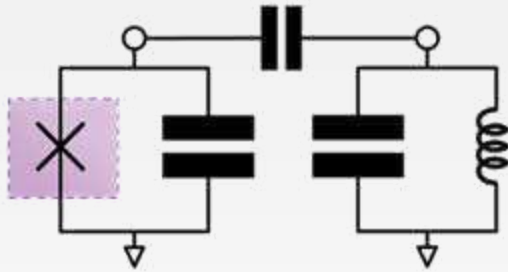
EXTRA SLIDES

JUMPS

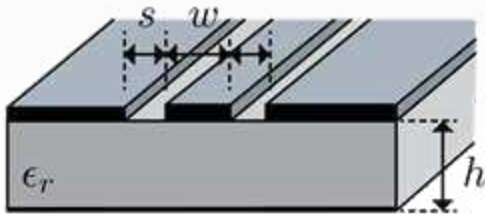
EPR

Circuit quantization

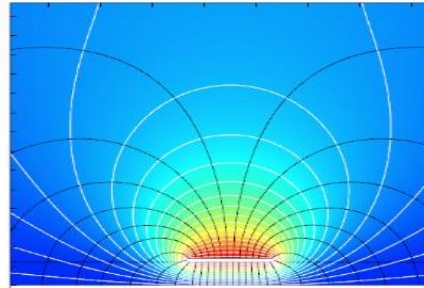
Lumped-element approximation



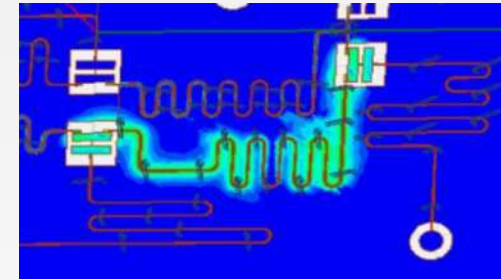
analytical



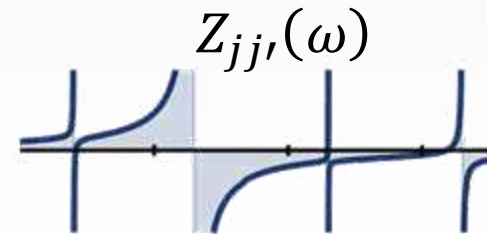
FE/boundary



Full Maxwell equations



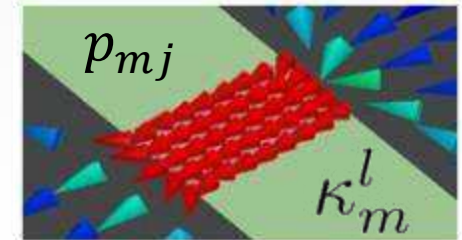
impedance*



eigen and driven

energy†

Image: Brann et al., QST (2018)



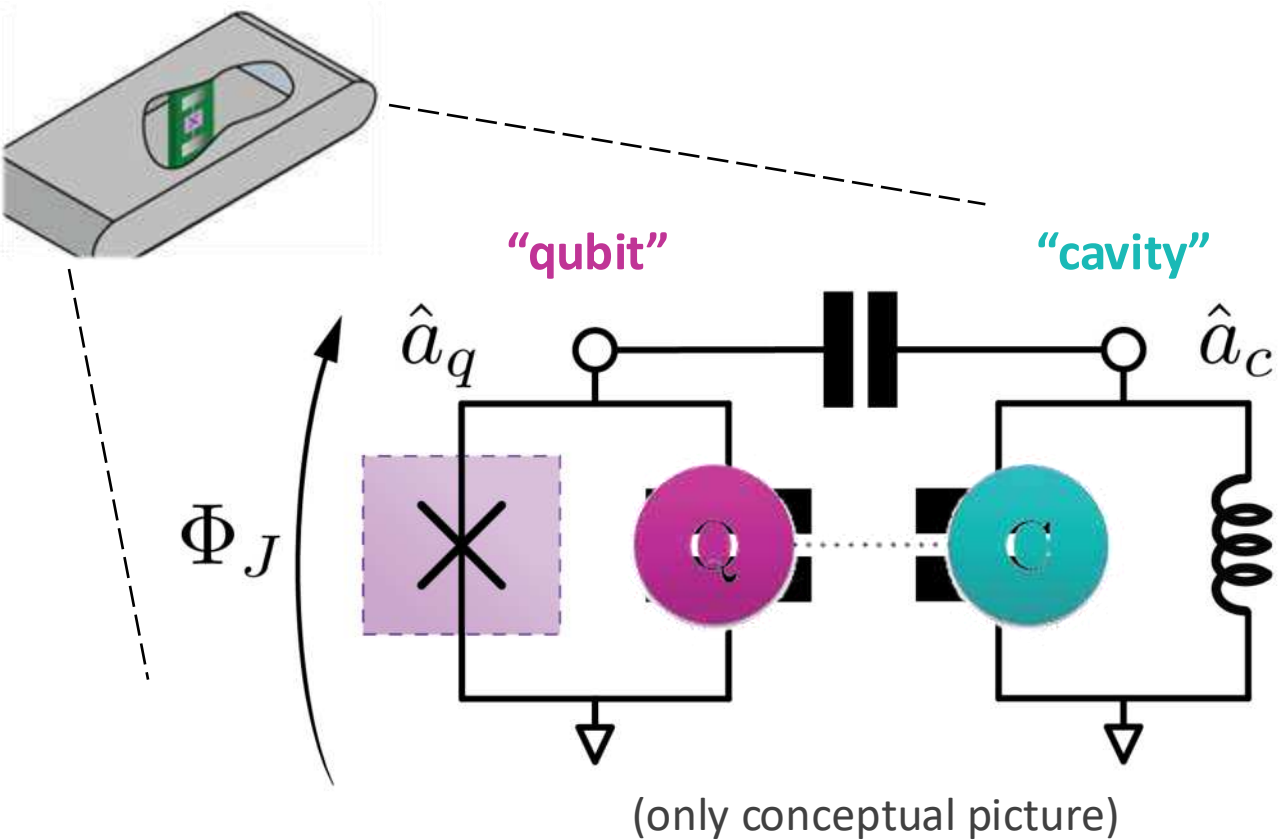
eigen

Captures more information,
complexity, accuracy

* Nigg, Paik, et al. (2012); Bourassa et al. (2012); Solgun et al. (2014, 2015, 2017) ...

† Mineev et al., arXiv:1902.10355, in prep. (2019)

Explanation of quantization



$$\hbar\omega_c \hat{a}_c^\dagger \hat{a}_c + \hbar\omega_q \hat{a}_q^\dagger \hat{a}_q$$

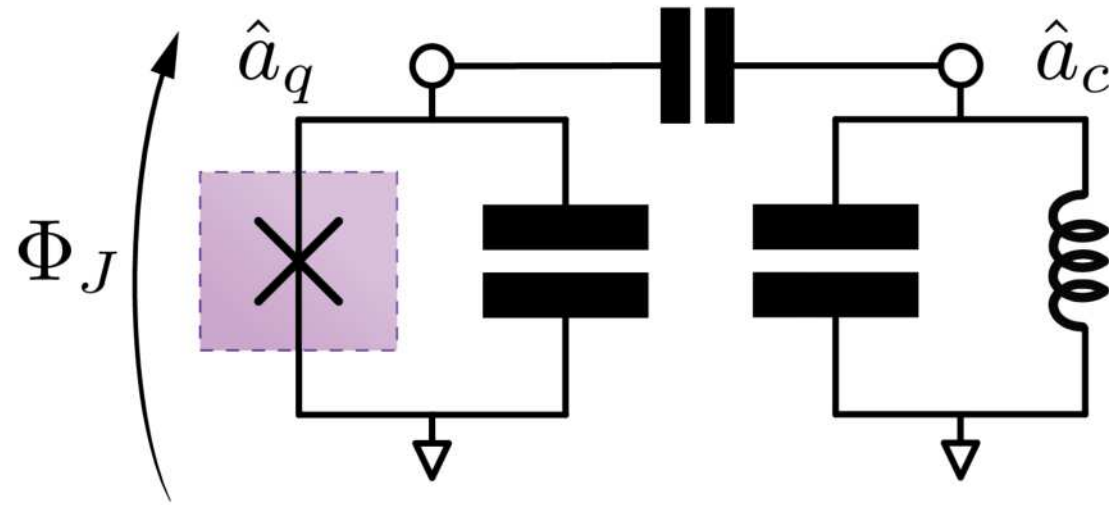
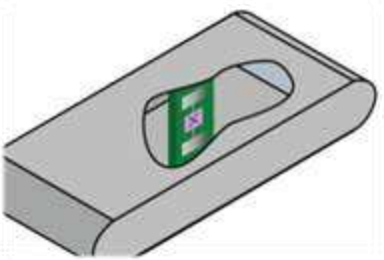
$$\hat{H}_{\text{full}} = \hat{H}_{\text{lin}} + \hat{H}_{\text{nl}}$$

$$-E_J \left(\cos(\hat{\phi}_J) + \hat{\phi}_J^2/2 \right)$$

$$\hat{\phi}_J = \phi_c^{\text{ZPF}} (\hat{a}_c^\dagger + \hat{a}_c) + \phi_q^{\text{ZPF}} (\hat{a}_q^\dagger + \hat{a}_q)$$

second quantization in eigen basis of linearized circuit

Energy-participation ratio (EPR) and the quantum bridge



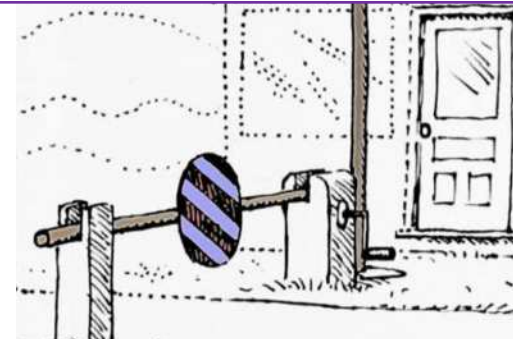
What fraction of the energy of a mode m is stored in the junction?

$$p_m = \frac{\text{Energy stored in junction}}{\text{Inductive energy stored in mode } m}$$

$$\frac{\langle n_m | : \frac{1}{2} E_J \phi_J^2 :_{\text{ZPF}} | n_m \rangle}{\langle n_m | \frac{1}{2} \hat{H}_{\text{lin}} | n_m \rangle} = \frac{\frac{1}{2} E_J \phi_J^2}{\frac{1}{2} \hbar \omega_m}$$

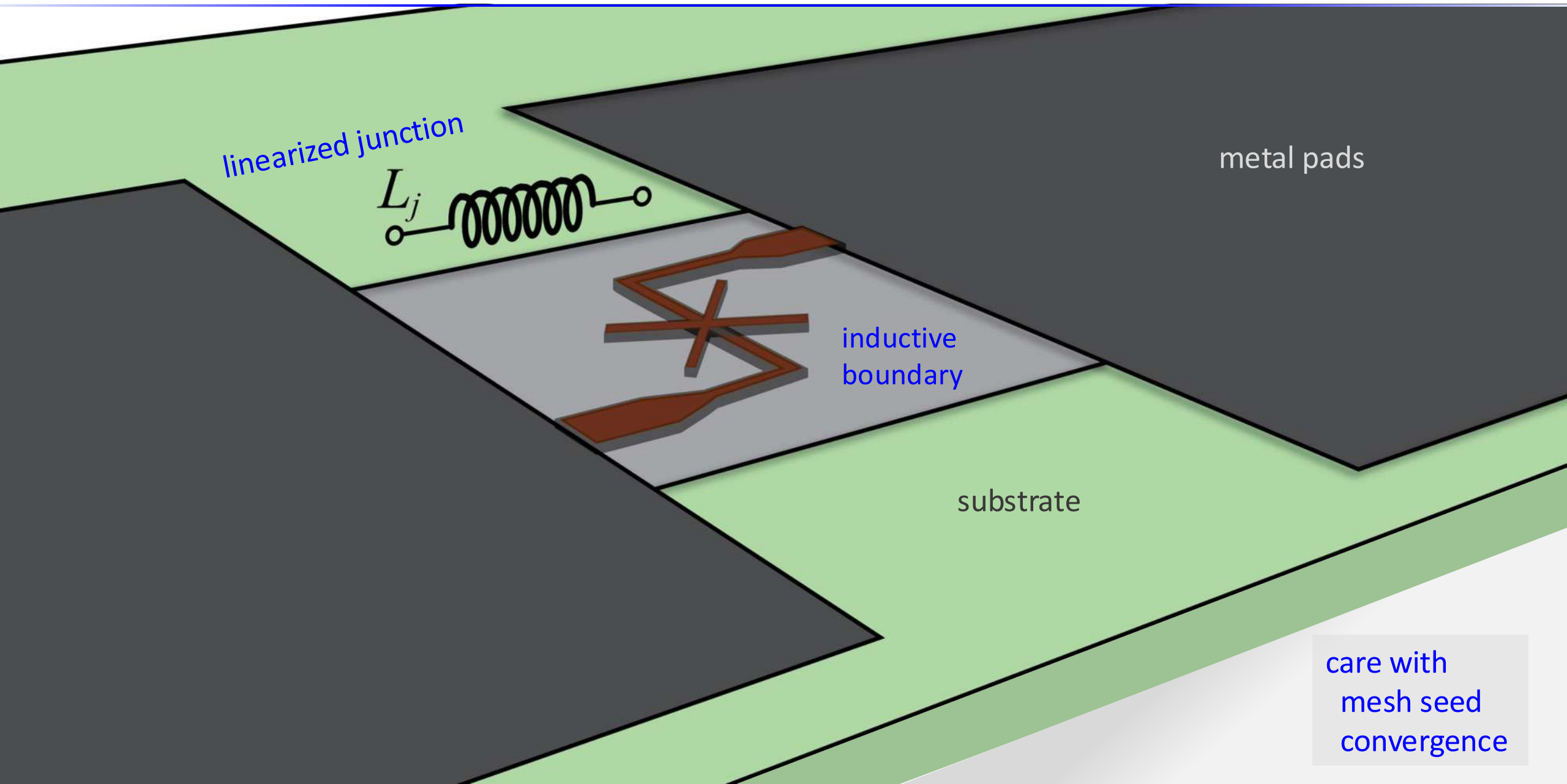


$$\hat{\phi}_J = \phi_c^{\text{ZPF}} (\hat{a}_c^\dagger + \hat{a}_c) + \phi_q^{\text{ZPF}} (\hat{a}_q^\dagger + \hat{a}_q)$$



Drawing:
Zurek, Physics Today (1991)

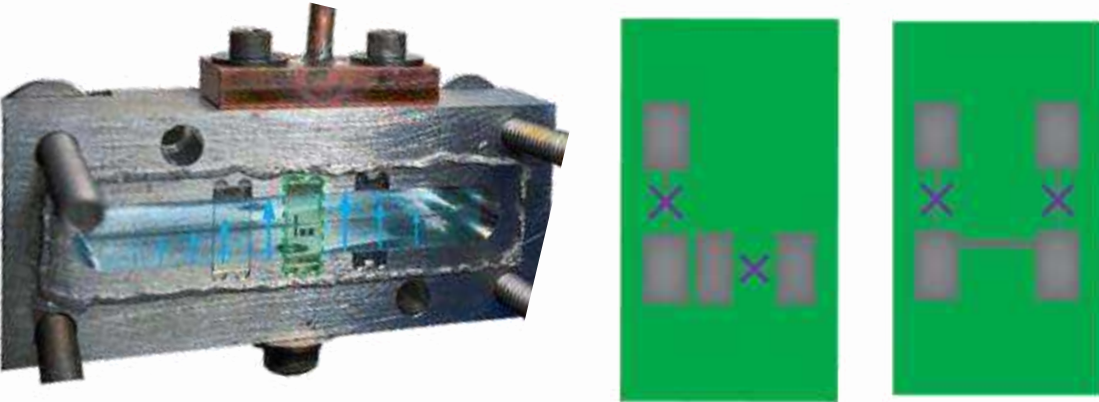
Finite-element model of linearized Junction



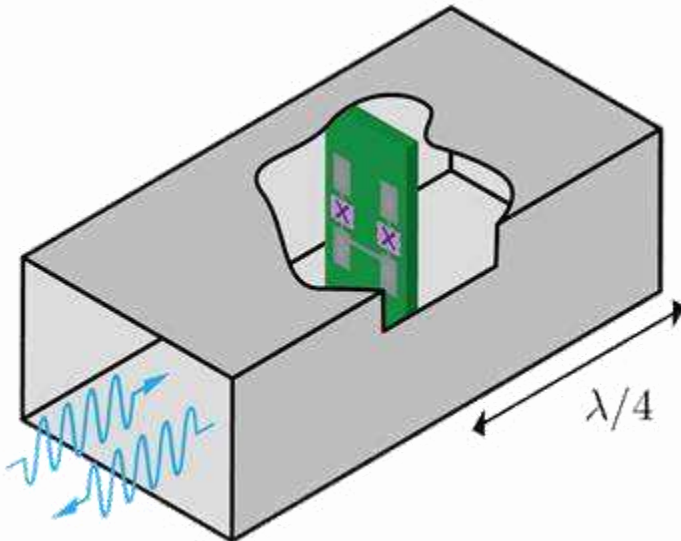
care with
mesh seed
convergence

Measured architectures and devices

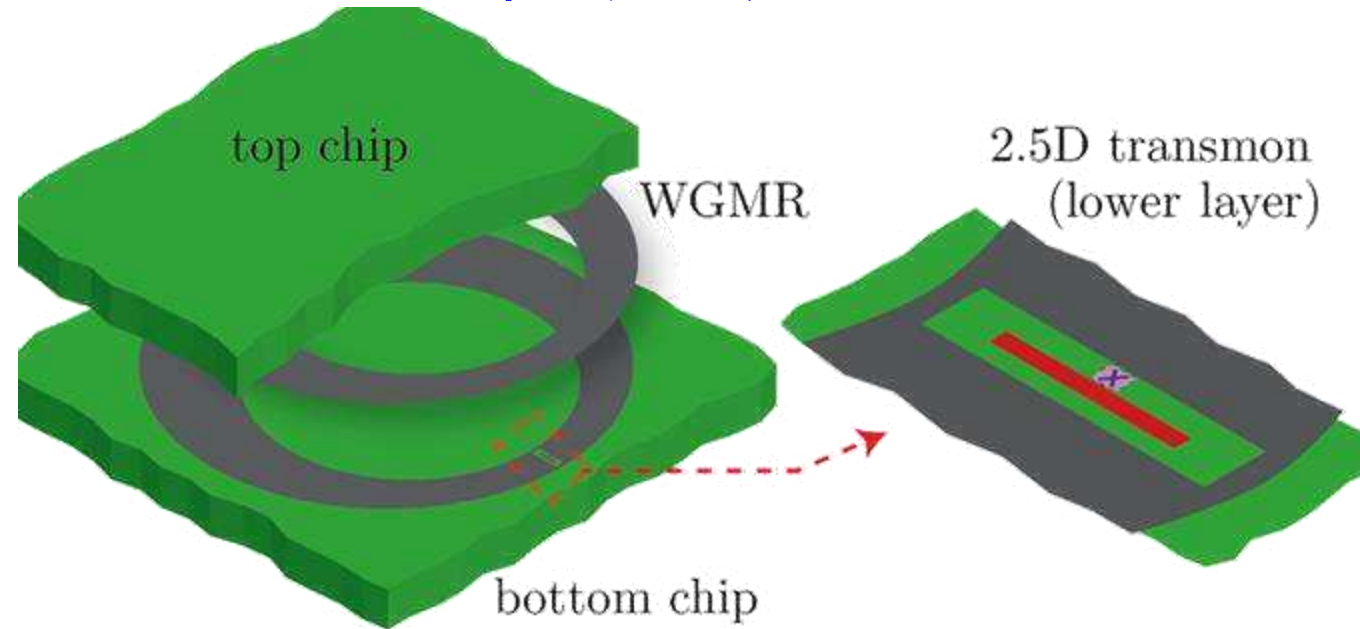
Three-dimensional (3D) *



Waveguide



Multilayer (2.5D) cQED



Minev *et al.*, APL (2013)

Minev *et al.*, WO/2016/138395 (2015)

Minev *et al.*, Phys. Rev. App. (2016)

* Minev *et al.*, arXiv:1803.00545 (2018); Related: Gambetta *et al.*, PRL (2011), Srinivasan *et al.*, PRL (2011), Dumur *et al.*, PRB (2015), Zhang *et al.*, Nature JQI (2017) ...

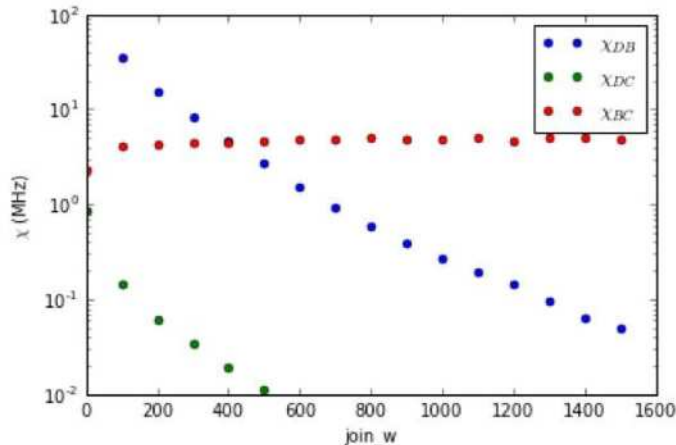
How to use the EPR approach

- Fully automated
 - control of HFSS
 - calculations
- Full numerical treatment
- Convergence checks

See my open source
project on github

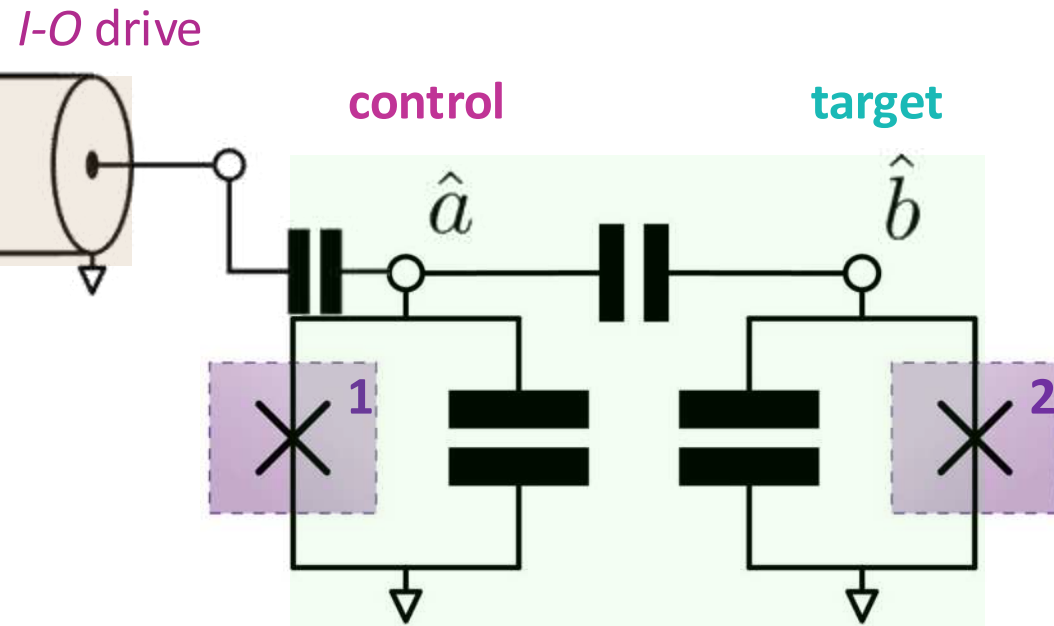


Used at Yale, ENS Paris, IBM,
Berkeley, Saclay, and others



```
1 from pyEPR import *
2
3 # 1. Project and design. Open link to HFSS controls.
4 project_info = Project_Info('c:/sims',
5                               project_name = 'two_qubit_one_cavity',
6                               design_name = 'Alice_Bob'
7                               )
8
9 # 2a. Junctions. Specify junctions in HFSS model
10 project_info.junctions['jAlice'] = {'Lj_variable': 'LJAlice', 'rect': 'qubitA'}
11 project_info.junctions['jBob'] = {'Lj_variable': 'LJBob', 'rect': 'qubitB'}
12
13 # 2b. Dissipative elements.
14 project_info.dissipative.dielectrics_bulk = ['si_substrate']
15 project_info.dissipative.dielectric_surfaces = ['interface']
16
17 # 3. Run analysis
18 epr_hfss = pyEPR_HFSS(project_info)
19 epr_hfss.do_EPR_analysis()
20
21 # 4. Hamiltonian analysis
22 epr = pyEPR_Analysis(epr_hfss.data_filename)#
23 epr.analyze_all_variations(cos_trunc = 8, fock_trunc = 7)
24 epr.plot_Hresults()
```

Cross-resonance (CR) entangling gate



$$\hat{H} = \omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} \quad \text{Eigen modes}$$

$$- \sum_{j=1}^2 \frac{E_j}{4!} \left(\phi_{aj} (\hat{a} + \hat{a}^\dagger) + \phi_{bj} (\hat{b} + \hat{b}^\dagger) \right)^4 + \dots \quad \text{Non-linear}$$

$$- i\epsilon_a(t) (\hat{a}^\dagger - \hat{a}) - i\epsilon_b(t) (\hat{b}^\dagger - \hat{b}) \quad \text{Drives}$$

$$- \frac{E_j}{4!} \frac{\partial^3 \phi}{\partial \phi_a^3} \left[\left(\frac{1}{2} \hat{a}^\dagger \hat{a} - \frac{1}{2} \right) \hat{a} \right] \left(\xi \hat{b} + \xi^* \hat{b}^\dagger \right)$$

Participation matrix

$$\begin{pmatrix} p_{a1} & p_{a2} \\ p_{b1} & p_{b2} \end{pmatrix} \approx \begin{pmatrix} 0.92 & 2 \times 10^{-4} \\ 2 \times 10^{-4} & 0.92 \end{pmatrix}$$

$$\omega_{zx} \approx \sqrt{p_{b1}} \times \frac{1}{2} \text{ GHz}$$

dispersive regime; χ_{qc} small